F79MB Statistical Models B

Lecture Notes

2026

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Chapter 1

Review of concepts of estimation and testing (and some new concepts)

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1.1 Review of estimation and testing

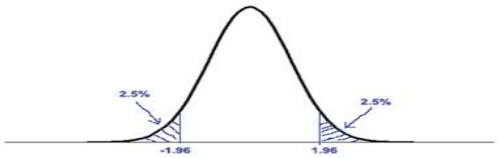
Example 1.1

The reaction time in athletics racing is measured as the time taken from the starting signal of the race to the moment an athlete has left the starting blocks. In a juniors championship, the reaction times of forty athletes racing in the 800 metres event were recorded and are given below (in seconds):

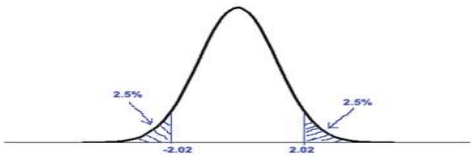
```
0.341  0.368  0.358  0.335  0.347  0.373  0.365  0.332
0.330  0.362  0.364  0.352  0.338  0.327  0.350  0.358
0.385  0.370  0.350  0.341  0.349  0.354  0.330  0.352
0.377  0.355  0.324  0.375  0.338  0.335  0.382  0.355
0.338  0.373  0.352  0.385  0.349  0.359  0.378  0.346
```

Determine a 95% confidence interval for the mean reaction time for this event. Sample mean = 0.3538, Sample standard deviation = 0.01685. Do we know the distribution of the data?

a) Since the sample size is 40, we can use critical values from the standard normal distribution employing the *Central Limit Theorem*: Find that 95% CI for μ is (0.3486, 0.3590).



b) If the sample comes from a normal distribution, we can also use critical values from the t_{39} distribution:



Find that 95% CI for μ is now (0.3484, 0.3592).

- Note: t-distribution CI slightly wider than normal distribution CI, because t-distribution takes into account extra uncertainty from estimating σ.
- Two ways that we can check whether or not the sample is normal are stem and leaf plots and QQ plots. (QQ plots covered in chapter 2)

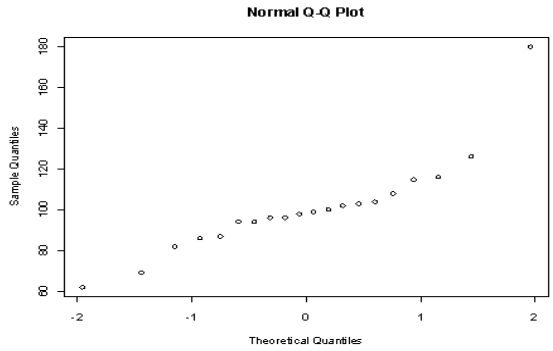
Example 1.2

A particular stretch of motorway (in a European country), has speed limit 100 kilometres per hour (kph). The speed of 20 vehicles travelling on this part of the motorway was recorded and gave the following measurements (in kph):

```
115  82   96   87  100
 69 108  116  180   99
 94  86  104   98  103
102  62   94  126   96
```

Calculate 95% confidence intervals for the true mean and s.d. of the speed of vehicles travelling on this part of the motorway. Sample mean = 100.85, Sample standard deviation = 23.8841.

Answer:
Are the data normally distributed? (QQ plots will be covered in chapter 2. For now, just need to know that for normal-distributed data, the normal QQ plot will be roughly a straight line.)



- The QQ plot shows that normality is plausible except for an outlier at 180.
- We will assume that it has been wrongly recorded and should, in fact, be 80. [Be very cautious of making this kind of assumption!]
- Calculations will be done twice, once with original value then with "corrected" value.

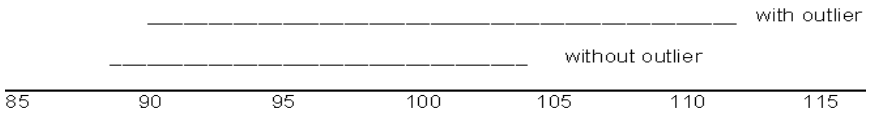
Using *original data*:

- 95% CI for mean is (89.67, 112.03)
- 95% CI for sd is (18.16, 34.88)

Now we will repeat the process using the *"corrected" value*: Sample mean = 95.85 Sample standard deviation = 15.40.

- 95% CI for mean (88.6,103.1)
- 95% CI for sd (11.71, 22.50)

95% CIs for mean



The outlier has caused increases in both location and spread.

1.2 A CI for the median

Resistant measure: One that is not influenced by extremely high or extremely low data values. The median tends to be more resistant than mean, and interquartile range tends to be more resistant than standard deviation.

Example 1.3:

Calculate a 95% confidence interval for the **median** of the results given in the '`eurodist`' dataset in `R`. The data give the road distances (in km) between 21 cities in Europe. The data are taken from a table in The Cambridge Encyclopaedia.

```
>data(eurodist)
```

Part of the table of distances is shown below:

	Ath	Bar	Bru	Cal	Cher	Col
Barcelona	3313					
Brussels	2963	1318				
Calais	3175	1326	204			
Cherbourg	3339	1294	583	460		
Cologne	2762	1498	206	409	785	
Copenhagen	3276	2218	966	1136	1545	760

General theory: A possible estimator of the population median (m) is the sample median $\hat{m}$. Asymptotically, i.e. for a large sample, it can be shown (we won't prove this) that approximately

$$\hat{m} \sim N\left(m,\ \left(\frac{1}{2f(m)\sqrt{n}}\right)^2\right)$$

where $f(m)$ is the density of the underlying population at the median m. For computation purposes $f(m)$ may need to be estimated, for instance by $f(\hat{m})$. The standard error of $\hat{m}$ is, from above,

$$s.e.(\hat{m}) = \frac{1}{2f(m)\sqrt{n}}.$$

For a *normally distributed population*, the standard error of $\hat{m}$ simplifies to

$$s.e.(\hat{m}) = \sigma\sqrt{\frac{\pi}{2n}}$$

Proof: See the extra file for Example 1.3 given on Canvas.

Note that

$$\text{s.e.(sample median)} = \sigma\sqrt{\frac{\pi}{2n}} = \sqrt{\frac{\pi}{2}}\frac{\sigma}{\sqrt{n}} \approx 1.25\frac{\sigma}{\sqrt{n}}$$

and recall that

$$\text{s.e.(sample mean)} = \frac{\sigma}{\sqrt{n}}$$

So, in the case of (approximately) normally distributed data, the median is a less efficient estimator than the sample mean (confidence intervals are wider). However, for longer-tailed distributions the sample median is a more efficient estimator of location than the sample mean.

Example 1.3 answer:

Pairwise distances between 21 cities, so sample size $n = \frac{21 \times 20}{2} = 210$.

Sample mean = 1505.148 Median = 1311.5 Standard deviation = 898.7842

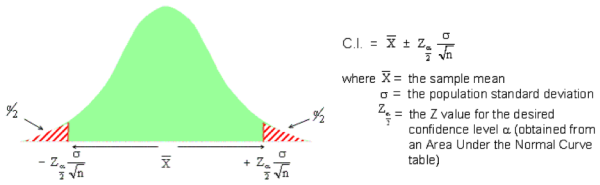
Approximate normality seems reasonably plausible, although with some evidence of skewness and outliers (check with stem and leaf plot). So the 95% confidence interval for the median is (1159.1, 1463.9). For comparison, a 95% CI for the mean is (1383.6, 1626.7). Note that CI for mean is narrower than CI for median.

1.3 Chapter 1 recap

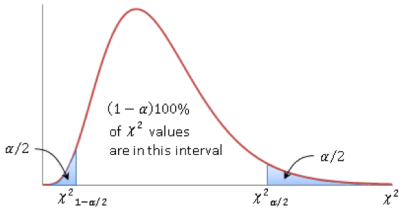
For Mean with n large ($n \geq 30$, say)	Use Z-distribution Confidence Interval	C.I. $= \bar{X} \pm Z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}$
For Mean with unknown variance, Normal data, and n small ($n < 30$, say)	Use t-distribution Confidence Interval	C.I. $= \bar{X} \pm t_{n-1,\frac{\alpha}{2}} \frac{S}{\sqrt{n}}$
For Variance	Use χ^2-distribution Confidence Interval	C.I. $= \left[\frac{(n-1)S^2}{\chi^2_{n-1,\frac{\alpha}{2}}}, \frac{(n-1)S^2}{\chi^2_{n-1,1-\frac{\alpha}{2}}} \right]$
For Proportion, if $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$	Use Z-distribution Confidence Interval	C.I. $= \hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

1.4 Summary

- $100(1-\alpha)\%$ CI for μ:



- $100(1-\alpha)\%$ CI for σ^2:



- **Median**:
  - Definition of the population median (m): If a continuous random variable $X \in (-\infty, \infty)$ follows a distribution with probability density function $f(x)$, then the median, m, is defined as the value such that

$$\int_{-\infty}^{m} f(x)\, dx \;=\; \frac{1}{2}$$

(i.e. it is the value that splits the probability distribution into 2 equal halves).

Chapter 2

Checking the fit of distributions to data

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2.1 Quantiles and Quantile function

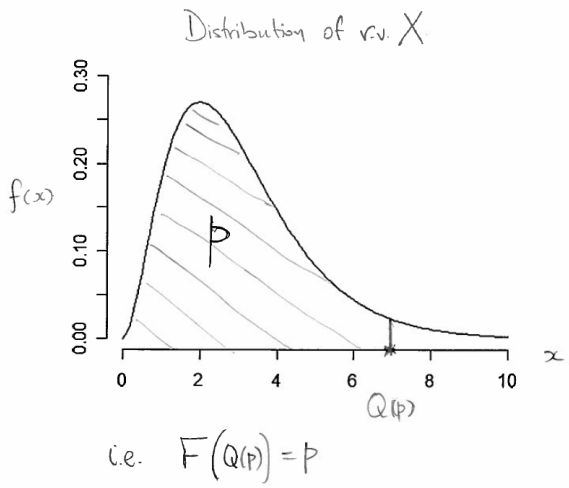
Recall that the pth **quantile** of the distribution of a random variable X is $Q(p)$ such that

$$P\{X \leq Q(p)\} = p, \quad \text{i.e.} \quad F\{Q(p)\} = p$$

with $F(\cdot)$ being the cdf of X. Then, the (theoretical) **quantile function** $Q(\cdot)$ of a distribution is given by

$$Q(p) = F^{-1}(p)$$

i.e. it is the inverse of the cdf. So, for example the pth quantile of the distribution of a random variable X is shown schematically below:



Idealised samples

Quantile functions can help produce 'idealised' samples from probability distributions. First define a sequence of n probability points, equally spaced between 0 and 1, i.e.

$$p_i = \frac{i - 0.5}{n}, \quad i = 1 \ldots, n.$$

For example, for $n = 20$:

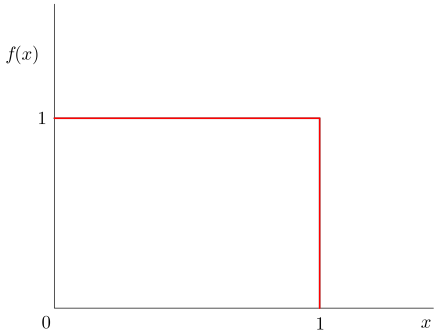
$$p_1 = 0.025, \;\; p_2 = 0.075, \ldots, \;\; p_{20} = 0.975.$$

Then, if a r.v. X has quantile function $Q(\cdot)$, we have (by definition):

$$P\{X \leq Q(p_i)\} = p_i \Rightarrow F\{Q(p_i)\} = p_i, \text{ for } i = 1. \ldots, n$$

and therefore, given $p_1, p_2, \ldots, p_n$, the quantiles $Q(p_1), Q(p_2), \ldots, Q(p_n)$ can be regarded as an idealised sample from the distribution with cdf $F(\cdot)$.

The Uniform $U(0, 1)$ Distribution



Remember

$$F(x) = x \Rightarrow F^{-1}(x) = x \quad \text{i.e.} \quad F(x) = F^{-1}(x)$$

which gives the quantile function as

$$Q(p) = p \quad \text{e.g.} \quad Q(0.2) = 0.2$$

Hence $p_1, \ldots, p_n$ may be regarded as a sample of size n from the $U(0,1)$ distribution.

The Normal $N(0,1)$ Distribution

For $Z \sim N(0,1)$

$$\Phi(z) = p \quad \text{and} \quad Q(p) = \Phi^{-1}(p) = z$$

with

$$\Phi(z) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-x^2/2}\, dx$$

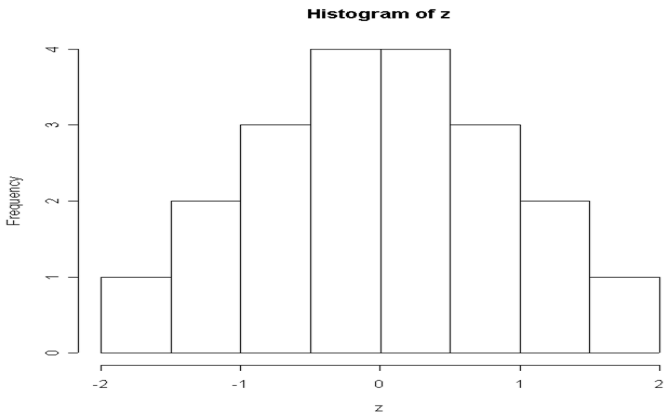
E.g. for $p = 0.025$, we have

$$\Phi(-1.96) = 0.025, \quad \text{so} \quad \Phi^{-1}(0.025) = Q(0.025) = -1.96$$

In R, `qnorm(ppoints(20))` gives the $N(0,1)$ quantiles for 20 equally spaced probability points in $(0,1)$.

```
> z=qnorm(ppoints(20))
> z
 [1] -1.95996398 -1.43953147 -1.15034938 -0.93458929 -0.75541503 -0.59776013
 [7] -0.45376219 -0.31863936 -0.18911843 -0.06270678  0.06270678  0.18911843
[13]  0.31863936  0.45376219  0.59776013  0.75541503  0.93458929  1.15034938
[19]  1.43953147  1.95996398
```

A histogram can be plotted using `hist(z)`



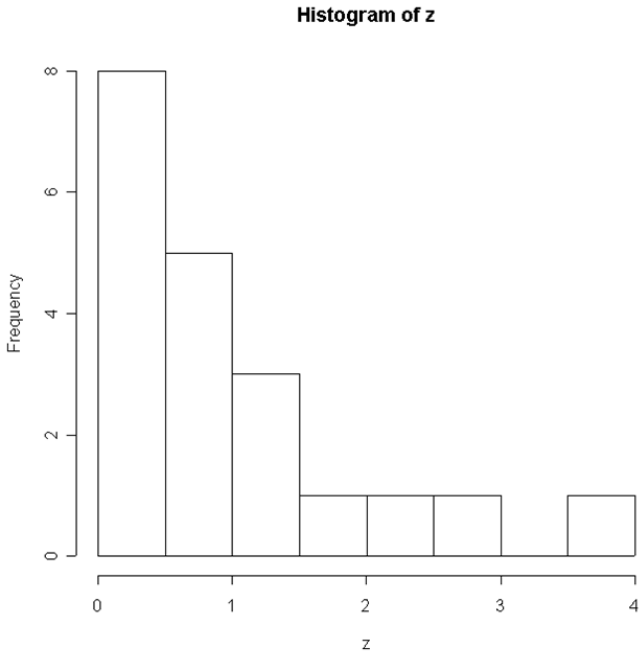
Exponential Distribution

The $Exp(\lambda)$ distribution has quantile function Q given by

$$Q(p) = -\ln(1-p)/\lambda \qquad \text{(exercise!)}$$

In R it is given by the function `qexp(p, rate = λ)` (with default $\lambda = 1$). Thus an $Exp(1)$ sample of size 20, together with its histogram, are obtained as follows:

```
> z=qexp(ppoints(20))
> z
 [1] 0.02531781 0.07796154 0.13353139 0.19237189 0.25489225 0.32158362
 [7] 0.39304259 0.47000363 0.55338524 0.64435702 0.74444047 0.85566611
[13] 0.98082925 1.12393010 1.29098418 1.49165488 1.74296931 2.07944154
[19] 2.59026717 3.68887945
> hist(z)
```



Empirical Quantile function

Suppose that we have a set of n observations $y_1, \ldots, y_n$ of a variable Y. Let

$$y_{(1)} \leq \ldots\ldots\ldots \leq y_{(n)}$$

be their **order** statistics (ordered sample). If we determine $p_1, p_2, \ldots, p_n$ as before, then we define the **empirical quantile function** Q_e of these data by

$$Q_e(p_i) = y_{(i)}$$

(*Note that alternative definitions are also given in the literature.*)

In general, $Q_e(p)$ for other values of p in the interval $(0, 1)$ is given with linear interpolation. Thus, for any probability p, $Q_e(p)$ is the value of the data that lies above a fraction, p, of the observations.

Remember: the three **quartiles** of the distribution of the data are given by $Q_e(1/4)$, $Q_e(1/2)$ and $Q_e(3/4)$

Example 2.1: Maximum daily ozone concentrations

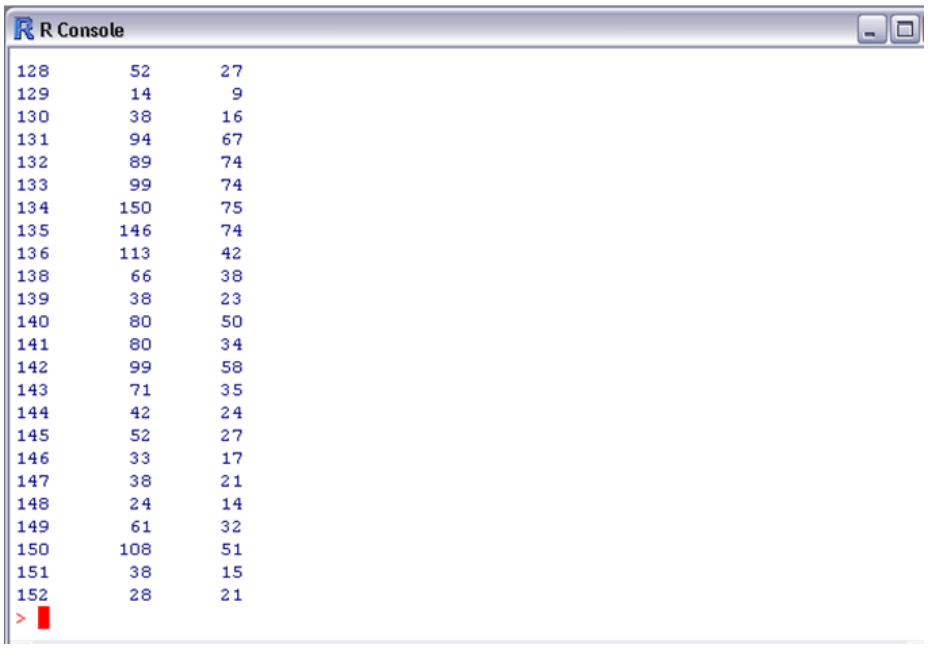
The R data frame **ozone** is available on Canvas. It gives maximum daily ozone concentrations, in parts per billion, measured at Stamford, Connecticut and Yonkers, New York, during the period May – September one year. Data are given only for the 132 days (out of a total of 152) on which readings were successfully obtained at both locations.



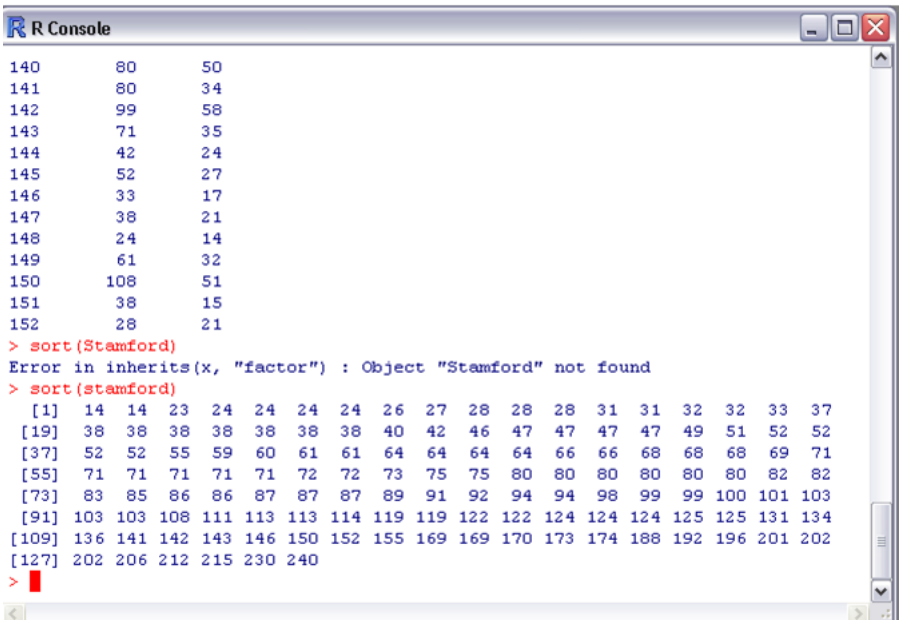
Here `ozone` is a 'data frame' in `R`. We use the command `attach()` to attach names to the variables (columns) of a dataframe.

```
> ozone=dget("ozone.R")
> attach(ozone)
```

```
> ozone
   stamford yonkers
1        66      47
2        52      37
7        49      45
8        64      52
9        68      51
10       26      22
11       86      27
12       52      25
14       75      55
15       87      72
16      188     132
18      103     106
19       82      42
20       71      45
21      103      80
22      240     107
23       31      21
24       40      50
25       47      31
26       51      37
27       31      19
28       47      33
29       14      22
```



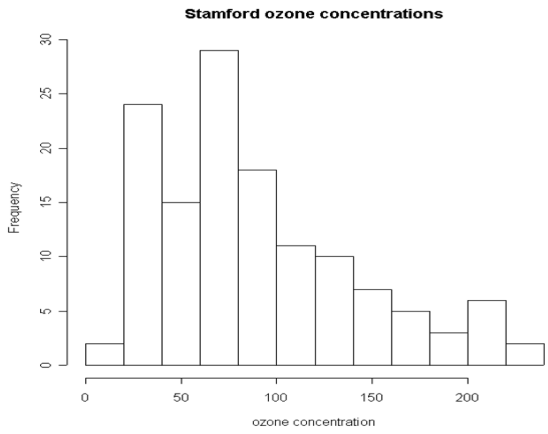
The sorted Stamford data are given by



The following R code

```
> hist(stamford, xlab="ozone concentration",
main="Stamford ozone concentrations")
```

produces the histogram shown below:



A simple summary is given by

```
summary(stamford)
```

Min	1st Qu.	Median	Mean	3rd Qu.	Max.
14.00	50.50	80.00	89.87	119.80	240.00

Much the same is achieved with the use of the R (empirical) quantile function, which by default evaluates this at $p = (0.00, 0.25, 0.50, 0.75, 1.00)$

```
quantile(stamford)
```

0%	25%	50%	75%	100%
14.00	50.50	80.00	119.80	240.00

Both the histogram and the summaries show the data to be quite positively skewed. This is quite typical of data constrained to be positive. We can specify any vector of points at which to evaluate the quantile function:

```
>p = seq(0,1,0.1)
>p
```

```
[1] 0.0  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
```

```
> p=seq(0,1,0.1)
> quantile(stamford,probs=p)
   0%   10%   20%   30%   40%   50%   60%   70%   80%   90%  100%
 14.0  31.1  42.8  59.3  69.8  80.0  88.2 106.5 125.0 169.9 240.0
```

2.2 Quantile Quantile plots

Q-Q Plots

We wish to investigate whether observations $y_1, \ldots, y_n$ may reasonably be regarded as a random sample from some theoretical distribution.

A Q-Q plot compares the *empirical* quantile function Q_e of the data with the *theoretical* quantile function Q of the distribution.

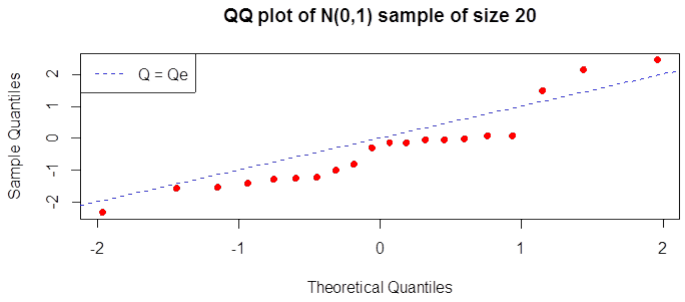
We plot $y_{(i)} (= Q_e(p_i))$ (on the y axis) against $Q(p_i)$ (on the x axis) for the probability points p_1, p_2 etc. as defined earlier. If the points on the graph form an approximately straight line with slope 1 and intercept 0, the theoretical distribution is said to be compatible with the observed values, i.e.

$$y_{(i)} \approx Q(p_i)$$

For example, data simulated from $N(0,1)$ distn:

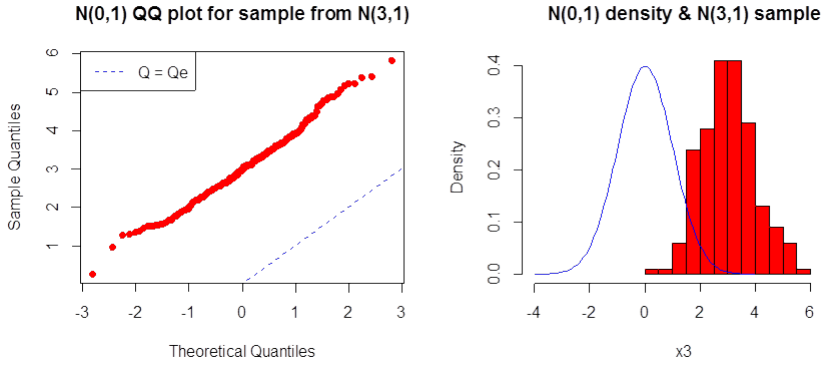


But, be careful when data set is small – plot will be far from perfect. E.g. with $n = 20$ observations from $N(0,1)$ distn:

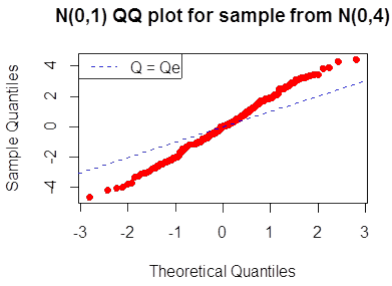


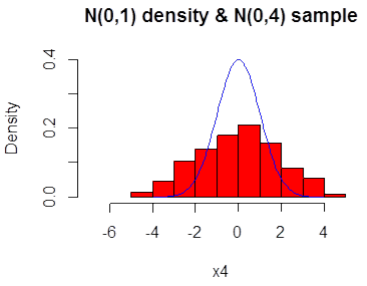
Q-Q plots can be used to identify how two distributions (e.g. data, theoretical) differ in location, spread or shape.

E.g. different location:

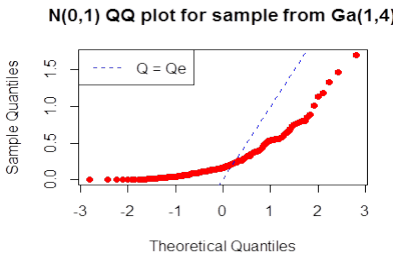


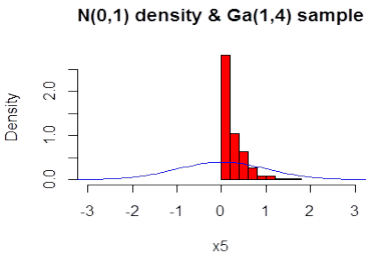
Different spread (same location):



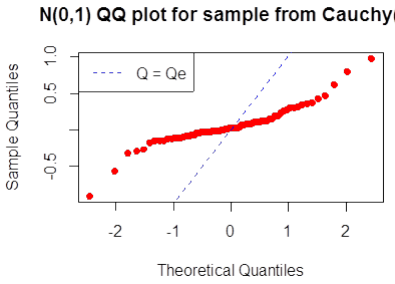


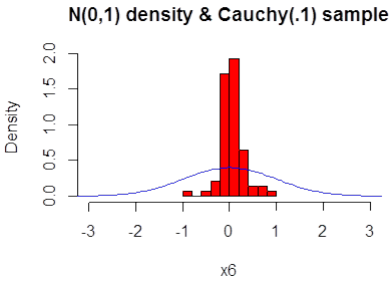
Different shape:



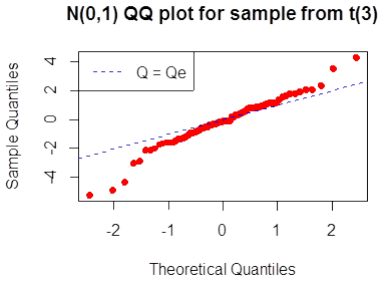


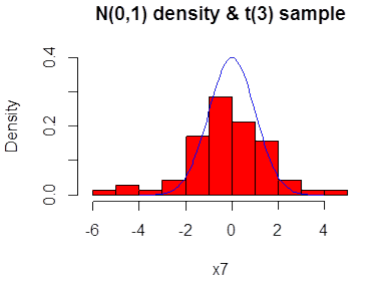
Short tails compared to normal distn:



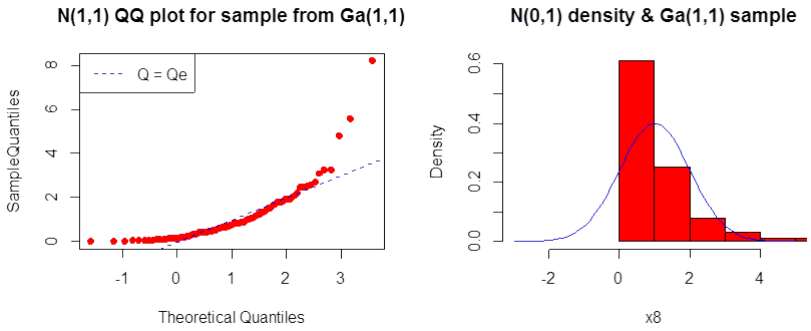


Long tails compared to normal distn:

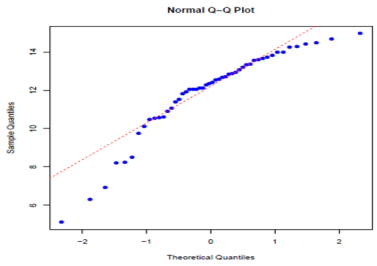




Positive skewness:



Negative skewness:



Refinement

If the data do not lie on the specific line $y_{(i)} = Q(p_i)$, but instead lie on a line with slope b and intercept a,

$$\text{i.e.} \quad y_{(i)} \approx a + bQ(p_i),$$

this indicates compatibility with the appropriate distribution scaled by b and shifted by a. Sometimes (not always!) the linear transformation allows us to identify (i.e. give the name of) the distribution. For example if Q is the quantile function of the $N(0,1)$ distribution, then the relation suggests compatibility with the $N(a, b^2)$ distribution.

Example 2.1 (cont.):Stamford Ozone Concentrations

We check for normality. A normal Q-Q plot compares empirical quantiles of the data with the quantiles of the $N(0,1)$ distribution. The `R` function `qqnorm` constructs the plot.

```
R Console

R : Copyright 2003, The R Development Core Team
Version 1.6.2  (2003-01-10)

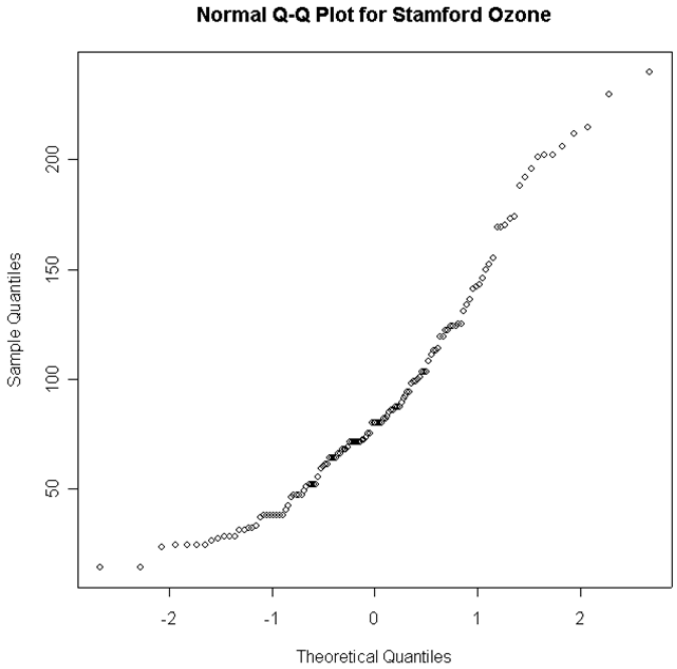
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type `license()' or `licence()' for distribution details.

R is a collaborative project with many contributors.
Type `contributors()' for more information.

Type `demo()' for some demos, `help()' for on-line help, or
`help.start()' for a HTML browser interface to help.
Type `q()' to quit R.

[Previously saved workspace restored]

> ozone=dget("ozone.R")
> attach(ozone)
> p=seq(0,1,0.1)
> quantile(stamford,probs=p)
   0%   10%   20%   30%   40%   50%   60%   70%   80%   90%  100%
 14.0  31.1  42.8  59.3  69.8  80.0  88.2 106.5 125.0 169.9 240.0
> qqnorm(stamford,main="Normal Q-Q Plot for Stamford Ozone")
>
```



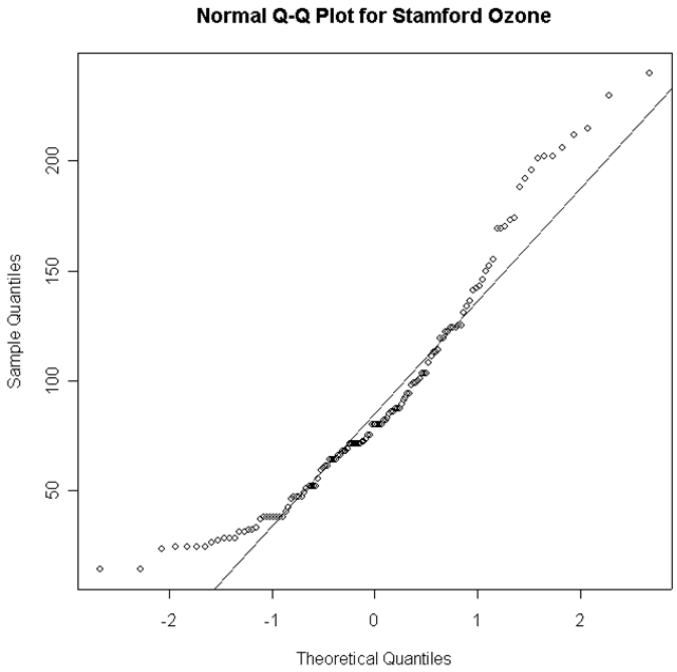
- Is there a straight line that (approximately) goes through the points?
- In any case, that line would not go through zero, so intercept $\neq 0$.
- Also, slope $\neq 1$.

The function `qqline` adds the straight line

$$y = a + bx$$

corresponding to the normal distribution $N(a, b^2)$ whose lower and upper quartiles match those of the data (see later). This is a resistant fit, unaffected by one or two extreme observations.

```
<environment: namespace:base>
> qqnorm(stamford,main="Normal Q-Q Plot for Stamford Ozone")
> qqline(stamford)
>
```

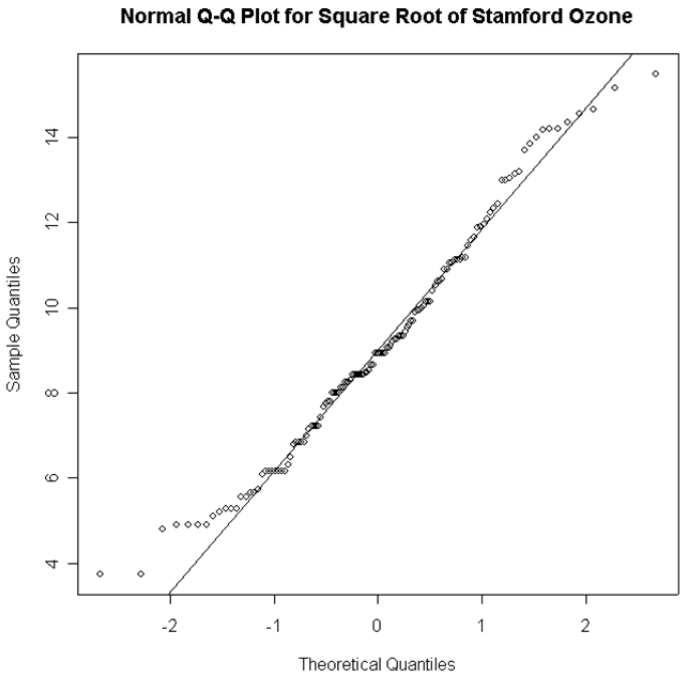


Plot indicates positive skewness.

The lack of linearity in the plotted quantiles suggests that the data are not well modelled by a normal distribution. Rather, the convex shape indicates **positive skewness** (as was seen in the histogram). We investigate whether a transformation would help. A check is carried out as to whether the **square roots** of the Stamford ozone concentrations can be modelled by a normal distribution.

```
>
>
>
> qqnorm(stamford^0.5,main="Normal Q-Q Plot for Square Root of Stamford Ozone")
> qqline(stamford^0.5)
>
```

Notice that, when using `R`, $x\hat{}y$ gives the yth power of x, and, of course, a power of 0.5 gives a square root.



The qqline suggests that the quantiles are reasonably well fitted by the straight line which has slope 2.844 and intercept 9.0 (see derivation on next slide). Thus the data are reasonably well modelled by the $N(9, 2.844^2)$ distribution. Note that the sample mean and standard deviation are 9.1 and 2.715 respectively.

The equation of the straight line is obtained (fitted to the data) by using the first and third quartiles to determine two points on the line, with co-ordinates (x_1, y_1) and (x_2, y_2) given by

$$(Q(0.25),\ Q_e(0.25)) \text{ and } (Q(0.75),\ Q_e(0.75))$$

Appropriate `R` commands give these as

$$(-0.675, 7.106) \quad \text{and} \quad (0.675, 10.940)$$

Therefore:

$$\text{Slope of line: } b = \tfrac{y_2 - y_1}{x_2 - x_1} = 2.844; \qquad \text{Intercept: } y_1 = a + bx_1 \Rightarrow a = 9.023.$$

So the line on the graph is

$$Q_e(p_i) = 9.023 + 2.844 Q(p_i)$$

implying that $\sqrt{\text{data}}$ follow $N(9.023, 2.844^2)$.

2.3 Checking data modelled by exponential distribution

Example 2.2

The `R` vector `abbey` in the package MASS gives 31 determinations of nickel content ($\mu g\ g^{-1}$) in a Canadian syenite rock.

```
> library(MASS)
> data(abbey)
> abbey
 [1]   5.2   6.5   6.9   7.0   7.0   7.0   7.4   8.0   8.0   8.0   8.0   8.5
[13]   9.0   9.0  10.0  11.0  11.0  12.0  12.0  13.7  14.0  14.0  14.0  16.0
[25]  17.0  17.0  18.0  24.0  28.0  34.0 125.0
>
```

We check whether the data are reasonably modelled by an exponential distribution.

There is no predefined function in R to construct an exponential Q-Q plot, so we have to work from first principles. We start with the Exp(1) distribution and need to plot $Q_e(p_i) = y_{(i)}$ against $Q(p_i)$, $i = 1, \ldots, 31$.

`qexp(ppoints(31))` gives the theoretical quantile values at 31 probability points for the Exp(1) distribution.

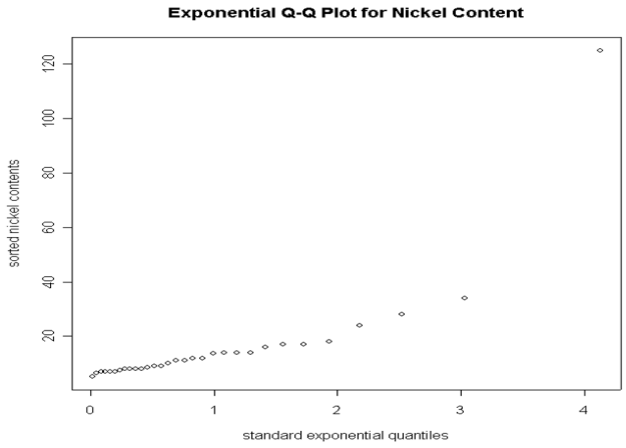
```
> qexp(ppoints(31))
 [1] 0.01626052 0.04959694 0.08408312 0.11980120 0.15684247 0.19530875
 [7] 0.23531409 0.27698678 0.32047190 0.36593427 0.41356232 0.46357274
[13] 0.51621647 0.57178632 0.63062682 0.69314718 0.75983856 0.83129752
[19] 0.90825856 0.99164017 1.08261195 1.18269541 1.29392104 1.41908418
[25] 1.56218503 1.72923911 1.92990981 2.18122424 2.51769647 3.02852210
[31] 4.12713439
>
```

`sort(abbey)` gives the sorted experimental values

```
> sort(abbey)
 [1]   5.2   6.5   6.9   7.0   7.0   7.0   7.4   8.0   8.0   8.0   8.0   8.5
[13]   9.0   9.0  10.0  11.0  11.0  12.0  12.0  13.7  14.0  14.0  14.0  16.0
[25]  17.0  17.0  18.0  24.0  28.0  34.0 125.0
>
```

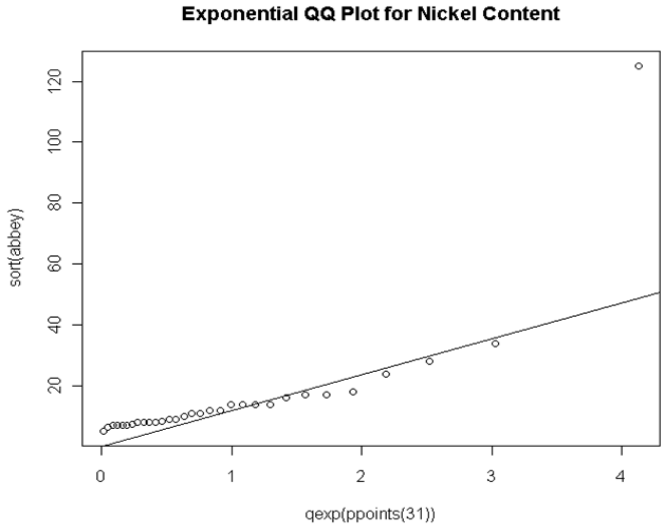
The following command produces a Q-Q plot with axes labelled accordingly.

```
Error: syntax error
> plot(qexp(ppoints(31)),sort(abbey),xlab="standard exponential quantiles",
+ ylab="sorted nickel contents",main="Exponential Q-Q Plot for Nickel Content")
>
```



Ignoring the extreme point, we can fit a straight line to the points (see later) and we find:

$$\hat{Q}_e(p) = 11.84Q(p)$$



So the original data (represented by r.v., say, X) seem to be described by the theoretical quantiles scaled by a factor of 11.84. Therefore, the data follow the distribution of the transformed variable $Y = 11.84X$. Since the QQ plot was produced assuming $X \sim$ Exp(1) we have

$$Y = 11.84X \sim \text{Exp}(1/11.84)$$

i.e. Exp(0.0845). If we ignore the highest observation, the data appear to be reasonably compatible with an Exp(0.0845) distribution (mean = 11.84).

The line has been drawn using a *resistant regression* technique (more later). We constrain it to go through the origin (hence the -1 in the code below)

```
> lqs(sort(abbey)~qexp(ppoints(31))-1)
Call:
lqs.formula(formula = sort(abbey) ~ qexp(ppoints(31)) - 1)

Coefficients:
qexp(ppoints(31))
            11.84

Scale estimates 4.853 3.529
```

Note that the line is then drawn on the plot that is already there using the `abline` command:

```
> plot(sort(abbey)~qexp(ppoints(31)))
> abline(0,11.84)
>
```

However, the probability that an Exp(0.0845) distribution would produce a value as extreme as the highest observation (125) is very small (2.6×10^{-5}). So there must be

considerable doubt about the suitability of an exponential model here. A further possibility is that some transformation of the data may be modelled by a normal distribution.

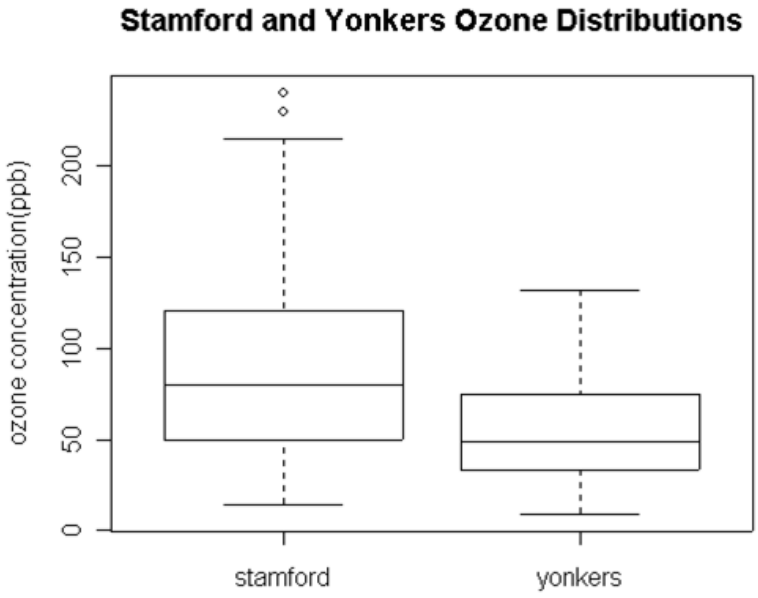
(*Try this at practicals*)

Boxplots

These are useful for comparing distributions of the same variable for different groups. The R code:

```
> boxplot(ozone,ylab="ozone concentration(ppb)",main="Stamford and Yonkers Ozone Distributions")
>
```

produces the following boxplot



Recall that the lower and upper ends of each box give the first and third quartiles of the corresponding distribution and the centre line indicates the median.

Chapter 3

Goodness of fit tests

Contents

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3.2 Kolmogorov-Smirnov goodness-of-fit tests 27

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3.1 χ^2 goodness of fit tests

These are used to test if a sample can be reasonably regarded as coming from a specified distribution.

Hypotheses:

Case (i): The parameters of the distribution are fully specified in the null hypothesis.

H_0 : The data come from a distribution with probability function $f(x;\theta_0)$, where θ_0 is a specified (vector of) parameter(s).

H_1 : The data don't follow this distribution.

Case (ii): The distribution is not fully specified.

H_0 : The data come from a distribution with probability function $f(x;\Theta)$

H_1 : The data don't follow this distribution.

The test is based on the χ^2 statistic given as

$$X^2 = \sum_{i=1}^{k} \frac{(f_i - e_i)^2}{e_i}$$

where f_i and e_i are observed and expected frequencies as determined below.

Discrete distributions

The random variable under consideration takes k discrete values, with observed frequencies f_i and expected frequencies (under the model specified in H_0) e_i, $i = 1, \ldots, k$.

Continuous distributions

Divide the range of the random variable into k cells; f_i are the observed data frequencies in cells $i = 1, \ldots, k$, and e_i are the corresponding expected frequencies under the hypothesised distribution (obtained using the cumulative distribution function).

In both cases, we have that under H_0 (approximately)

$$X^2 \sim \chi^2_{k-d-1}$$

where d is the number of model parameters estimated using the data (depending on H_0 in case (i) or (ii) above). This sampling distribution can be used to obtain p-values (or critical values) for testing. Reject H_0 if the differences between observed and expected frequencies are large, i.e. if the value of the χ^2 statistic is large with reference to its sampling distribution.

Note: The value of X^2 is sensitive to small values of e_i, in which case the approximate χ^2_{k-d-1} sampling distribution is less valid. In practice, we combine adjacent cells so that $e_i \geq 5$ for all i.

Example 2.1 (cont.): Test the hypothesis that the Stamford ozone data come from a normal distribution.

Solution: ...

Test also whether a normal distribution fits adequately a suitable transformation of the data. [*Exercise*]

Example 2.1 (cont.) χ^2-test

Cell	f_i	e_i	$(f_i - e_i)^2/e_i$
[0,25)	7	8.531	...
[25,50)	26	15.246	
[50,75)	29	21.785	
[75,100)	25	24.886	
[100,125)	17	22.729	
[125,150)	9	16.597	
[150,175)	8	9.689	
[175,200)	3	4.522	6.712
[200,225)	6	1.687	
[225,250)	2	0.503	

$\chi^2 = 31.458$

With last 3 cells grouped: $\chi^2 = 18.205$

3.2 Kolmogorov-Smirnov goodness-of-fit tests

The Kolmogorov-Smirnov (K-S) test can be used to decide if a sample comes from a population with a specific distribution. In this case it is also referred to as the "one-sample" K-S test. It is a nonparametric test, i.e. it does not depend on the underlying distribution being tested. The test is based on the empirical cumulative distribution function, $F_n(x)$, of the sample. For a sample of n i.i.d. variables X_i, this is defined as

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} I\{X_i \leq x\}$$

where $I\{\}$ is the indicator function. Given n ordered data points (from smallest to largest) $x_{(1)}, x_{(2)}, \ldots, x_{(n)}$, the function $F_n(x)$ can be written as

$$F_n(x_{(i)}) = \frac{i}{n}$$

i.e. a step function that increases by $1/n$ at the value of each ordered data point.

To test H_0: the data follow a specified distribution, against H_1: the data don't follow the specified distribution, we use a test statistic based on the figure:

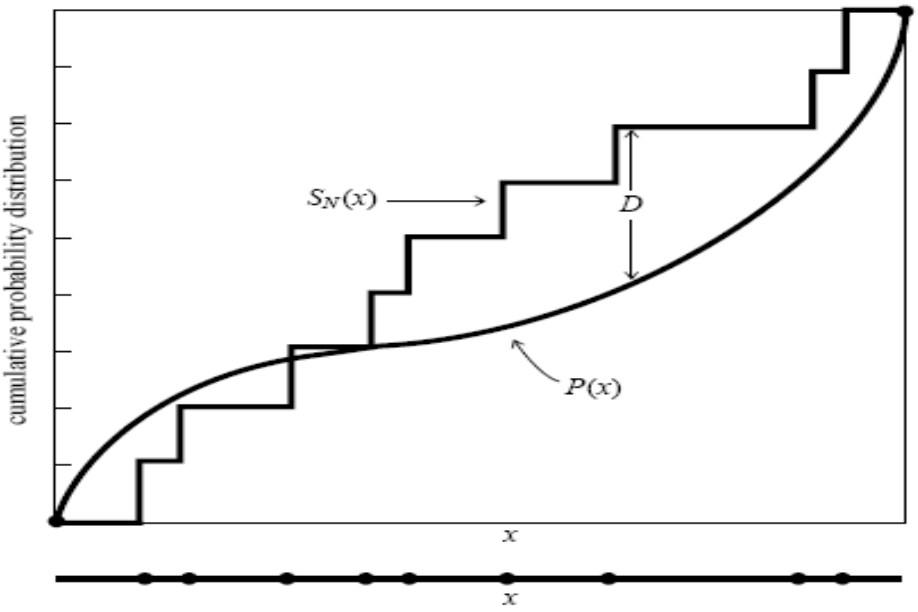


Figure: KS statistic D_n. Compare $P(x) \equiv F(x)$ to $F_n(x) \equiv S_n(x)$.
D_n is the greatest distance between the two CDFs.

$$\begin{aligned} D_n &= \sup_x |F_n(x) - F(x)| \\ &= \max_{1 \leq i \leq n} \left(F(x_i) - \frac{i-1}{n}, \frac{i}{n} - F(x_i) \right) \end{aligned}$$

We reject H_0 at significance level α if $D_n > D_n^{\alpha}$, where D_n^{α} is a critical value associated with the sampling distribution of D_n, and is obtained by tables (or use p-value provided in `R`).

- K-S test (in this form) only applies to continuous distributions that have been fully specified. It is not valid if the parameters of the distribution have been estimated from the data.
- Also, the test is not very sensitive at the tails of the distribution.

There is a plethora of statistical tests available for checking whether a sample of data comes from a population with a specific distribution. The Kolmogorov-Smirnov test is one useful such test. We will consider it in more detail at the practical sessions.

Chapter 4

Non-parametric Estimation

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The dataset chem in library MASS is made available in R by typing:

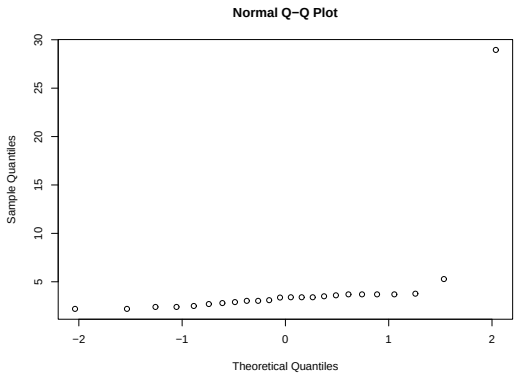
```
>library(MASS)
>data(chem)
```

It contains 24 determinations of copper ($\mu g\ g^{-1}$) in wholemeal flour. We wish to investigate the distribution of these data and obtain confidence intervals for the mean.

```
2.90  3.10  3.40  3.40  3.70  3.70  2.80   2.50  2.40
2.40  2.70  2.20  5.28  3.37  3.03  3.03  28.95  3.77
3.40  2.20  3.50  3.60  3.70  3.70
```

Sample mean = 4.280, sample sd = 5.297

The Q-Q plot reveals an outlier if the data are considered to be normally distributed:

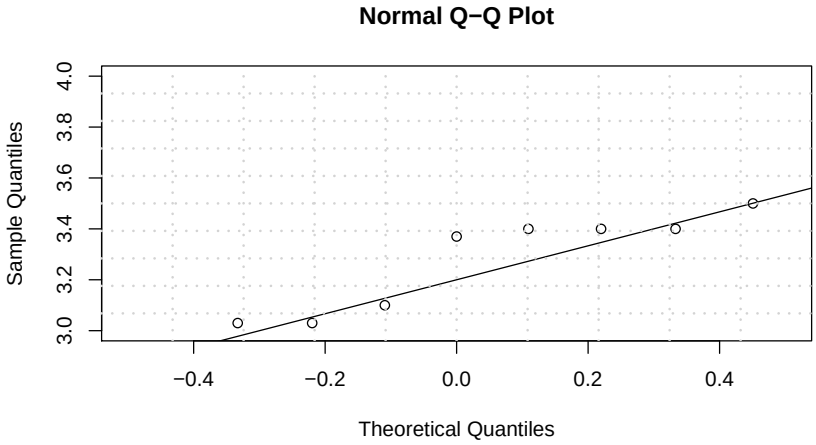


The extreme value, the 17th in the list, can be removed using the command:

```
>chem1=chem[-17]
```

This produces a rather more linear Q-Q plot. So we will now consider the revised data set, chem1.

A zoomed in version of the Q-Q plot is obtained below.



The commands to produce this are:

```
> qqnorm(chem1,xlim=c(-0.5,0.5),ylim=c(3,4))
> qqline(chem1)
> grid(10,10,lwd=2)
```

It can be seen that the intercept is about 3.21. This can be used as an estimate for the population mean. We can also obtain confidence intervals for the mean or median. Issues to consider:

- Sample size is *not* large for CLT
- Standard deviation σ must be estimated (sample s.d.s) this will be reflected in more uncertainty (wider CI)
- If population distn is approximately normal we can use t_{n-1} distn for CI.

- But, can normality be assumed?
- CI for median can also be obtained, but this may also rely on normality assumption.

The data in the `chem` sample are from a larger study which suggests that the mean copper level is $3.68\mu g\ g^{-1}$. If we ignore the extreme observation, we can obtain a 95% CI for the mean:

```
> t.test(chem1)
```

giving (2.911, 3.505). This implies that the data without the extreme point **are not compatible with the larger study**. However, using the original data CI is (2.044, 6.517) which is compatible. This CI may not be valid – **need to use alternative non-parametric techniques!**

More general:

- Extreme observation adds great uncertainty about what is going on
- It is possible that copper is very heterogeneously distributed
- Much more knowledge of the experimental situation, more data, and alternative methodology are needed

4.1 Bootstrap Estimation

- Methodology took its name in a paper by Efron (1979).
- Computer intensive method.
- Mainly distinguished in two types: **parametric** and **non-parametric** bootstrap.

Suppose that we want to make inferences about parameter θ using observed data $y_1, y_2, \ldots, y_n$, which follow a distribution with cdf $F(y;\theta)$. Usually inference will be based on the **sampling distribution** of an estimate $\hat{\theta}$. A sampling distribution is obtained either by theoretical results, or is based on many samples from $F(y;\theta)$.

Alternatively, bootstrap constructs samples by **repeatedly sampling with replacement** from the available sample $y_1, y_2, \ldots, y_n$.

4.2 Bootstrap Estimation (continued)

Basic idea (non-parametric bootstrap):

Construct an empirical distribution, $\hat{F}_n$:

$$\hat{F}_n(y) = \frac{1}{n}\{\text{Number of } y_i \leq y\}$$

1. Draw a sample of size n from $\hat{F}$

- This is the bootstrap sample $y_1^*, y_2^*, \ldots, y_n^*$ with y^*s selected **with replacement** from $y_1, y_2, \ldots, y_n$
- This is the *non-parametric (full) bootstrap*.

2. Obtain an estimate $\hat{\theta}^*$ from the bootstrap sample.
   - This is done in the same way as $\hat{\theta}$ is obtained from the original sample.
3. Repeat steps 1 and 2 (say B times).

Repetition of the above steps will provide the empirical distribution of $\hat{\theta}^*$, which serves as an approximation of the distribution of $\hat{\theta}$, and is referred to as the **bootstrap (sampling) distribution** of $\hat{\theta}$

4.3 Bootstrap Estimation – Examples

Suppose that we have a random sample $y_1, y_2, ..., y_n$ with $Y_i \sim \text{Exp}(\lambda)$. We wish to make inferences about λ.

From CLT (asymptotically) $\bar{Y} \sim N(\lambda^{-1}, (n\lambda^2)^{-1})$

We can use this to estimate quantities of interest (e.g. for confidence intervals or tests for λ).

What if assumptions or asymptotic results may not hold - e.g.

- small sample?
- and/or unknown/non-symmetric distribution?

Bootstrap allows us to avoid making assumptions about the sampling distribution of a statistic of interest, by instead **forming an empirical sampling distribution of the statistic**. This is generally achieved with resampling from the available sample.

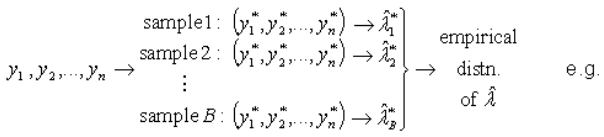
4.3.1 (a) Non-parametric (full) bootstrap

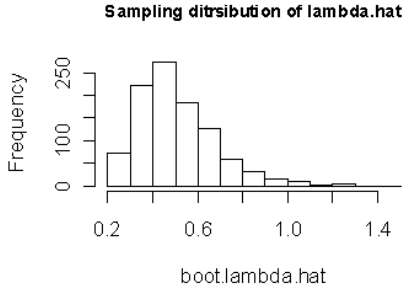
We **resample with replacement from the original sample**:

`for j in (1 to B)`

1. `obtain` j^{th} `sample` $y_1^*, y_2^*, ..., y_n^*$ `by resampling with replacement from` $y_1, y_2, ..., y_n$
2. `compute` $\bar{y}_j^*$ `and obtain` $\hat{\lambda}_j^* = 1/\bar{y}_j^*$

With `B` sufficiently large, the output set of estimates $(\hat{\lambda}_1^*, \ldots, \hat{\lambda}_B^*)$ can be used to **provide an empirical sampling distribution of** $\hat{\lambda}$. Schematically, this can be depicted as





Hence we can:

- estimate the mean of $\hat{\lambda}$ by

$$\hat{E}(\hat{\lambda}) \quad = \quad \frac{1}{B}\sum_{j=1}^{B}\hat{\lambda}_j^*$$

- estimate its median, using the empirical quantiles of the bootstrap values λ_j^*, i.e.

$$\widehat{\text{median}}(\hat{\lambda}) \quad = \quad Q_e(0.5)$$

- estimate its variance:

$$\widehat{\text{Var}}(\hat{\lambda}) \quad = \quad \frac{1}{B-1}\left(\sum_{j=1}^{B}\left(\hat{\lambda}_j^*\right)^2 - \frac{1}{B}\left(\sum_{j=1}^{B}\hat{\lambda}_j^*\right)^2\right)$$

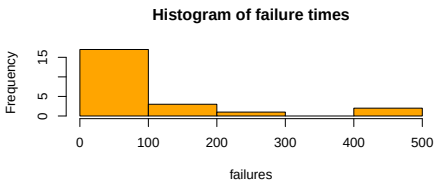
- Obtain $100(1-\alpha)\%$ confidence intervals:

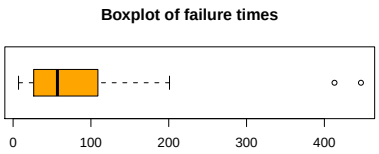
$$[Q_e(\alpha/2),\, Q_e(1-\alpha/2)]$$

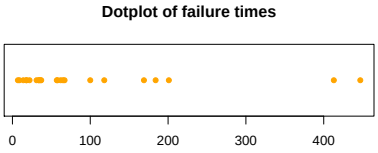
- Compute p-values for hypothesis tests (see later) etc.

Example 4.1: The numbers of operating hours between successive failures of the air-conditioning equipment aboard a Boeing 720 aircraft were as follows:

```
413  14  58  37  100  65  9  169  447  184  36  201
118  34  31  18  18  67 57  62  7  22  34
```







Data distribution has long tail to the right. We estimate the population median, m, by the sample median, $\hat{m} = 57$.

How **accurate** is this estimate? Can we use the estimator provided by the sample median to construct a CI for m?

- Asymptotic normality *doubtful* ($n = 23$)
- Use bootstrap!

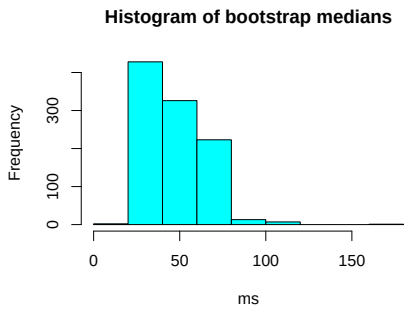
We would like the sampling distribution of $\hat{m}$ ...

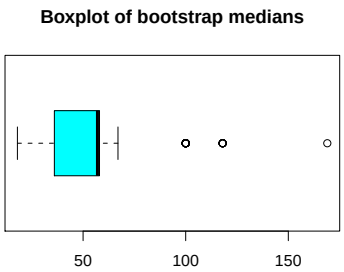
... but this requires knowledge of the (unknown) underlying population distribution. Use instead the empirical distribution of the data (obtained with bootstrap!)

Set up an R vector, `median.b`, of size `B=1000` and use it to store the results of 1000 simulations of $\hat{m}$.

```
> B = 1000
> median.b = numeric(B)
> for(i in 1:B) median.b[i] = median(sample(failures, size=23, replace=TRUE))
> summary(median.b)
```

```
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
18.00   36.00   57.00   50.54   62.00  169.00
```





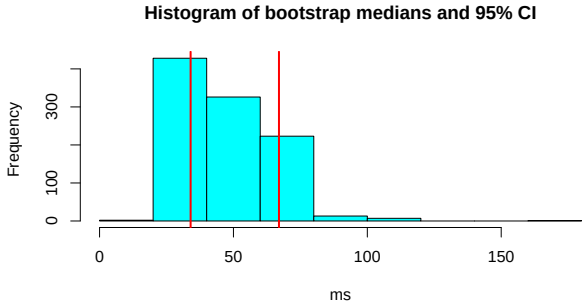
Recall that each component of `median.b` is the median of a random sample of size 23. Hence the variability in `median.b` is much less than the variability in `failures` itself. A reasonable 95% CI for the original population median m is given by the so-called **percentile bootstrap CI**:

$$(Q_e(0.025), Q_e(0.975))$$

where Q_e is the empirical quantile function of the bootstrap simulations `median.b` of $\hat{m}$.

```
> quantile(median.b,c(0.025,0.975))
```

so that (34, 67) is a reasonable 95% confidence interval for m.



4.3.2 (b) Parametric bootstrap

If we are *prepared to assume* that the sample is considered to come from Exp(λ), then start by obtaining an estimate of λ: $\hat{\lambda} = 1/\bar{y}$. Then **use the Exp$\left(\hat{\lambda}\right)$ distribution to draw the bootstrap samples**. Once the bootstrap samples are available, we proceed as with the non-parametric method before.

Algorithm:

`for j in (1 to B)`

1. `obtain new` (j^{th}) `sample` $y_1^*, y_2^*, ..., y_n^* \sim$ `Exp` $(\hat{\lambda})$
2. `compute` $\bar{y}_j^*$ `and obtain` $\hat{\lambda}_j^* = 1/\bar{y}_j^*$

Various inferences can then be made as before.

4.4 Hypothesis tests using bootstrap

For hypothesis testing it is important that the **empirical sampling distribution is generated under the null** H_0. This means that usually **parametric bootstrap should be used!** .(Or data rescaling if suitable – see later.)

Example 4.1 (cont.) – ***Parametric bootstrap***:
We will use bootstrap to test

H_0: the true mean failure time (say μ) is 140 hours

v.

H_1: the true mean failure time is *less* than 140 hours

Now the empirical sampling distribution (and hence the bootstrap samples) must come from a distribution with $\mu = 140$.

We therefore **assume** that data, under H_0, come from Exp(1/140).

We first obtain an estimate of μ ($\hat{\mu} = 95.7$)

then obtain B bootstrap values $\hat{\mu}^*_1, \ldots, \hat{\mu}^*_B$ and finally compute p-value as

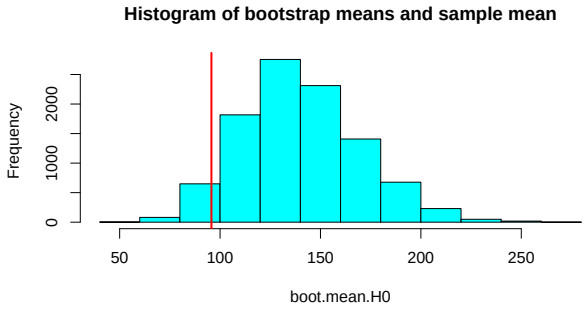
$$p = \frac{\sum_{j=1}^{B} I(\hat{\mu}^*_j \leq \hat{\mu}) + 1}{B + 1}, \text{ where } I(\cdot) \text{ is the indicator function.}$$

In R this can be done as follows:

```
##  Perform test for mean = mean.hyp v mean < mean.hyp
##  Parametric bootstrap is well suited for hypothesis testing
B = 10000
mean.hyp = 140
boot.mean.H0 = numeric(B)  # Create vector for bootstrap values
for (i in 1: B){
  y.boot = rexp(23,rate = (1/mean.hyp)) # Notice this is under H0
  boot.mean.H0[i] = mean(y.boot)
}

# Obtain sampling distn under H0 and p-value
hist(boot.mean.H0, col="cyan",
     main="Histogram of bootstrap means and sample mean")
abline(v = mu.hat, col="red", lwd=2.0)
boot.pval =
   (1 + length(boot.mean.H0[boot.mean.H0 <= mu.hat])) / (B + 1)
boot.pval
```

This code will give



and p-value $= 0.049$. Note that a *test based on asymptotic normality* will give

```
t.test(failures,mu=140, alternative="less")
```

```
... t = -1.7812, df = 22, p-value = 0.04435
```

Example 4.1 (cont.) – ***Non-parametric bootstrap***:
Again, for `failures` data, test

H_0: the true mean failure time (say μ) is 140 hours

v.

H_1: the true mean failure time is *less* than 140 hours

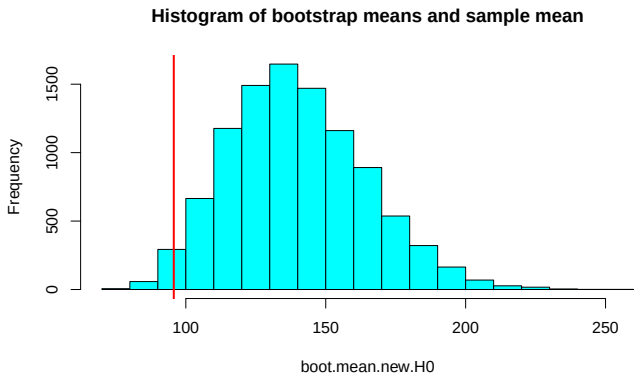
- First rescale data so that H_0 is true:
  $ynew_i = y_i - \bar{y} + 140$
- Then resample new data (with replacement) and proceed with bootstrap test as before.
- Note that suitable rescaling not always straightforward!

R Code:

```
# Rescale data so that H0 is true
failures.new = failures - mean(failures) + mean.hyp
mean(failures.new)

boot.mean.new.H0 = numeric(B)  #Create vector for bootstrap values
for (i in 1: B){
  boot.new.y = sample(failures.new,23,replace=T) #This is under H0
  boot.mean.new.H0[i] = mean(boot.new.y)
}
```

Then sampling distribution and p-value obtained as before:



- This is the empirical distribution of the sample mean from a population with mean 140.
- Notice that it is not symmetric - bootstrap justified!
- Observed sample mean = 95.7. How likely is such an extreme (small) value, under H_0? -p-value = 0.02

Chapter 5

Regression review and diagnostics

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5.1 Regression - brief revision

- We study the relationship between a **response** (or **dependent**) random variable and one or more **explanatory** (or **independent**) variables.

- The value assumed by the response variable Y is, in part, explained by the values of the other variables, X_1, $X_2, \ldots$

  For example:

  - Height of mother (x) and height of daughter (Y)
  - Selling price of a house (Y) may depend upon total floor area (x_1), number of bedrooms (x_2), age of the house (x_3), etc.

- We are only concerned here with the theory of *linear* relationships.

- Data: n pairs (x_i, y_i) , $i = 1, 2, \ldots, n$ [*Simple* Linear Regression]

- Basic display is a *scatterplot* – this may exhibit pattern, indicative of an underlying relationship between the variables.

It is convenient to define the **summary statistics**

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad S_{yy} = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

$$S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

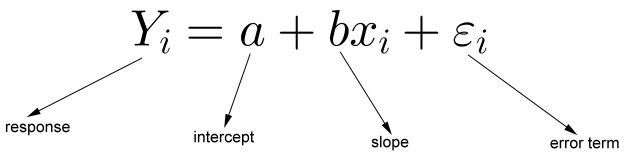
[Note: for calculations,

$$S_{xx} = \sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2, \qquad S_{yy} = \sum y_i^2 - \frac{1}{n}\left(\sum y_i\right)^2,$$

$$S_{xy} = \sum x_i y_i - \frac{1}{n}\left(\sum x_i\right)\left(\sum y_i\right)\Bigg]$$

Linear regression analysis

The basic linear model with a single explanatory variable is:



- a, b are non-random parameters.
- x_i, the explanatory variable values, are non-random.
- ε_i is a random variable, the "error term", a "residual" random variable with mean 0 and variance σ^2. The ε_i's are *uncorrelated*.

The response variable values Y_i are random because of the randomness in the ε_i values.

- Another way to express the model is:

  $$E[Y_i|x_i] = a + bx_i$$

  with uncorrelated errors with constant variance σ^2
- It follows that $V[Y_i] = \sigma^2$ and the Y_i's are uncorrelated.
- 'Homoscedastic' means that the Y_i all have the same variance, σ^2

Estimation of parameters

Use *Least Squares* – i.e. minimise

$$S = \sum_{i=1}^{n} (y_i - a - bx_i)^2$$

to obtain estimators

$$\boxed{\hat{b} = \frac{S_{xy}}{S_{xx}} \quad \textbf{and} \quad \hat{a} = \bar{y} - \hat{b}\bar{x}}$$

The **fitted value** at $X = x$ is:

$$\hat{y} = \hat{a} + \hat{b}x$$

and we note that the fitted line goes through $(\bar{x}, \bar{y})$. The **residual** *from the fit* for observation (x_i, y_i) is the estimated error:

$$e_i = \hat{\varepsilon}_i = y_i - \hat{y}_i$$

Notice that

$$\begin{aligned} \epsilon_i &= y_i - (a + bx_i) \\ e_i &= y_i - \left(\hat{a} + \hat{b}x_i\right) \end{aligned}$$

The sum of squared residuals is given as

$$\sum_i e_i^2 = \sum (y_i - \hat{y}_i)^2 = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$$

Properties of Parameter estimates

$$E\left[\hat{b}\right] = b, \quad V\left[\hat{b}\right] = \frac{1}{S_{xx}}\sigma^2$$

$$E\left[\hat{a}\right] = a, \quad V\left[\hat{a}\right] = \left(\frac{\sum x_i^2}{nS_{xx}}\right)\sigma^2$$

(Notice that estimates are *unbiased*.)

$$Cov\left[\hat{a}, \hat{b}\right] = -\left(\frac{\sum x_i}{nS_{xx}}\right)\sigma^2$$

Note: $Cov = 0$ if $\sum x_i = 0$ – and this is easily achieved (centre x values by defining a new variable $x_i' = x_i - \bar{x}$, and notice $\sum x_i' = \sum (x_i - \bar{x}) = 0$).

Also:

$$\widehat{\sigma^2} = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n-2}\left(S_{yy} - \frac{S_{xy}^2}{S_{xx}}\right)$$

and

$$E\left[\widehat{\sigma^2}\right] = \sigma^2 \qquad \text{(unbiased)}$$

Partitioning the total variability in the responses

$$\begin{aligned} \sum (y_i - \bar{y})^2 &= \sum (y_i - \hat{y}_i)^2 + \sum (\hat{y}_i - \bar{y})^2 \\ SS_T &= SS_{RES} + SS_{REG} \end{aligned}$$

- SS_T : "total sum of squares" – total variability in the responses
- SS_{RES} : "residual (or error) sum of squares" – variability remaining "unexplained" by the model
- SS_{REG}: "regression sum of squares" – variability "explained" by the regression model

Coefficient of determination

The proportion of the total variability in the responses "explained" by a model is called the *coefficient of determination*, R^2:

$$\boxed{R^2 = \frac{SS_{REG}}{SS_T}} = \frac{S_{xy}^2}{S_{xx} S_{xy}}$$

For Simple Linear Regression, $R^2 = r^2$, the sample correlation between x and y

Inference on parameters: full normal model

- Now further assume: $\varepsilon_i \sim N(0, \sigma^2)$ for all i. i.e. the errors are i.i.d. *normal* random variables.
- Then the response r.v.s Y_i are independent with

$$Y_i \sim N(a + bx_i, \sigma^2)$$

- We can then use Maximum Likelihood (ML) estimation: estimates of a and b same as with Least Squares (LS), but

$$MLE\left(\sigma^2\right) = \frac{\sum e_i^2}{n} = \frac{n-2}{n}\,\widehat{\sigma^2} \qquad (\textit{biased})$$

Inference on parameters

- Under the *normal model* we have:

$$\hat{b} \sim N\left(b, \frac{\sigma^2}{S_{xx}}\right) \quad \text{giving} \quad \frac{\hat{b} - b}{\left(\sigma^2/S_{xx}\right)^{1/2}} \sim N(0,1)$$

- With unknown variance:

$$\frac{\hat{b} - b}{\left(\widehat{\sigma^2}/S_{xx}\right)^{1/2}} \sim t_{n-2}$$

- Similarly for parameter a. *Exercise*: check details, using estimated standard error of $\hat{a}$ given earlier.
- Also $\hat{a}$, $\hat{b}$ independent of $\widehat{\sigma^2}$ and

$$\frac{(n-2)\widehat{\sigma^2}}{\sigma^2} \sim \chi^2_{n-2}$$

Inference on responses

- **For mean response:**

  At x_0, the *mean (expected)* response μ_0 is estimated as

$$\hat{\mu}_0 = \hat{E}(Y|x_0) = \hat{a} + \hat{b}x_0$$

  and under the *normal model*, we have

$$\frac{\hat{\mu}_0 - \mu_0}{e.s.e.(\hat{\mu}_0)} \sim t_{n-2}$$

  where

$$e.s.e.\,(\hat{\mu}_0) = \left[\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\widehat{\sigma^2}\right]^{1/2}$$

- **For individual response:**

  At x_0, the *individual* response y_0 is estimated as before

$$\hat{y}_0 = \hat{a} + \hat{b}x_0$$

  but *uncertainty* about this estimate is now *greater*.

  Have

$$\frac{\hat{y}_0 - y_0}{e.s.e.\,(\hat{y}_0)} \sim t_{n-2}$$

  with

$$e.s.e. = \left[\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\widehat{\sigma^2}\right]^{1/2}$$

5.2 Regression illustration using R

Enter a small example data set:

```
> x=c(3.1,3.4,3.0,3.2,3.9,2.8,2.2)
> y=c(3.5,3.2,3.0,2.9,4.0,2.5,2.3)
```

Take a look at the data:

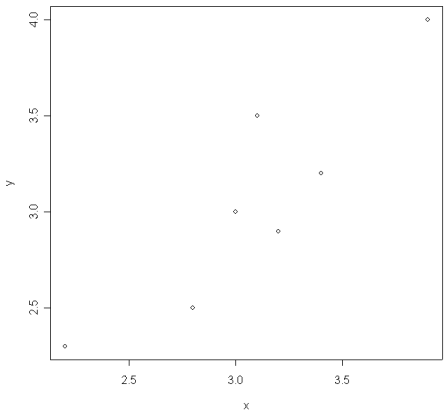
```
> x
[1] 3.1 3.4 3.0 3.2 3.9 2.8 2.2
> y
[1] 3.5 3.2 3.0 2.9 4.0 2.5 2.3
```

Now take a look at a scatterplot:

> plot(x, y)

or

> plot(y ~ x)



Now use the `lm` command to fit the regression model, and examine the result.

> example1 = lm(y ~ x)

> example1

```
Call:
lm(formula = y ~ x)

Coefficients:
(Intercept)            x
  -0.004506     0.992201
```

So the fitted model is

$$\hat{y} = -0.004506 + 0.992201x$$

Can get a lot more information using `summary(example1)`:

> summary(example1)

```
Call:
lm(formula = y ~ x)

Residuals:
      1       2       3       4       5       6       7
```

```
 0.4287 -0.1690  0.0279 -0.2705  0.1349 -0.2737  0.1217

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.004506   0.683082  -0.007  0.99499
x            0.992201   0.218681   4.537  0.00618 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2808 on 5 degrees of freedom
Multiple R-squared:  0.8046,Adjusted R-squared:  0.7655
F-statistic: 20.59 on 1 and 5 DF,  p-value: 0.006185
```

Examine fitted values:

```
> yhat = fitted.values(example1)
> yhat
```

```
       1        2        3        4        5        6        7
3.071317 3.368977 2.972097 3.170537 3.865078 2.773657 2.178336
```

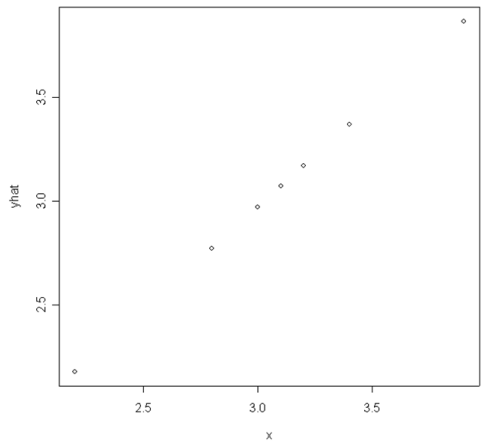
Alternatively:

```
> ahat = -0.004506
> bhat = 0.992201
> yhat = ahat + bhat*x
> yhat
```

```
[1] 3.071317 3.368977 2.972097 3.170537 3.865078 2.773657 2.178336
```

Plot fitted values, `yhat`, against explanatory variable values, x. This in fact gives points along the fitted line.

```
> plot(yhat ~ x)
```



Next examine residuals:

```
> epsilon = residuals(example1)
> epsilon
```

```
          1           2           3           4           5
 0.42868284 -0.16897747  0.02790295 -0.27053726  0.13492201
          6           7
-0.27365685  0.12166378
```

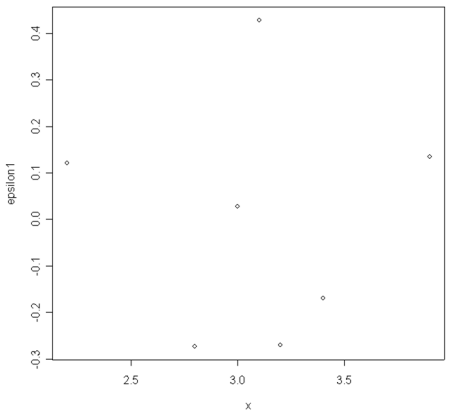
Alternatively:

```
> epsilon = y - yhat
> epsilon
```

```
[1]  0.42868284 -0.16897747  0.02790295 -0.27053726  0.13492201
[6] -0.27365685  0.12166378
```

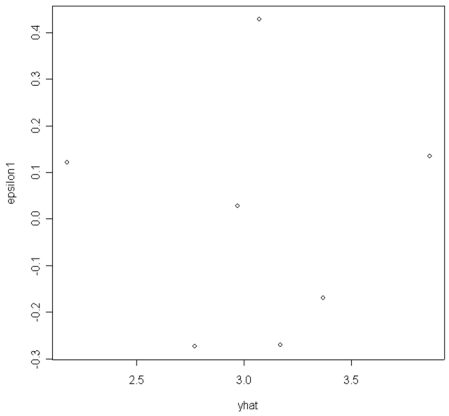
Plot residuals against the explanatory variable x:

```
> plot(epsilon ~ x)
```



Plot residuals against the fitted values `yhat`:

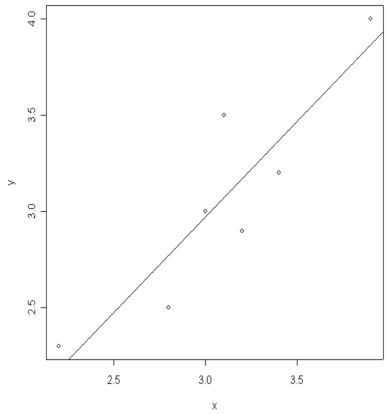
```
> plot(epsilon ~ yhat)
```



Both graphs show the same thing — the residuals follow a random pattern. *Note:* Since the fitted line is approximately $\hat{y} = x$, both graphs are extremely similar in this case.

Plot y against x with fitted regression line:

```
> plot(y ~ x)
> abline(example1)
```



Prediction in R

Predict y at $x = 3.1$, and obtain 95% Confidence Interval (CI) for *mean response*:

```
> predict(example1, data.frame(x=3.1), se.fit=T,
interval="confidence")


$fit
       fit      lwr      upr
1 3.071317 2.798397 3.344237

$se.fit
[1] 0.1061706

$df
[1] 5

$residual.scale
[1] 0.2807794
```

Take a look at the R help for this command using ?predict.lm

Now obtain 95% Prediction Interval at $x = 3.1$, sometimes called CI for *individual response*:

```
> predict(example1, data.frame(x=3.1), interval="prediction")
```

```
       fit      lwr     upr
1 3.071317 2.299675 3.84296
```

Notice that the point estimate $\hat{y} = 3.0713$ is the same as before, but that the Prediction Interval $(2.300, 3.843)$ for an individual response is wider than the Confidence Interval $(2.798, 3.344)$ for the mean response.

Sum of squares

'Analysis of Variance'

```
> summary.aov(example1)

            Df Sum Sq Mean Sq F value  Pr(>F)
x            1 1.6230  1.6230   20.59 0.00618 **
Residuals    5 0.3942  0.0788
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$SS_{REG} = 1.6230$

$SS_{RES} = 0.3942$

5.3 Regression Model Diagnostics : Residuals and Influence

Consider again the problem of fitting the model

$$y_i = a + bx_i + \varepsilon_i \quad i = 1, \ldots, n$$

Assume again a single continuous response variable y. How do we assess the quality of a given fit? While summary statistics are helpful, they are not sufficient. Good diagnostics are typically based on case analysis, i.e. *an examination of each observation in turn in relation to the fitting procedure.*

This leads to an examination of residuals and influence.

Residuals

The residuals should be thought of as *what is left* of the values of the response variable *after the fit has been subtracted.* Ideally they should show no further dependence on x. Remember: fitted equation

$$\hat{y} \quad = \quad \hat{a} + \hat{b}x$$

should capture (most of) the dependence of y upon x. In general this should be investigated graphically by plotting residuals against the explanatory variable(s) x. For linear models, it can also be very useful to plot residuals against fitted values $\hat{y}$ (especially when we have more than one x variable).

Example: Anscombe's Artificial Data

The R data frame `anscombe` is made available by

```
> data(anscombe)
> attach(anscombe)
```

This contains 4 artificial datasets created by Frank Anscombe, each of 11 observations of a continuous response variable y and a continuous explanatory variable x. A full discussion is given in *Applied Linear Regression*, by S. Weisberg (1985, p107-108).

We fit a least squares linear model to each data set. Take a look at the data:

```
> anscombe
```

```
   x1 x2 x3 x4    y1   y2    y3    y4
1  10 10 10  8  8.04 9.14  7.46  6.58
2   8  8  8  8  6.95 8.14  6.77  5.76
3  13 13 13  8  7.58 8.74 12.74  7.71
4   9  9  9  8  8.81 8.77  7.11  8.84
5  11 11 11  8  8.33 9.26  7.81  8.47
6  14 14 14  8  9.96 8.10  8.84  7.04
7   6  6  6  8  7.24 6.13  6.08  5.25
8   4  4  4 19  4.26 3.10  5.39 12.50
9  12 12 12  8 10.84 9.13  8.15  5.56
10  7  7  7  8  4.82 7.26  6.42  7.91
11  5  5  5  8  5.68 4.74  5.73  6.89
```

Model 1

```
> model1 = lm(y1 ~ x1)
> summary(model1)

Call:
lm(formula = y1 ~ x1)

Residuals:
     Min       1Q   Median       3Q      Max
-1.92127 -0.45577 -0.04136  0.70941  1.83882

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0001     1.1247   2.667  0.02573 *
x1            0.5001     0.1179   4.241  0.00217 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 1.237 on 9 degrees of freedom
Multiple R-squared:  0.6665,Adjusted R-squared:  0.6295
F-statistic: 17.99 on 1 and 9 DF,  p-value: 0.00217
```

Model 2

```
> model2 = lm(y2 ~ x2)
> summary(model2)

Call:
lm(formula = y2 ~ x2)

Residuals:
    Min      1Q  Median      3Q     Max
-1.9009 -0.7609  0.1291  0.9491  1.2691

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    3.001      1.125   2.667  0.02576 *
x2             0.500      0.118   4.239  0.00218 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.237 on 9 degrees of freedom
Multiple R-squared:  0.6662,Adjusted R-squared:  0.6292
F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002179
```

Model 3

```
> model3 = lm(y3 ~ x3)
> summary(model3)

Call:
lm(formula = y3 ~ x3)

Residuals:
    Min      1Q  Median      3Q     Max
-1.1586 -0.6146 -0.2303  0.1540  3.2411

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0025     1.1245   2.670  0.02562 *
x3            0.4997     0.1179   4.239  0.00218 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.236 on 9 degrees of freedom
```

```
Multiple R-squared:  0.6663,Adjusted R-squared:  0.6292
F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002176
```

Model 4

```
> model4 = lm(y4 ~ x4)
> summary(model4)

Call:
lm(formula = y4 ~ x4)

Residuals:
   Min     1Q Median     3Q    Max
-1.751 -0.831  0.000  0.809  1.839

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0017     1.1239   2.671  0.02559 *
x4            0.4999     0.1178   4.243  0.00216 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.236 on 9 degrees of freedom
Multiple R-squared:  0.6667,Adjusted R-squared:  0.6297
F-statistic:    18 on 1 and 9 DF,  p-value: 0.002165
```

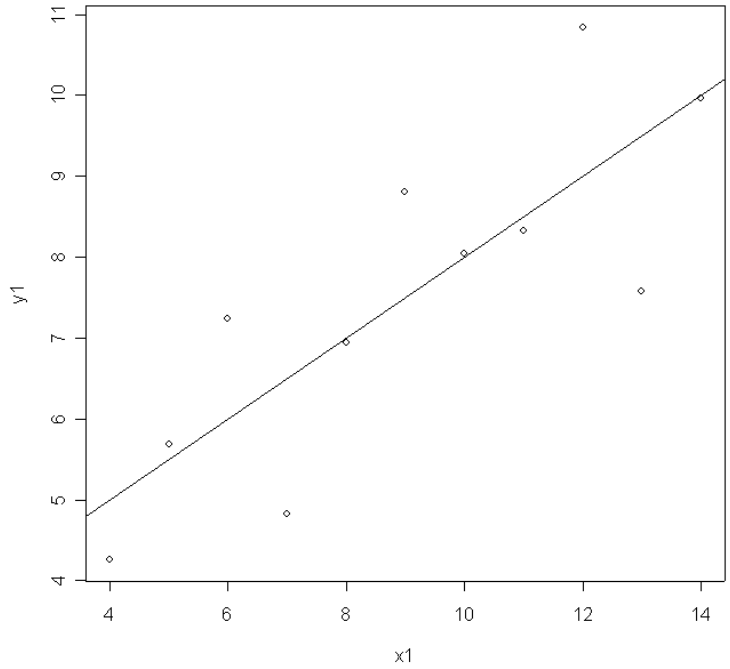
In each case, the fitted line is $\hat{y} = 3.00 + 0.50x$. We also find that the standard errors of $\hat{a}, \hat{b}$, the residual standard error, and the coefficient of determination R^2 are each *almost identical* across the 4 datasets.

Judging by everything we've seen so far, it appears that all four data sets are equally well described by the same linear regression model. But now we'll look at some graphs, and find that very *different* judgements should be made about the appropriateness of the fitted model to each of the 4 cases. We look at scatterplots of the data, and also plots of residuals.

Recall model assumption: $\epsilon_i \sim N(0,\ \sigma^2)$, independent. So we want plotted residuals e_i to be

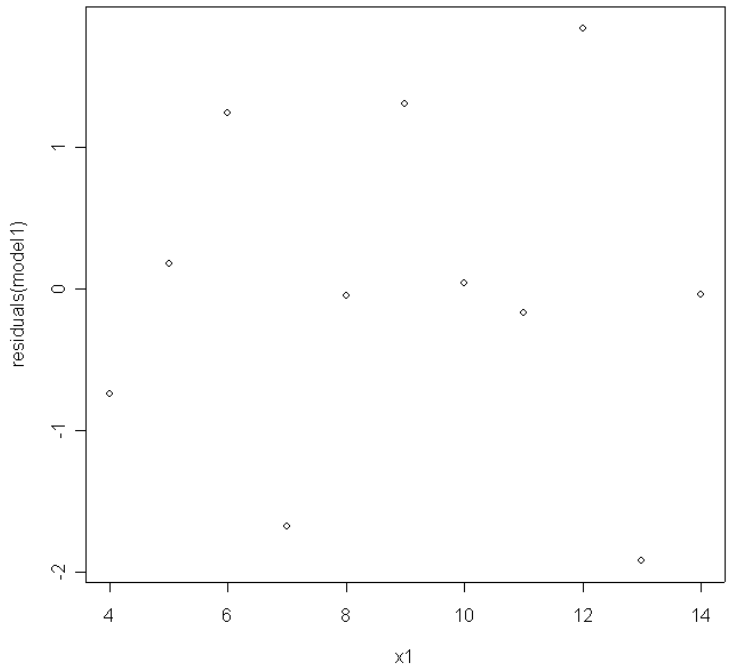
- centred about 0
- with constant variance (so the amount of vertical spread in the residuals does not change as we move along the horizontal axis)

Model 1: $y1$ vs $x1$ and fitted line



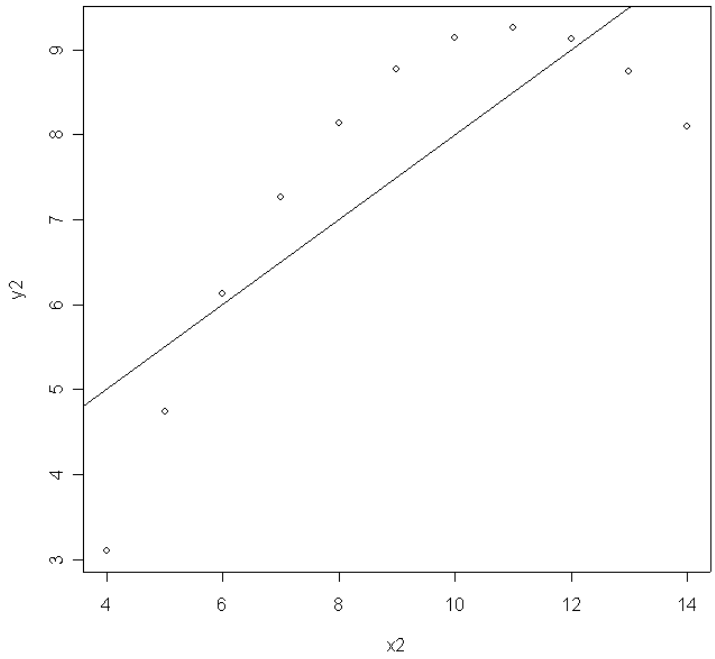
Reasonably good fit.

Model 1: residuals



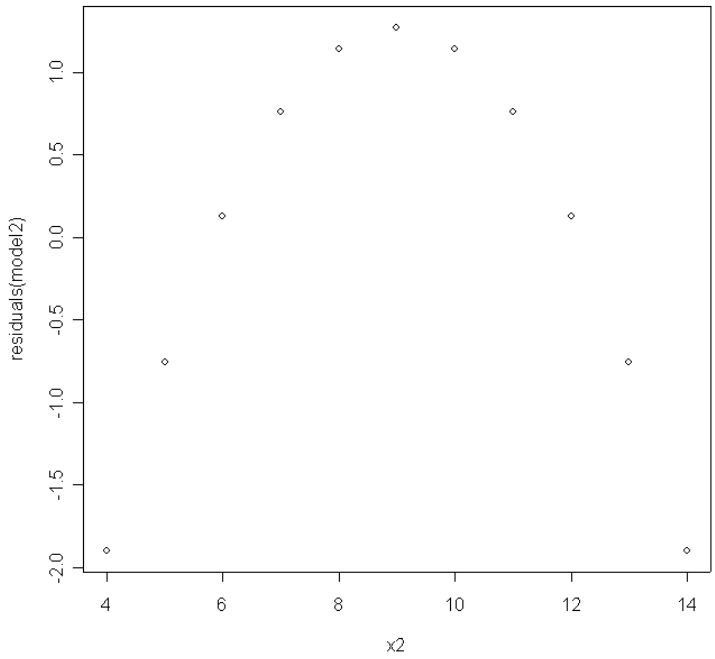
No particular pattern; model is fine.

Model 2: $y2$ vs $x2$ and fitted line



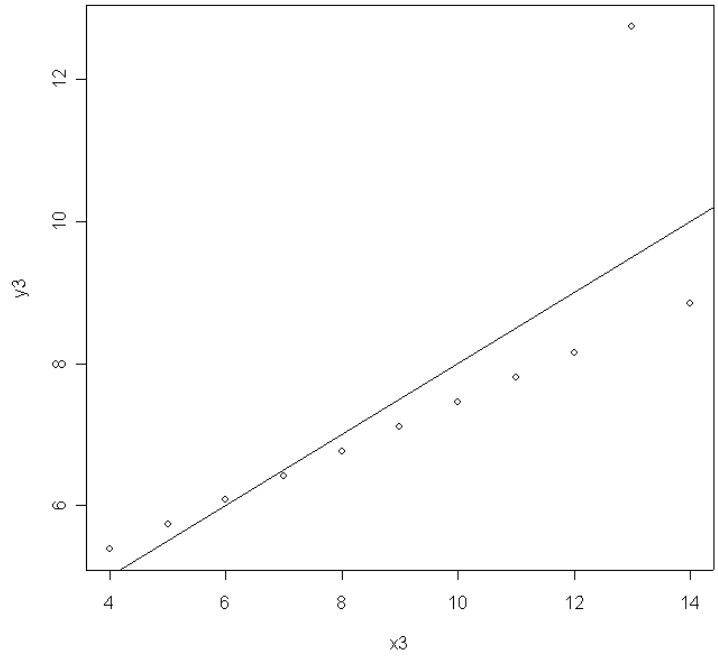
Non-linear relationship; need a non-linear model.

Model 2: residuals



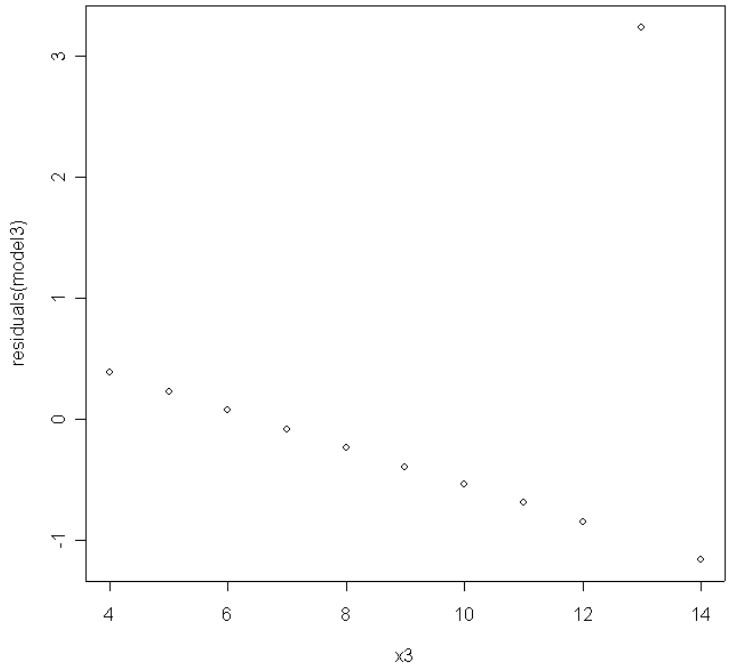
Residuals depend systematically on x (possibly quadratic relationship).
Fitted line $\hat{y} = \hat{a} + \hat{b}x$ does not capture all dependence of y on x

Model 3: $y3$ vs $x3$ and fitted line



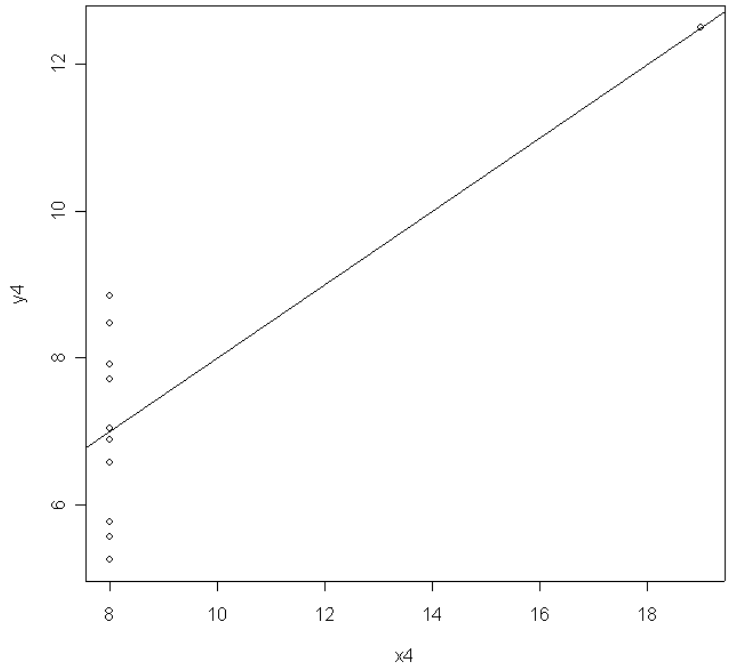
Perfect linear relationship except for one *outlier*. Should treat this point separately from the rest.

Model 3: residuals



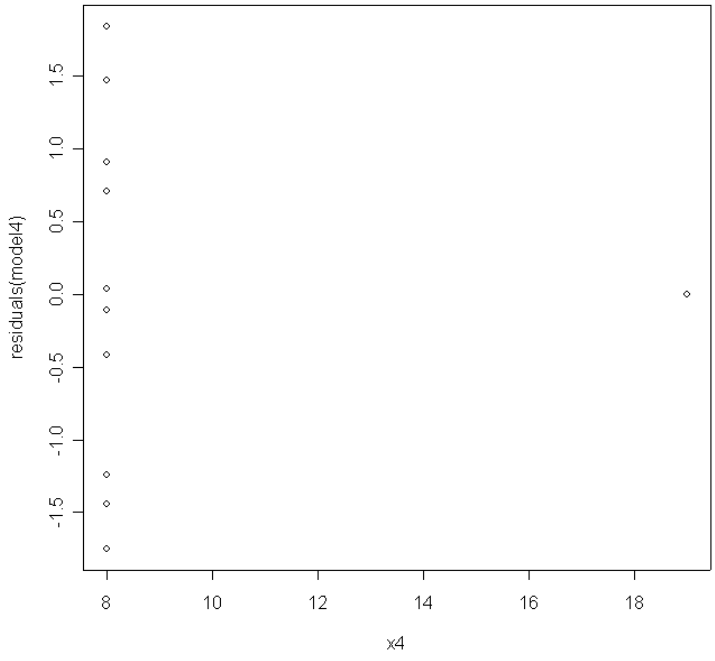
Trend in residuals (decrease as x increases), and *outlier*.

Model 4: $y4$ vs $x4$ and fitted line



Only two x values. Can't tell whether the true relationship is linear or not.

Model 4: residuals

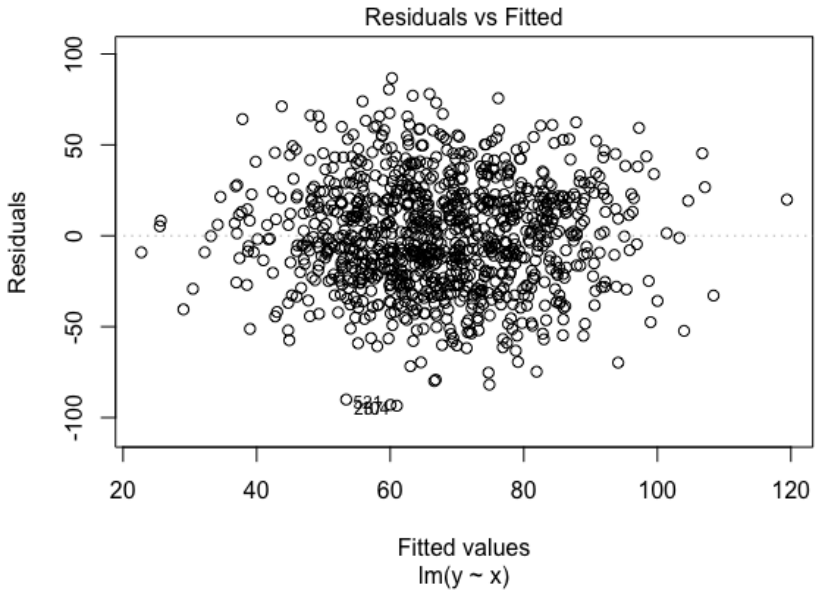


In general, residuals provide information about:

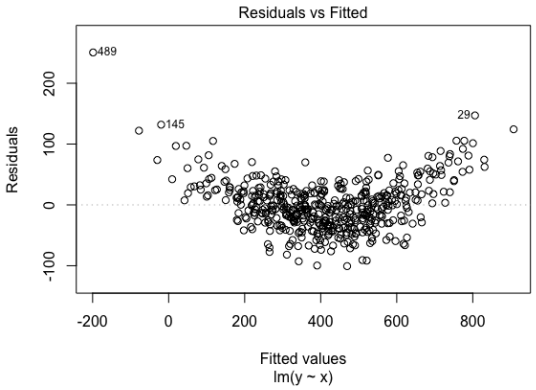
- whether the *best relation* has been fitted
- the relative merits of *different fits*
- mild, but non-random, *departures from the hypothesised fit*
- the magnitude of the residual variation
- the identification of *outliers*
- possible further dependence on x, other than through location, of the conditional distribution of y given x – in particular *heterogeneity of spread* of the residuals.

Plotting residuals against fitted values can show up problems with the model.

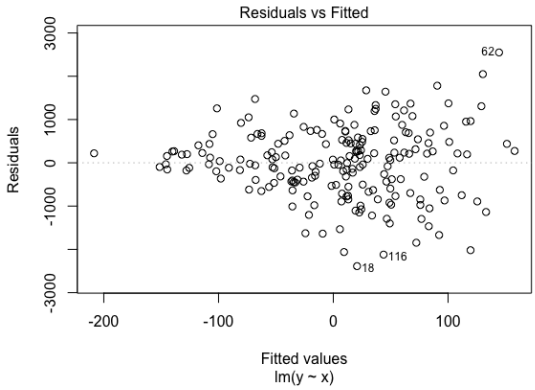
A satisfactory residuals plot:



No pattern. Random scatter, residuals centred about 0. Problematic residuals plots:



Non-linear relationship.



Heteroscedasticity.

Heteroscedasticity: vertical spread of residuals changes as fitted values increase.

Standardised residuals

$$r_i = \frac{e_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}$$

where h_{ii} is the i-th diagonal element of the "hat" matrix (see later).

In simple linear regression:

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$$

Under model assumptions:

- $E(r_i) = 0$ (as for e_i)
- But all have same variance, $var(r_i) = 1$. *Note that* $var(e_i) = \sigma^2 (1 - h_{ii})$

Influence

Influence measures the extent to which a fit is affected by individual observations.

A possible formal definition is the following: the influence of any observation is a measure of the difference between (i) the fit and (ii) the fit which would be obtained if that observation were omitted.

Observations with large influences require more careful checking. Especially for linear models, influence is often measured by Cook's distance.

Cook's Distance

$$D_i = \frac{\sum_{j=1}^{n} \left(\hat{y}_j^{(i)} - \hat{y}_j \right)^2}{p\, \widehat{\sigma^2}}$$

- $\hat{y}_j$ is the fitted value of observation j where the fit is obtained using all the data
- $\hat{y}_j^{(i)}$ is the fitted value of observation j where the fit is obtained using all the data except observation i
- p is the number of regression parameters in the model (so $p = 2$ for simple linear regression)

This will require running $(n + 1)$ models to obtain all D_i values. Instead we can use the equivalent expression:

$$D_i = \frac{r_i^2}{p} \frac{h_{ii}}{1 - h_{ii}} = \frac{e_i^2}{p\, \widehat{\sigma^2}} \frac{h_{ii}}{(1 - h_{ii})^2}$$

where e_i and r_i are the usual and standardised residuals respectively, and h_{ii} is the ith diagonal element of the hat matrix H (defined later).

As a rule of thumb, observations for which $\boxed{D_i > 1}$ make a noticeable difference to the parameter estimates, and should be examined carefully for the appropriateness of their use in fitting the model.

An observation with a large residual *usually* has a large influence, but not always. However, an observation with an unusual value of its explanatory variable(s) x can pull a fit towards it and have a large influence though a small residual. We need to look at *both* residuals and influence to check the regression.

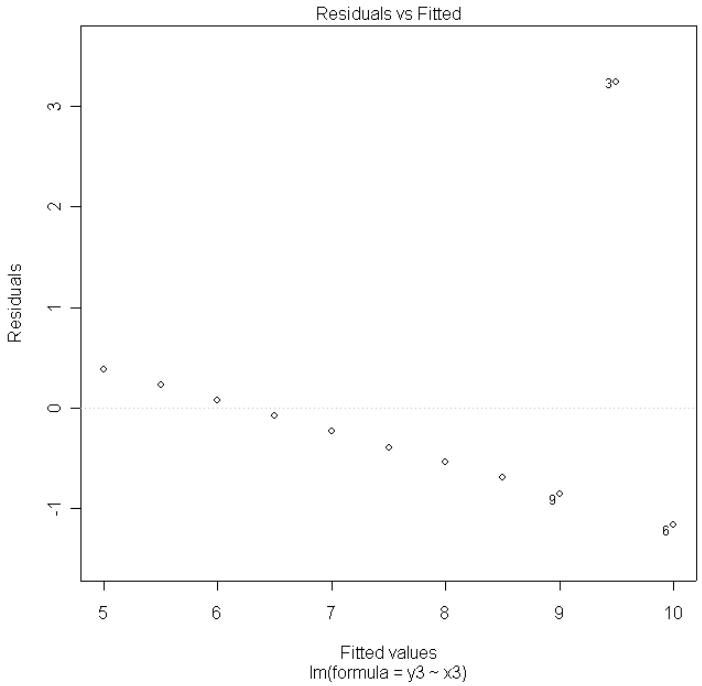
Example: Anscombe's third data set.

```
> plot(model3, which = 1:4)
```

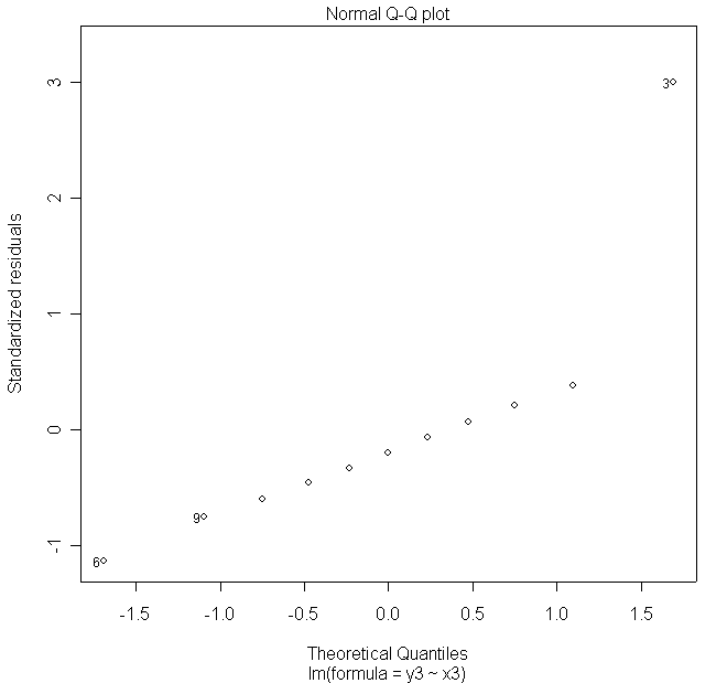
Take a look at R help using `?plot.lm`

Produces the following 4 graphs:

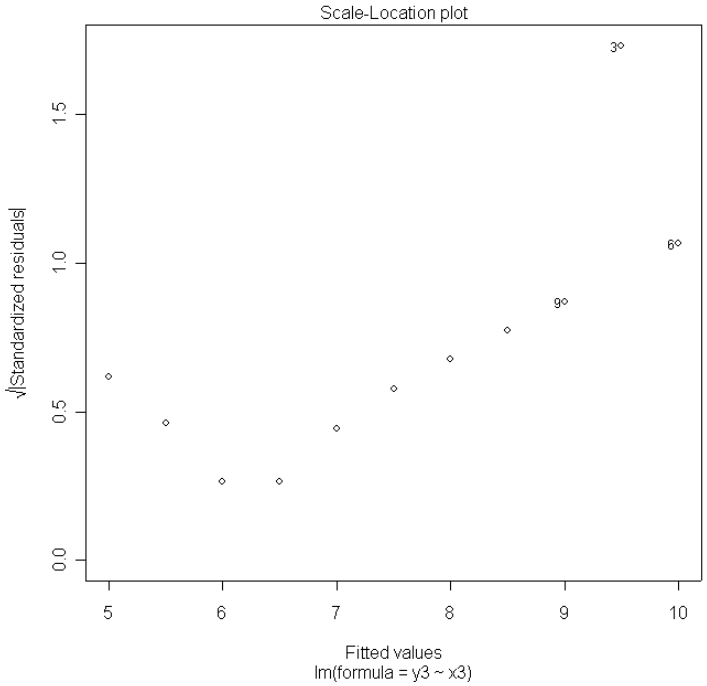
Model 3: residuals



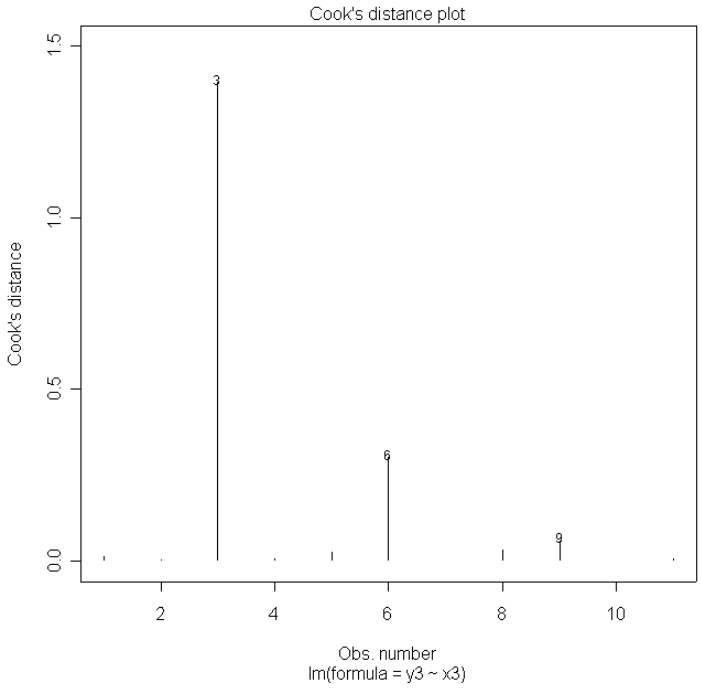
Model 3: normal Q-Q plot



Model 3: standardised residuals



Model 3: Cook's distance plot



The last graph produced by the plot function shows that observation number 3 has an unusually large value of Cook's distance

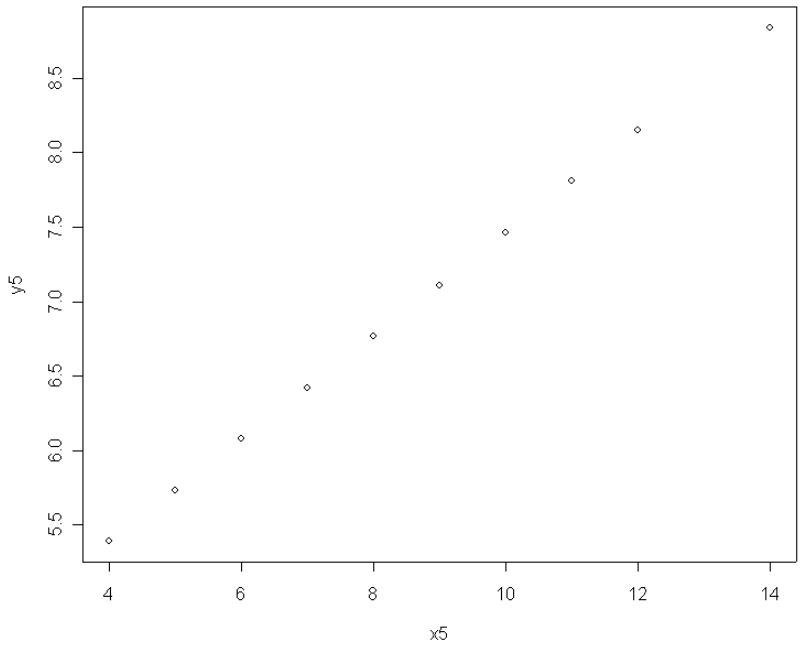
$$D_3 = 1.39$$

We now refit the data omitting this observation:

```
> x5=x3[-3]
> y5=y3[-3]
> model5=lm(y5~x5)
```

Note: we investigate the effect of omitting this data point; we do not simply throw it away and forget about it.

Model 5: $y5$ vs $x5$



Model 5

```
> summary(model5)

Call:
lm(formula = y5 ~ x5)

Residuals:
       Min         1Q     Median         3Q        Max
-0.0041558 -0.0022240  0.0000649  0.0018182  0.0050649

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.0056494  0.0029242    1370   <2e-16 ***
x5          0.3453896  0.0003206    1077   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.003082 on 8 degrees of freedom
Multiple R-squared:      1,Adjusted R-squared:      1
F-statistic: 1.161e+06 on 1 and 8 DF,  p-value: < 2.2e-16
```

For the full data set, the fitted model 3 was

$$\hat{y} \;=\; 3.0025 + 0.4997x$$

With observation 3 omitted, the new fitted model 5 is

$$\hat{y} \quad = \quad 4.0056 + 0.3454x$$

Note substantial changes in estimated intercept and estimated slope, confirming that observation 3 is *influential*.

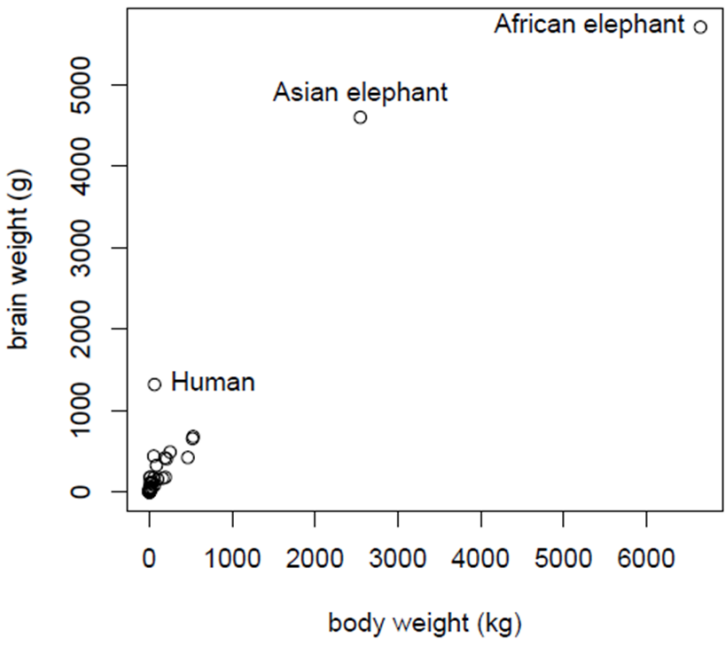
Further regression example: Brain and Body Weights

Brain weights (g) and body weights (kg) for 62 species of mammal ranging in size from the lesser short-tailed shrew (0.005 body weight) to the African elephant (6654 kg body weight). The data are taken from a study by Allison and Cicchetti (1976), and are available in `R` by:

```
> library(MASS)
> data(mammals)
```

Here brain weight is treated as the **response variable** and body weight as the **explanatory variable**. The following `R` code produces the plot of brain weight against body weight. Note that the `identify` command requires appropriate user interaction, to identify interesting points in the plot.

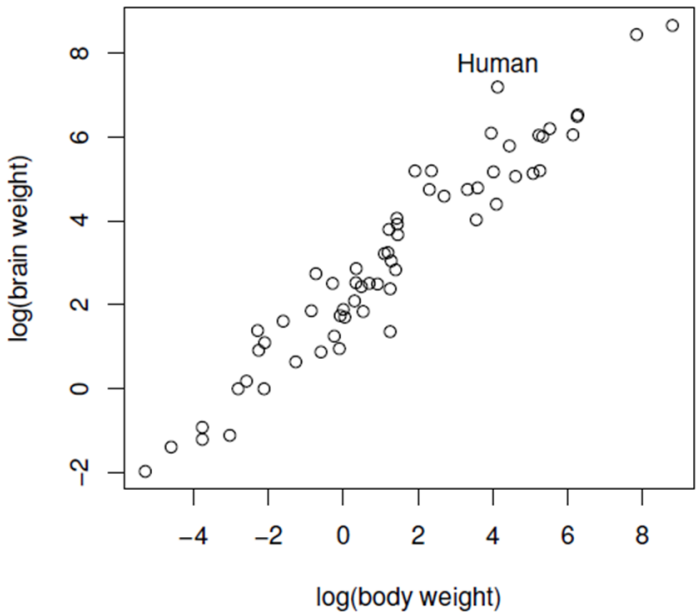
```
> plot(brain ~ body, data=mammals, xlab = "body weight (kg)",
 ylab = "brain weight (g)")
> attach(mammals)
> identify(brain ~ body, labels = row.names(mammals))
```



The plot suggests transformation of the data will be helpful. Severe positive **skewness** of the distributions of both variables suggests **log transformation** for both.

Plot transformed data; remember that `identify` command requires user interaction.

```
> plot(log(brain) ~ log(body), data=mammals,
xlab = "log(body weight)", ylab = "log(brain weight)")
> identify(log(brain) ~ log(body), labels = row.names(mammals))
```



Fit the linear regression model to transformed data:

$$\log(\text{brainweight}) = a + b \times \log(\text{bodyweight}) + \varepsilon$$

In R:

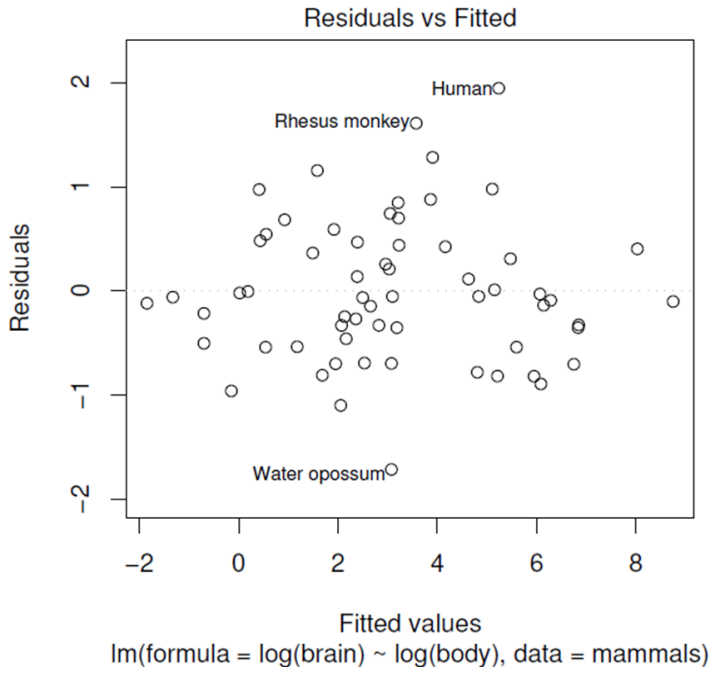
```
> mammals.model.1 = lm(log(brain) ~ log(body))
> summary(mammals.model.1)
> plot(mammals.model.1)
```

The fitted model is

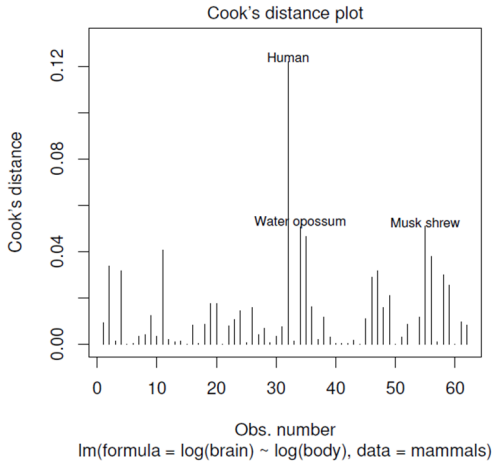
$$\log(\text{brainweight}) = 2.13 + 0.752 \log(\text{bodyweight})$$

In terms of the untransformed variables,

$$\text{brainweight} = 8.46 \times \text{bodyweight}^{0.752}$$



Model fit seems satisfactory



Cook's distance shows no unduly influential points (none with $D > 1$).

Observation with greatest influence on the fit (also with largest residual in this case) is that corresponding to 'Human'.

5.4 Correlation – revision

Our data are (x_i, y_i) , $i = 1, 2, \ldots, n$, a random sample from (X, Y)

The distinction between the roles of X and Y in regression analysis is abandoned and we require X to be a **random variable**. The strength of the association between X and Y (i.e. the strength of the linear relationship between X and Y) is summarised by the **sample correlation coefficient**, r, given as

$$
\begin{aligned}
r &= \frac{S_{xy}}{(S_{xx}S_{yy})^{1/2}} \\
&= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}
\end{aligned}
$$

Notes:

i. $|r| \leq 1$

ii. When data points fall on a straight line

$$r = +1 \quad \text{or} \quad -1$$

depending on whether the line has a positive or negative slope

iii. In simple linear regression,

$$
\begin{aligned}
\text{Coefficient of determination} &= \text{Sample correlation between } x \text{ and } y \\
\text{That is,} \qquad R^2 &= r^2
\end{aligned}
$$

i.e. $100r^2$ is the percentage of the total variation of the y's which is accounted for by the relationship with x

iv. the value of r gives **no information** about "cause and effect"

v. beware "spurious correlations" – usually the result of the presence of other variables, which you may not even know about.

Inference on $\rho = Corr(X, Y)$

Suppose (X, Y) have a *bivariate normal* distribution with correlation coefficient ρ

Recall definition of population correlation:

$$\rho = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

There are two particularly useful results

(i) Result 1 : Testing for **no association**

Under $H_0 : \rho = 0$,

$$\frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$$

This enables us to test $H_0 : \rho = 0$ by comparing the computed test statistic to critical values of t_{n-2} distribution. Equivalent to testing "$b = 0$" in the simple linear regression context. However, this result cannot be used to test $H_0 : \rho = \rho_0$ for non-zero ρ_0, and cannot be used to find CI for ρ.

(ii) Result 2 : Fisher's transformation of r

$$\text{Let} \quad W = \tanh^{-1}(r) \qquad \left[= \frac{1}{2} \log \left(\frac{1+r}{1-r} \right) \right]$$

Then

$$W \sim N \left(\tanh^{-1}(\rho), \frac{1}{n-3} \right)$$

This is a more general result than in (i). It can be used to test $H_0 : \rho = \rho_0$ for any ρ_0, and to find CIs for ρ.

To find a 95% CI for ρ, first notice that 95% CI for $\tanh^{-1}(\rho)$ is

$$W \pm 1.96 \sqrt{\frac{1}{n-3}}$$

So a 95% CI for ρ is

$$\left(\tanh \left(W - 1.96 \sqrt{\frac{1}{n-3}} \right), \tanh \left(W + 1.96 \sqrt{\frac{1}{n-3}} \right) \right)$$
$$= \left(\tanh \left(\tanh^{-1}(r) - 1.96 \sqrt{\frac{1}{n-3}} \right), \tanh \left(\tanh^{-1}(r) + 1.96 \sqrt{\frac{1}{n-3}} \right) \right)$$

Note: have used here the fact that $\tanh^{-1}(r)$ is a monotonic increasing function for $-1 < r < 1$.

More generally, $100(1-\alpha)\%$ CI for ρ is

$$\left(\tanh \left(\tanh^{-1}(r) - z_{\frac{\alpha}{2}} \sqrt{\frac{1}{n-3}} \right), \tanh \left(\tanh^{-1}(r) + z_{\frac{\alpha}{2}2} \sqrt{\frac{1}{n-3}} \right) \right)$$

Chapter 6

Multiple regression

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6.1 Multiple regression: Parameter estimation

Given response variables Y_i $(i = 1, 2, \ldots, n)$ and explanatory variables/predictors $X_1, X_2, \ldots, X_k$, the linear regression model is

$$Y_i \;=\; \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \epsilon_i \qquad (i = 1, 2, \ldots, n)$$

with $E[\epsilon_i] = 0$ and $\mathrm{Var}[\epsilon_i] = \sigma^2$ for $i = 1, \ldots, n$. There are $k+2$ parameters: $\beta_0, \beta_1, \ldots, \beta_k$ and σ^2. Note that β_0 is the intercept term, formerly denoted a.

In matrix notation, the model can be written as

$$\boldsymbol{Y} \;=\; X\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where

$$\boldsymbol{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad X = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix},$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \qquad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}.$$

X is called the **design matrix**. Note: intercept is embedded in the vector $\boldsymbol{\beta}$. To estimate $\beta_0, \ldots, \beta_k$ we use OLS (Ordinary Least Squares).

Minimise square of residuals S:

$$\begin{aligned} S &= (\boldsymbol{Y} - X\boldsymbol{\beta})^T (\boldsymbol{Y} - X\boldsymbol{\beta}) \\ &= \left(\boldsymbol{Y}^T - \boldsymbol{\beta}^T X^T\right)(\boldsymbol{Y} - X\boldsymbol{\beta}) \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - \boldsymbol{\beta}^T X^T \boldsymbol{Y} - \boldsymbol{Y}^T X\boldsymbol{\beta} + \boldsymbol{\beta}^T X^T X\boldsymbol{\beta} \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - 2\boldsymbol{\beta}^T X^T \boldsymbol{Y} + \boldsymbol{\beta}^T X^T X\boldsymbol{\beta} \\ \Rightarrow \quad \frac{\partial S}{\partial \boldsymbol{\beta}} &= -2X^T\boldsymbol{Y} + 2X^T X\boldsymbol{\beta} \end{aligned}$$

At a stationary point,

$$\begin{aligned} \frac{\partial S}{\partial \boldsymbol{\beta}} &= \mathbf{0} \\ \Rightarrow \quad 2X^T\boldsymbol{Y} &= 2X^T X\hat{\boldsymbol{\beta}} \\ \Rightarrow \quad \boxed{\hat{\boldsymbol{\beta}} &= \left(X^T X\right)^{-1} X^T\boldsymbol{Y}} \end{aligned}$$

provided $X^T X$ is non-singular.

$$\begin{aligned} \text{Check for bias:} \qquad \hat{\boldsymbol{\beta}} &= \left(X^T X\right)^{-1} X^T\boldsymbol{Y} \\ &= \left(X^T X\right)^{-1} X^T (X\boldsymbol{\beta} + \boldsymbol{\epsilon}) \\ &= \boldsymbol{\beta} + \left(X^T X\right)^{-1} X^T\boldsymbol{\epsilon} \\ \Rightarrow \quad E\left[\hat{\boldsymbol{\beta}}\right] &= \boldsymbol{\beta} + \left(X^T X\right)^{-1} X^T\, E[\boldsymbol{\epsilon}] \\ &= \boldsymbol{\beta} \end{aligned}$$

That is, $\hat{\boldsymbol{\beta}}$ is unbiased for $\boldsymbol{\beta}$. Fitted values are given by

$$\hat{\boldsymbol{Y}} = X\hat{\boldsymbol{\beta}} = X\left(X^T X\right)^{-1} X^T\boldsymbol{Y} = H\boldsymbol{Y}$$

where $H = X\left(X^T X\right)^{-1} X^T$ is called the **hat matrix**.
The Residual (or Error) Sum of Squares is

$$\begin{aligned} SS_{RES} &= \left(\boldsymbol{Y} - X\hat{\boldsymbol{\beta}}\right)^T \left(\boldsymbol{Y} - X\hat{\boldsymbol{\beta}}\right) \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - 2\hat{\boldsymbol{\beta}}^T X^T\boldsymbol{Y} + \hat{\boldsymbol{\beta}}^T X^T X\hat{\boldsymbol{\beta}} \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - 2\hat{\boldsymbol{\beta}}^T X^T\boldsymbol{Y} + \hat{\boldsymbol{\beta}}^T X^T X\left(X^T X\right)^{-1} X^T\boldsymbol{Y} \\ &= \boldsymbol{Y}^T\boldsymbol{Y} - \hat{\boldsymbol{\beta}}^T X^T\boldsymbol{Y} \end{aligned}$$

We can estimate σ^2 based on this:

$$\boxed{\widehat{\sigma^2} = \frac{1}{n - (k+1)}\, SS_{RES}}$$

(n observations, k explanatory variables.) Note that $\widehat{\sigma^2}$ is unbiased (unlike the MLE).

6.2 Multiple regression: Example

Example 6.1

Fit a linear regression model to the data below.

y	x_1	x_2
3.5	3.1	30
3.2	3.4	25
3.0	3.0	20
2.9	3.2	30
4.0	3.9	40
2.5	2.8	25
2.3	2.2	30

The model is $\boldsymbol{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where

$$\boldsymbol{Y} = \begin{pmatrix} 3.5 \\ 3.2 \\ 3.0 \\ 2.9 \\ 4.0 \\ 2.5 \\ 2.3 \end{pmatrix}, \qquad X = \begin{pmatrix} 1 & 3.1 & 30 \\ 1 & 3.4 & 25 \\ 1 & 3.0 & 20 \\ 1 & 3.2 & 30 \\ 1 & 3.9 & 40 \\ 1 & 2.8 & 25 \\ 1 & 2.2 & 30 \end{pmatrix}$$

OLS estimator is $\hat{\boldsymbol{\beta}} = \left(X^T X\right)^{-1} X^T \boldsymbol{Y}$. To calculate this for our example, can use R

First, input the design matrix.

```
> X = matrix(c(1,1,1,1,1,1,1,3.1,3.4,3.0,3.2, 3.9,2.8,2.2,30,25,20,30,40,25,30),7,3)
```

Then compute $\left(X^T X\right)^{-1}$:

```
> XT = t(X)
> XTX = XT %*% X
> XTXinverse = solve(XTX)
```

Also need to compute $X^T \boldsymbol{Y}$:

```
> Y = matrix(c(3.5,3.2,3.0,2.9,4.0,2.5,2.3),7,1)
> XTY = XT %*% Y
```

Finally, compute $\hat{\boldsymbol{\beta}} = \left(X^T X\right)^{-1} X^T \boldsymbol{Y}$:

```
> betahat = XTXinverse %*% XTY
> betahat
             [,1]
[1,] -0.21381852
[2,]  0.89843390
[3,]  0.01745279
```

So the fitted model is

$$\hat{Y} = -0.2138 + 0.8984x_1 + 0.01745x_2$$

Can produce the result directly using the `lm` command in R

First, copy the appropriate columns of the matrix X into vectors x_1, x_2

```
> x1 = X[,2]
> x2 = X[,3]
```

Then fit the regression model and print a summary.

```
> regression = lm(Y ~ x1 + x2)
> summary(regression)


Call:
lm(formula = Y ~ x1 + x2)

Residuals:
       1        2        3        4        5        6        7
 0.40509 -0.07718  0.16946 -0.28475  0.01181 -0.23812  0.01368

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.21382    0.75024  -0.285   0.7898
x1           0.89843    0.25300   3.551   0.0238 *
x2           0.01745    0.02116   0.825   0.4558
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2902 on 4 degrees of freedom
Multiple R-squared:  0.833,     Adjusted R-squared:  0.7495
F-statistic: 9.975 on 2 and 4 DF,  p-value: 0.02789
```

May also want to examine the hat matrix $H = X\left(X^TX\right)^{-1}X^T$

```
> H = X %*% XTXinverse %*% XT
> H
          [,1]       [,2]        [,3]      [,4]        [,5]        [,6]        [,7]
[1,] 0.15269461  0.1077844  0.08383234 0.1497006  0.20059880  0.12574850  0.17964072
[2,] 0.10778443  0.3498388  0.35329341 0.1418701  0.10087517  0.14532474 -0.19898664
[3,] 0.08383234  0.3532934  0.49700599 0.1017964 -0.20359281  0.24550898 -0.07784431
[4,] 0.14970060  0.1418701  0.10179641 0.1543068  0.22984800  0.11423307  0.10824505
[5,] 0.20059880  0.1008752 -0.20359281 0.2298480  0.80953478 -0.07461999 -0.06264394
[6,] 0.12574850  0.1453247  0.24550898 0.1142331 -0.07461999  0.21441732  0.22938738
[7,] 0.17964072 -0.1989866 -0.07784431 0.1082450 -0.06264394  0.22938738  0.82220175
```

Can also find the fitted values $\hat{\boldsymbol{Y}} = H\boldsymbol{Y}$

```
> Yhat = H %*% Y
> Yhat
         [,1]
[1,] 3.094910
[2,] 3.277176
[3,] 2.830539
[4,] 3.184754
[5,] 3.988185
[6,] 2.738116
[7,] 2.286320
```

Again, these may be found from the output of the `lm` command (previously stored with the name `regression`)

```
> fitted.values(regression)
```

```
       1        2        3        4        5        6        7
3.094910 3.277176 2.830539 3.184754 3.988185 2.738116 2.286320
```

And can examine residuals $\boldsymbol{e} = \boldsymbol{Y} - \hat{\boldsymbol{Y}}$

```
> Y-Yhat     [,1]
[1,]  0.40508982
[2,] -0.07717642
[3,]  0.16946108
[4,] -0.28475357
[5,]  0.01181483
[6,] -0.23811608
[7,]  0.01368033
```

Or from output of `lm`

```
> residuals(regression2)
          1           2           3           4           5           6           7
 0.40508982 -0.07717642  0.16946108 -0.28475357  0.01181483 -0.23811608  0.01368033
```

6.3 Multiple regression: Hypothesis tests and Confidence intervals

Consider now the normal model. That is,

$$\boldsymbol{\epsilon} \sim N\left(\mathbf{0}, \sigma^2 I_n\right)$$

It can be shown that

$$\frac{\hat{\beta}_i - \beta_i}{s.e.\left(\hat{\beta}_i\right)} \sim t_{n-k-1}$$

where

$$s.e.\left(\hat{\beta}_i\right) = \hat{\sigma}\sqrt{c_{ii}}$$

and c_{ii} is the ith diagonal element of $\left(X^T X\right)^{-1}$

Notation: rows/columns are numbered from zero, so the top left diagonal element is c_{00}

Example 6.2

For the data of example 6.1, test whether the intercept is non-zero. First, recall that

$$\widehat{\sigma^2} = \frac{1}{n-k-1}\left(\boldsymbol{Y}^T\boldsymbol{Y} - \hat{\boldsymbol{\beta}}^T X^T \boldsymbol{Y}\right)$$

Using `R`, compute

$$\boldsymbol{Y}^T\boldsymbol{Y} = 67.44, \qquad \hat{\boldsymbol{\beta}}^T X^T \boldsymbol{Y} = 67.1031,$$
$$\widehat{\sigma^2} = \frac{1}{7-2-1}(67.44 - 67.1031) = 0.08422$$

$$\Rightarrow \quad \hat{\sigma} = 0.2902$$

We previously computed $\left(X^T X\right)^{-1}$

```
> XTXinverse
```

```
             [,1]        [,2]         [,3]
[1,]  6.68309535 -1.52924919 -0.063749424
[2,] -1.52924919  0.76001842 -0.028558268
[3,] -0.06374942 -0.02855827  0.005315523
```

So that

$$c_{00} = 6.683, \quad c_{11} = 0.760, \quad c_{22} = 0.0053$$

and then

$$s.e.\left(\hat{\beta}_0\right) \;=\; \hat{\sigma}\sqrt{c_{00}} \;=\; 0.2902 \times \sqrt{6.683} \;=\; 0.7502$$

We have (from earlier) $\hat{\beta}_0 = -0.2138$. Want to test

$$\begin{aligned} H_0 &: \beta_0 = 0 \\ \text{versus } H_1 &: \beta_0 \neq 0 \end{aligned}$$

Test statistic is

$$t \;=\; \frac{\hat{\beta}_0}{s.e.\left(\hat{\beta}_0\right)} \;=\; \frac{-0.2138}{0.7502} \;=\; -0.285$$

4 degrees of freedom, so for two-sided test at 10% level tables give critical value $t_4(95\%) = 2.132$. Cannot reject H_0 at 10% level.

No evidence of non-zero intercept. Or directly from output of `lm` shown earlier.

```
> summary(regression)

Call:
lm(formula = Y ~ x1 + x2)

Residuals:
      1       2       3       4       5       6       7
 0.40509 -0.07718  0.16946 -0.28475  0.01181 -0.23812  0.01368

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.21382    0.75024  -0.285   0.7898
x1           0.89843    0.25300   3.551   0.0238 *
x2           0.01745    0.02116   0.825   0.4558
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2902 on 4 degrees of freedom
Multiple R-squared:  0.833,     Adjusted R-squared:  0.7495
F-statistic: 9.975 on 2 and 4 DF,  p-value: 0.02789
```

Hypothesis tests and Confidence intervals

Can go for through same process for slope parameters β_1, β_2. Confidence intervals can be computed in the usual way. Eg 95% CI for β_1 is

$$\begin{aligned} & \hat{\beta}_1 \pm t_{n-k-1}(2.5\%) \times s.e.\left(\hat{\beta}_1\right) \\ = \quad & 0.8984 \pm 2.776 \times 0.2530 \quad = \quad [0.196, 1.601] \end{aligned}$$

Or in R:

```
> confint(regression, level=0.95)
                  2.5 %     97.5 %
(Intercept) -2.29680563 1.86916860
x1           0.19599296 1.60087484
x2          -0.04129219 0.07619777
```

Variance estimation

Recall

$$\widehat{\sigma^2} \quad = \quad \frac{SS_{RES}}{n-k-1} \quad = \quad \frac{1}{n-k-1}\left(\boldsymbol{Y}^T\boldsymbol{Y} - \hat{\boldsymbol{\beta}}^T X^T \boldsymbol{Y}\right)$$

Similarly to simple linear regression, we have

$$\frac{(n-k-1)\widehat{\sigma^2}}{\sigma^2} \quad \sim \quad \chi^2_{n-k-1}$$

So a 95% CI for σ^2 is

$$\left(\frac{(n-k-1)\,\widehat{\sigma}^2}{\chi^2_{0.025,\,n-k-1}}, \quad \frac{(n-k-1)\,\widehat{\sigma}^2}{\chi^2_{0.975,\,n-k-1}}\right)$$

In our example, $\widehat{\sigma^2} = 0.08422$ and $n-k-1 = 4$
95% CI for σ^2 is

$$\begin{aligned} & \left(\frac{4 \times 0.08422}{\chi^2_{0.025,\,4}}, \frac{4 \times 0.08422}{\chi^2_{0.975,\,4}}\right) \\ = \quad & \left(\frac{4 \times 0.08422}{11.14}, \frac{4 \times 0.08422}{0.4844}\right) \quad = \quad (0.030, 0.695) \end{aligned}$$

95% CI for σ is

$$(\sqrt{0.030},\ \sqrt{0.695}) \quad = \quad (0.174,\ 0.834)$$

ANOVA

We can also test

$$\begin{aligned} H_0 &: \beta_1 = \beta_2 = \cdots = \beta_k = 0 \\ \text{versus } H_1 &: \beta_i \neq 0 \text{ for at least one } i \in \{1, 2, \ldots, k\} \end{aligned}$$

This tests the overall fit of the model. Under H_0 (and the normal model) we have

$$F_0 = \frac{SS_{REG}/k}{SS_{RES}/(n-k-1)} \sim F_{k,n-k-1}$$

and the p-value for H_0 is given by $\Pr(F_{k,n-k-1} \geq F_0)$

Note that for simple linear regression the hypothesis reduces to $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$, and the test turns out to be the same as the t-test.

6.4 Multiple regression: Interpretation

- Some caution needed when interpreting multiple regression analysis.
- Consider importance/effect of an explanatory variable *in the presence of other variables* already in the model.
- The coefficients/parameters and associated standard errors and p-values do not tell the whole story.
- Need to weigh up all the evidence – all the plots and summary statistics, including the contributions to the total Sum of Squares and the overall coefficient of determination R^2

Example 6.3

Return to data of example 6.1 and fit only one explanatory variable. First fit x_1 alone.

```
> model1 = lm(Y ~ x1)
> summary(model1)

Call:
lm(formula = Y ~ x1)

Residuals:
      1       2       3       4       5       6       7
 0.4287 -0.1690  0.0279 -0.2705  0.1349 -0.2737  0.1217

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.004506   0.683082  -0.007  0.99499
x1           0.992201   0.218681   4.537  0.00618 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2808 on 5 degrees of freedom
Multiple R-squared:  0.8046,    Adjusted R-squared:  0.7655
F-statistic: 20.59 on 1 and 5 DF,  p-value: 0.006185
```

Coefficient β_1 is highly significant, with p-value 0.00618.

$R^2 = 0.8046$, variable x_1 explains 80.5% of the variation.

```
> aov(model1)
Call:
   aov(formula = model1)

Terms:
                       x1 Residuals
Sum of Squares  1.6229574 0.3941854
Deg. of Freedom         1         5

Residual standard error: 0.2807794
Estimated effects may be unbalanced
```

Total variation is $S_{yy} = 1.623 + 0.394 = 2.017$.

Variable x_1 accounts for 1.623 of this, which is 80.5% of the total.

Recall original analysis, with both x_1 and x_2

```
> regression = lm(Y ~ x1 + x2)
> summary(regression)
Call:
lm(formula = Y ~ x1 + x2)

Residuals:
       1        2        3        4        5        6        7
 0.40509 -0.07718  0.16946 -0.28475  0.01181 -0.23812  0.01368

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.21382    0.75024  -0.285   0.7898
x1           0.89843    0.25300   3.551   0.0238 *
x2           0.01745    0.02116   0.825   0.4558
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2902 on 4 degrees of freedom
Multiple R-squared:  0.833,     Adjusted R-squared:  0.7495
F-statistic: 9.975 on 2 and 4 DF,  p-value: 0.02789
```

Together the variables explain 83.3% of the variation (not much more than x_1 alone).

```
> aov(regression)
Call:
   aov(formula = regression)

Terms:
                        x1        x2 Residuals
Sum of Squares  1.6229574 0.0573038 0.3368816
Deg. of Freedom         1         1         4

Residual standard error: 0.2902075
Estimated effects may be unbalanced
```

In the presence of x_1, adding x_2 accounts for only a further 0.0573 of the total variation.

Now try fitting x_2 alone.

```
> model2 = lm(Y ~ x2)
> summary(model2)
```

```
Call:
lm(formula = Y ~ x2)

Residuals:
      1       2       3       4       5       6       7
 0.3697  0.3258  0.3818 -0.2303  0.3576 -0.3742 -0.8303

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.59394    1.00446   1.587    0.173
x2           0.05121    0.03445   1.486    0.197

Residual standard error: 0.5289 on 5 degrees of freedom
Multiple R-squared:  0.3065,    Adjusted R-squared:  0.1678
F-statistic:  2.21 on 1 and 5 DF,  p-value: 0.1973
```

```
> aov(model2)
```

```
Call:
   aov(formula = model2)

Terms:
                       x2 Residuals
Sum of Squares  0.6182035 1.3989394
Deg. of Freedom         1         5

Residual standard error: 0.5289498
Estimated effects may be unbalanced
```

Total variation is $S_{yy} = 0.618 + 1.399 = 2.017$, as before.

Variable x_2 accounts for 0.618 of this, which is 30.6% of the total.

Coefficient β_2 is not significantly different from zero (even at 10% level), with p-value 0.197.

What happens if we fit x_2 first, then x_1?

```
> model21 = lm(Y ~ x2 + x1)
> summary(model21)

Call:
lm(formula = Y ~ x2 + x1)

Residuals:
       1        2        3        4        5        6        7
 0.40509 -0.07718  0.16946 -0.28475  0.01181 -0.23812  0.01368

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.21382    0.75024  -0.285   0.7898
x2           0.01745    0.02116   0.825   0.4558
x1           0.89843    0.25300   3.551   0.0238 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2902 on 4 degrees of freedom
Multiple R-squared:  0.833,     Adjusted R-squared:  0.7495
F-statistic: 9.975 on 2 and 4 DF,  p-value: 0.02789
```

```
> aov(model21)
```

```
Call:
   aov(formula = model21)

Terms:
                       x2        x1 Residuals
Sum of Squares  0.6182035 1.0620578 0.3368816
Deg. of Freedom         1         1         4

Residual standard error: 0.2902075
Estimated effects may be unbalanced
```

The variation explained by x_2 here is 0.618, the same as when only x_2 was fitted.

Once x_2 has been fitted, the remaining variation explained by x_1 is 1.062, *not* the same as when only x_1 was fitted.

Together the two variables explain 83.3% of the variation, as before.

Multiple regression: Interpretation

- The **order** in which variables are fitted **does make a difference** in how much variation each explains.
- `summary` t-tests indicate *marginal* significance of the variable in question, given the presence of *all* the other variables.
- `anova` F-tests are sequential — they indicate the significance of the variable in question, given the presence of those variables that appear *above* it in the `anova` output.
- It is clearer and simpler if we fit the variable which explains most of the variation in the responses first, then the variable of next importance, and so on, as measured by SS values.

6.5 Multiple regression: Example

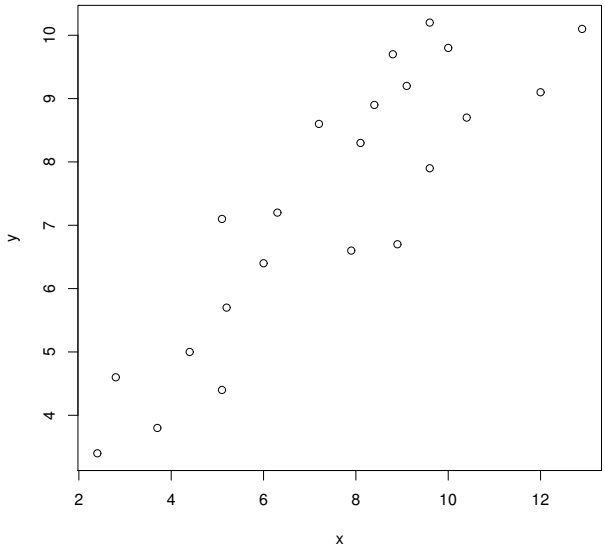
Example 6.4

A model with a *qualitative* explanatory variable (factor).

- Data: $n = 22$ pairs (x_i, y_i) where y is the response.
- Data arise under two different conditions (type = 1 or 2).
- Presented below sorted by x within type.

Row	y	x	Type
1	3.4	2.4	1
2	4.6	2.8	1
3	3.8	3.7	1
4	5.0	4.4	1
⋮	⋮	⋮	⋮
19	9.7	8.8	2
20	9.2	9.1	2
21	10.2	9.6	2
22	9.8	10.0	2

```
> plot(y ~ x)
```



First, fit model ignoring type.

```
> mod3 = lm(y ~ x); summary(mod3);

Call:
lm(formula = y ~ x)

Residuals:
     Min       1Q   Median       3Q      Max
-1.58460 -0.83189 -0.07654  0.79318  1.48079

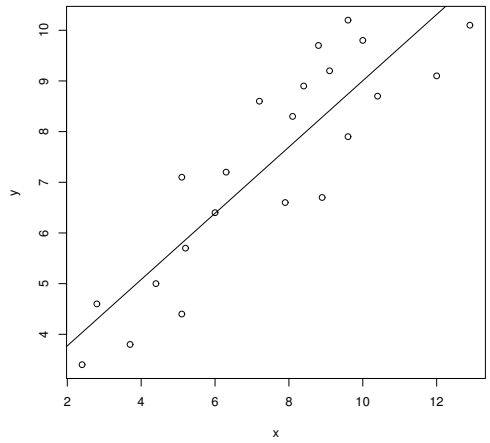
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.4644     0.6249   3.944 0.000803 ***
x             0.6540     0.0785   8.331  6.2e-08 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.033 on 20 degrees of freedom
Multiple R-squared:  0.7763,    Adjusted R-squared:  0.7651
F-statistic:  69.4 on 1 and 20 DF,  p-value: 6.201e-08

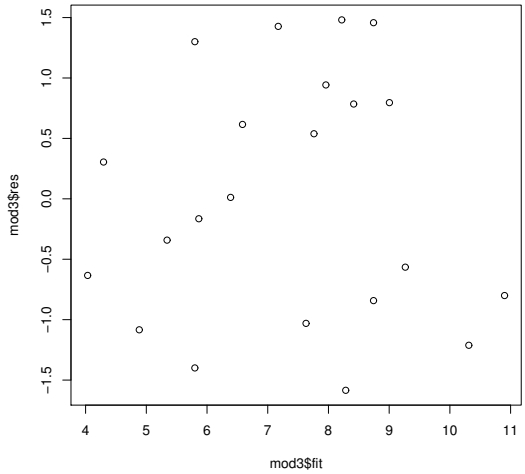
> summary.aov(mod3)

            Df Sum Sq Mean Sq F value  Pr(>F)
x            1  74.03   74.03    69.4 6.2e-08 ***
Residuals   20  21.34    1.07
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

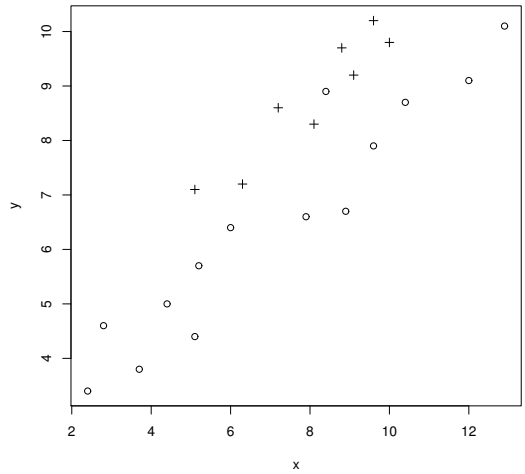
Plot data with fitted line:



Residuals versus fitted values:



Distinguish the two types:



Now fit a model which includes the qualitative variable 'type'

To make sure that R treats the qualitative variable correctly, declare `type` as a factor:

```
> type = factor(type)
> mod4 = lm(y ~ x + type); summary(mod4);
```

```
Call:
lm(formula = y ~ x + type)

Residuals:
     Min       1Q   Median       3Q      Max
-0.90463 -0.39486 -0.03586  0.34657  1.59988

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.18426    0.37348   5.848 1.24e-05 ***
x            0.60903    0.04714  12.921 7.36e-11 ***
type2        1.69077    0.27486   6.151 6.52e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6127 on 19 degrees of freedom
Multiple R-squared:  0.9252,    Adjusted R-squared:  0.9173
F-statistic: 117.5 on 2 and 19 DF,  p-value: 2.001e-11
```

```
> summary.aov(mod4)

            Df Sum Sq Mean Sq F value   Pr(>F)
x            1  74.03   74.03  197.22 1.74e-11 ***
type         1  14.20   14.20   37.84 6.52e-06 ***
Residuals   19   7.13    0.38
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Fitted model is

$$\hat{y} \quad = \quad 2.18426 + 0.60903x \, (+1.69077 \text{ if type } 2)$$

eg Observation 1, with $x = 2.4$ and $type = 1$,

$$\hat{y}_1 \quad = \quad 2.18426 + 0.60903 \times 2.4 \quad = \quad 3.64593$$

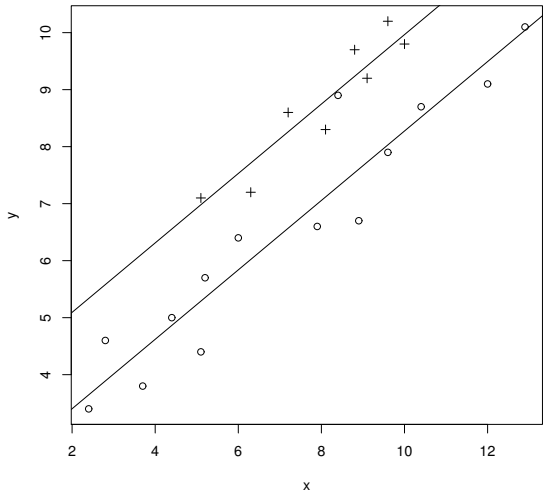
eg Observation 20, with $x = 9.1$ and $type = 2$,

$$\hat{y}_{20} \quad = \quad 2.18426 + 0.60903 \times 9.1 + 1.69077 \quad = \quad 9.41720$$

```
> fitted(mod4)

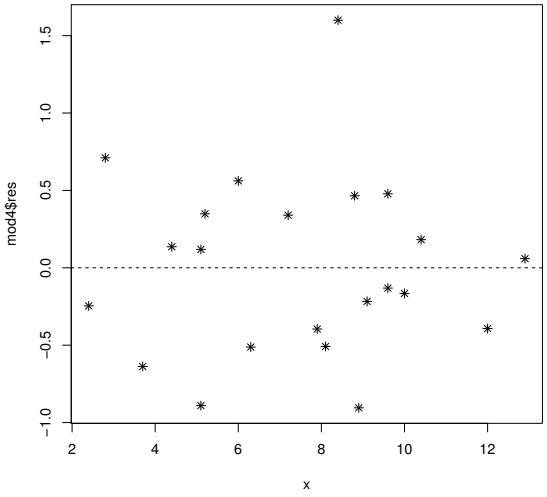
       1        2        3        4        5        6        7        8
3.645930 3.889543 4.437671 4.863993 5.290315 5.351218 5.838443 6.995603
       9       10       11       12       13       14       15       16
7.300119 7.604634 8.030956 8.518181 9.492632 10.040760 6.981083 7.711921
      17       18       19       20       21       22
8.260049 8.808177 9.234499 9.417209 9.721724 9.965337
```

Fitted lines for type 1 and type 2 have same slope, different intercepts.



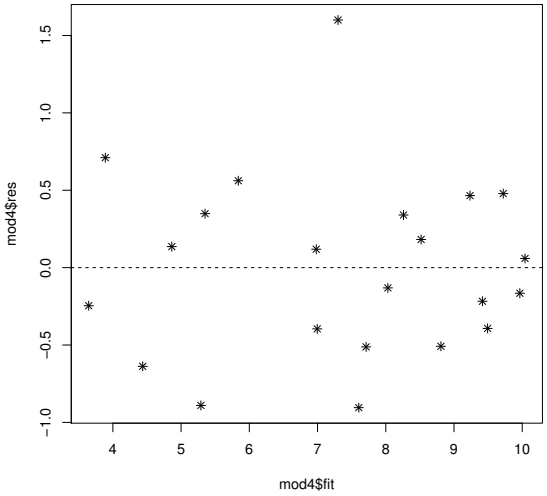
Plot residuals versus x values.

```
> plot(x, mod4$res, pch=8)
> abline(0,0,lty="dashed")
```



Plot residuals versus fitted values.

```
> plot(mod4$fit, mod4$res, pch=8)
> abline(0,0,lty="dashed")
```



Interaction

- Finally, we can extend the model to allow different slopes (as well as different intercepts) for the two types.
- That is, allow an *interaction* between the explanatory variables x and *type*.
- An interaction exists between two explanatory variables when **the effect of one on the response variable is different at different values/levels of the other**.
- For example consider the effect of policyholder's *age* and *gender* on a response variable *claim rate*.

  If the effect of age on *claim rate* is different for males and females, then there is an interaction between *age* and *gender*.

Using data of example 6.4, fit model with interaction between x and *type*.

```
> mod5 = lm(y ~ x * type); summary(mod5);
Call:
lm(formula = y ~ x * type)

Residuals:
     Min       1Q   Median       3Q      Max
-0.90080 -0.38551 -0.01445  0.36309  1.60651

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.22119    0.40345   5.506 3.15e-05 ***
x            0.60385    0.05152  11.721 7.36e-10 ***
type2        1.35000    1.20826   1.117    0.279
x:type2      0.04305    0.14843   0.290    0.775
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.628 on 18 degrees of freedom
Multiple R-squared:  0.9256,    Adjusted R-squared:  0.9132
F-statistic:  74.6 on 3 and 18 DF,  p-value: 2.388e-10


> summary.aov(mod5)
            Df Sum Sq Mean Sq F value   Pr(>F)
x            1  74.03   74.03 187.715 5.81e-11 ***
type         1  14.20   14.20  36.014 1.12e-05 ***
x:type       1   0.03    0.03   0.084    0.775
Residuals   18   7.10    0.39
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- In this particular case, an interaction term is not helpful.
- In a case where an interaction term does improve the fit and the coefficient is significant, then both variables and the interaction between them should be included in the model.

Chapter 7

Other regression techniques

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7.1 Weighted Least Squares

In fitting models of the form

$$y_i = f(x_i) + \epsilon_i \qquad i = 1, \ldots, n$$

Ordinary Least Squares is *optimal* under the condition

$$\epsilon_1, \ldots, \epsilon_n \text{ are i.i.d. } N\left(0, \sigma^2\right)$$

and is a reasonable fitting method when this condition is at least approximately satisfied. Most importantly we require here that there should be **no significant outliers**.

In the case where we have instead

$$\epsilon_1, \ldots, \epsilon_n \text{ are independent } N\left(0, \sigma_i^2\right)$$

it is natural to use instead Weighted Least Squares (WLS):

Choose $\hat{f}$ from within the permitted class of functions f to minimise

$$\sum_i w_i \left(y_i - f(x_i)\right)^2$$

where we can take w_i proportional to $1/\sigma_i^2$
(Note: only relative weights matter)

Example 7.1

Return to hill races data of example 6.6. Seems reasonable to assume greater variability in the times for longer races —- with the variability perhaps proportional to the distance. Refit hills.model.2 with weights proportional to $1/\text{distance}^2$

```
hills.model.3 = lm(time ~ -1 + dist + I(dist^2) + climb
                   +I(climb^2), data = hills[-18,], weights = 1/dist^2)
> summary(hills.model.3)

Call:
lm(formula = time ~ -1 + dist + I(dist^2) + climb + I(climb^2),
    data = hills[-18, ], weights = 1/dist^2)

Weighted Residuals:
     Min       1Q   Median       3Q      Max
-2.39289 -0.39879  0.09138  0.43629  1.79996

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
dist       4.943e+00  5.567e-01   8.879 6.76e-10 ***
I(dist^2)  5.485e-02  3.817e-02   1.437  0.16114
climb      3.494e-03  2.082e-03   1.678  0.10367
I(climb^2) 1.338e-06  4.205e-07   3.182  0.00339 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9236 on 30 degrees of freedom
Multiple R-squared:  0.9855,    Adjusted R-squared:  0.9836
F-statistic: 509.4 on 4 and 30 DF,  p-value: < 2.2e-16
```

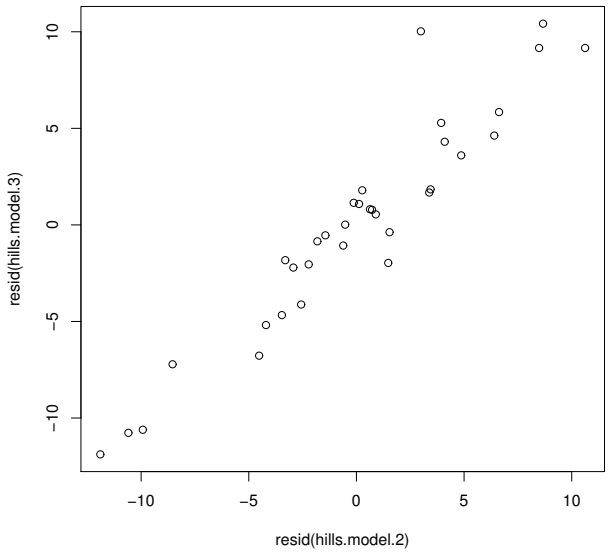
The fitted model now is:

$$\begin{aligned}\text{time} &= 4.94 \times \text{dist} + 0.0549 \times \text{dist}^2 + 0.00349 \times \text{climb} \\ &\quad + 0.00000134 \times \text{climb}^2\end{aligned}$$

Note that:

- Residuals are on "reweighted" scale -– cannot be compared directly to previous model.
- Although coefficients have changed from Model 2 (unweighted), the fitted model is not very different.
- Confirm this by plotting residuals from two models against each other.

```
plot(resid(hills.model.3) ~ resid(hills.model.2))
```



7.2 Resistant regression

As already observed, Least Squares fitting is very sensitive to outlying observations. However, there are also a large number of resistant fitting techniques available. One such is Least Trimmed Squares (LTS): choose $\hat{f}$ from within the permitted class of functions f to minimise

$$\sum_{i=1}^{q} \left(y_i - \hat{f}\left(x_i\right) \right)^2$$

where the sum is taken only over the q smallest squared residuals, with (usually) $q = \lfloor (n+p-1)/2 \rfloor$. This requires iterative refitting of the model until satisfactory convergence is achieved.

Example 7.2: Phones data

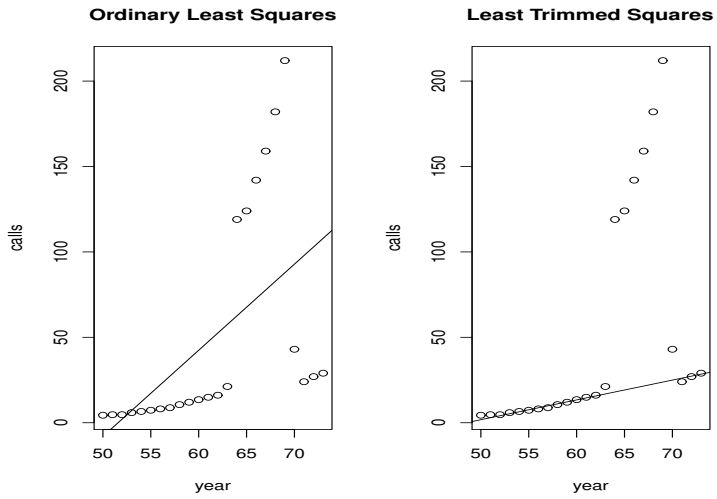
The `R` dataset `phones` in the package MASS gives the annual number of phone calls (millions) in Belgium over the period 1950–73.

Consider the model

$$\text{calls} \quad = \quad a + b \times \text{year}$$

The following two graphs plot the data and show the result of fitting the model by Ordinary Least Squares and then fitting the same model by Least Trimmed Squares.

```
> par(mfrow=c(1,2))
> plot(calls~year,main="Ordinary Least Squares")
> phones.ls=lm(calls~year)
> abline(phones.ls)
> plot(calls~year,main="Least Trimmed Squares")
> phones.lts=lqs(calls~year)
> abline(phones.lts)
```



There's clearly something strange about the data between 1964 and 1970. The explanation is that for a period of time total **length** of all phone calls was accidentally recorded, instead of **number** of calls. The incorrect data drastically affect the OLS fit, but not the LTS fit, which follows the correct data.

7.3 Nonparametric regression

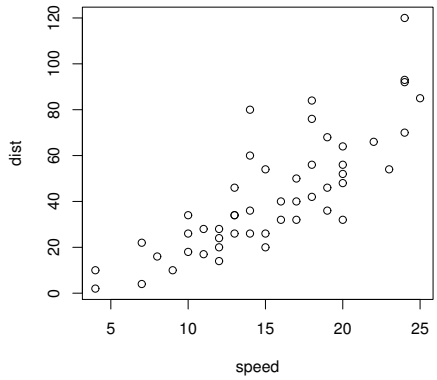
Sometimes we simply wish to fit a smooth model without specifying any particular functional form for f. Again there are very many techniques here. One such is called `loess`. This constructs the fitted value $\hat{f}(x_i)$ for each observation i by performing a local regression using only those observations with x values in the neighbourhood of x_i (and attaching most weight to the closest observations).

Example 7.3: Cars data

The `R` data frame `cars` (in the base package) records 50 observations of *speed* (mph) and *stopping distance* (ft). These observations were collected in the 1920s. We treat stopping distance as the response variable and seek to model its dependence on speed.

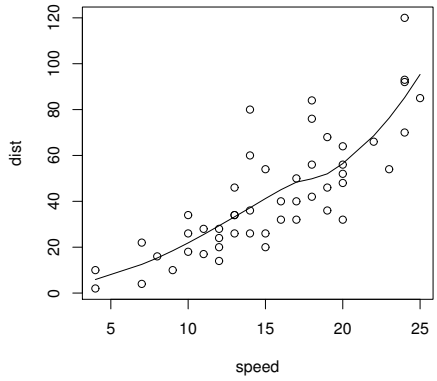
```
> detach(hills)
> data(cars)
> attach(cars)
```

```
> plot(dist ~ speed)
```



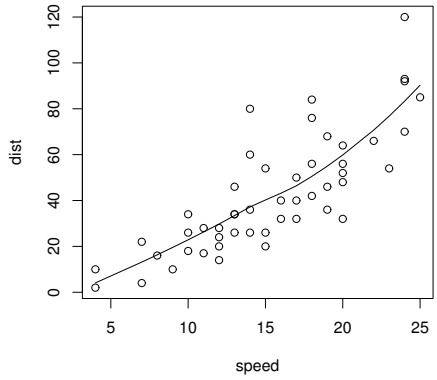
To fit a model using `loess`, possible `R` code is

```
> cars.lo = loess(dist ~ speed)
> plot(cars)
> lines(fitted(cars.lo) ~ speed)
```



The optional argument `span` can be increased from its default value of 0.75 to give more smoothing (use `?loess` for info).

```
> cars.lo2 = loess(dist ~ speed, span=1)
> plot(cars)
> lines(fitted(cars.lo2) ~ speed)
```



More robust and resistant fits can be given by specifying the further optional argument `family="symmetric"`. Again, use `?loess` for more details.

Chapter 8

Generalised linear models

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8.1 Introduction

- The **normal linear regression** model for a response variable Y_i on a single explanatory variable X_i is described by

$$\mathrm{E}\left[Y_i \mid X_i\right] \;=\; \mu_i, \quad i = 1, \ldots, n$$

with

$$\mu_i \;=\; \beta_0 + \beta_1 x_i, \quad Y_i \sim N\left(\mu_i,\, \sigma^2\right)$$

- Generalized Linear Model extends this idea to cases where:
  - the **normal** distribution is **not appropriate**
  - and/or the **identity relationship** between the linear function of x and μ_i **does not hold**.

Example 8.1

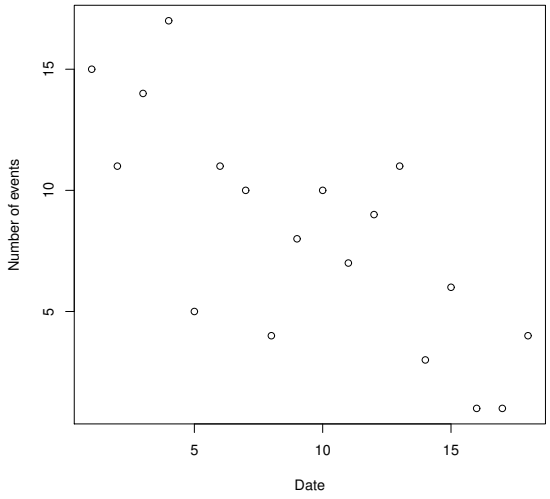
The table below shows numbers of stressful events reported by a sample of people for the previous 18 months, classified by 'date' = number of months before interview.

Date i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Total
Events N_i	15	11	14	17	5	11	10	4	8	10	7	9	11	3	6	1	1	4	147

Here the response variable N_i is a non-negative integer, so the simplest model is

$$N_i \sim \text{Poisson}(\mu_i)$$

where the N_i are assumed independent. Plot the response variable N_i against the explanatory variable i:



It can be seen that the data show a decreasing trend. If we fitted a linear model, then extrapolating to the right could lead us to predict negative numbers of events. Instead, we can fit a (decreasing) exponential curve, called a *log-linear trend model*.

Log-linear trend model

Consider a 'trend model' in which N_x is the number of events in time period numbered x (or for a group indexed by x in some other circumstances) and

$$N_x \sim \text{Poisson}\,(\mu_x)$$

where the N_x are independent.

- Note that $E\,[N_x] = \mu_x$
- Suppose that these means have a trend described as

  $$E\,[N_x] = \mu_x = \exp(\beta_0 + \beta_1 x)$$

  and note that this ensures that μ_x is non-negative.
- Can write the above relationship as

  $$\log(\mu_x) = \beta_0 + \beta_1 x$$

Example 8.2

Suppose D_x is the number of deaths in a year in a group aged x with E_x giving the number exposed to risk, and let q_x denote the mortality rate.

Age x	E_x	d_x/E_x
80	2858	0.110
81	2503	0.122
82	2391	0.133
...		
98	171	0.345
99	124	0.298

- Here D_x is a non-negative integer bounded by E_x, so the simplest model is

$$D_x \sim \text{Binomial}(E_x, q_x)$$

  where the D_x are independent.

- Crude mortality estimates are given by

$$\hat{q}_x = \frac{D_x}{E_x}$$

  and notice that $\mathrm{E}[\hat{q}_x] = q_x$

- We want to model the mortality rate q_x as a function of age x

- A function of the form

$$\log(q_x) = \beta_0 + \beta_1 x$$

  would only guarantee positive q_x

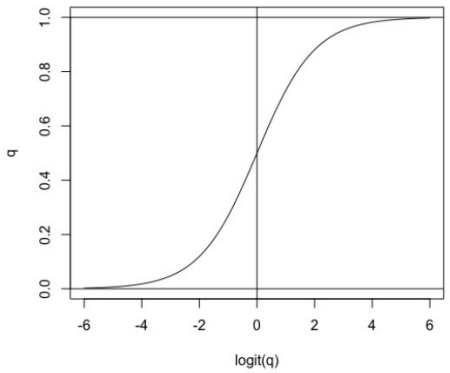
- We want $q_x \in [0, 1]$
  One option is to use the *logistic transformation*

$$\text{logit}(q) = \log\left(\frac{q_x}{1-q_x}\right) = \beta_0 + \beta_1 x$$

Exercise: Show that if $\text{logit}(q) = \eta$, then

$$q = \frac{\mathrm{e}^{\eta}}{1+\mathrm{e}^{\eta}}$$

The (inverse) logistic function



'Sigmoid' shape.

Logistic function as log-odds

Consider the odds for death (vs. survival):

$$\frac{P(\text{death})}{P(\text{survival})} \quad = \quad \frac{q_x}{1-q_x} \quad = \quad \exp(\beta_0 + \beta_1 x)$$

In terms of the logarithm of the odds,

$$\log\left(\frac{P(\text{death})}{P(\text{survival})}\right) \quad = \quad \log\left(\frac{q_x}{1-q_x}\right)$$

So the logistic function applied to the death probability q_x gives the log odds for death.

Logistic GLM

Suppose D_x is the number of successes in E_x trials, and we assume

$$D_x \quad \sim \quad \text{Binomial}(E_x,\, q_x)$$

where the D_x are independent. Suppose the trend can be described through the logistic relationship

$$\log\left(\frac{q_x}{1-q_x}\right) \quad = \quad \beta_0 + \beta_1 x$$

This is the *logistic GLM*.

8.2 Generalised Linear Models (GLMs) :Definition of GLM

A GLM is defined by 3 components:

1. The linear predictor $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$, where $\boldsymbol{\beta}$ is the $p \times 1$ parameter vector and $X = \begin{pmatrix} \mathbf{x}_1^T \\ \vdots \\ \mathbf{x}_n^T \end{pmatrix}$ is the $n \times p$ matrix of explanatory variables.

2. The link function $g(\cdot)$ that relates the mean of the response variable, $\mu_i = \mathrm{E}[Y_i \mid X_i]$, with the linear predictor

$$\begin{aligned} g(\mu_i) &= \eta_i = \mathbf{x}_i^T \boldsymbol{\beta} \\ \Rightarrow \mu_i &= g^{-1}(\eta_i) \end{aligned}$$

3. The random component specifying the distribution of the response variable Y (e.g. normal, Poisson, binomial etc).

Examples

We have already seen the following examples.

- Normal linear regression: $\mu_i = \mathrm{E}[Y_i \mid X_i]$ and
  - $g(\mu_i) = \mu_i = \beta_0 + \beta_1 x_i$ (identity link)
  - $Y_i \sim N(\mu_i, \sigma^2)$

- Poisson trend: $\mu_x = \mathrm{E}[D_x]$ and
  - $g(\mu_x) = \log(\mu_x) = \beta_0 + \beta_1 x$ (log link)
  - $D_x \sim \text{Poisson}(\mu_x)$

- Logistic regression: $\mu_x = \mathrm{E}[\hat{q}_x] = q_x$ and
  - $g(q_x) = \text{logit}(q_x) = \beta_0 + \beta_1 x$ (logit link)
  - $D_x \sim \text{Binomial}(E_x, q_x)$

Definition of GLM continued

- The distribution of the response variable Y must be a member of the exponential family, i.e. have a pdf (or pmf) of the form

$$f(y;\theta) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}$$

  where $a(\cdot), b(\cdot)$ and $c(\cdot,\cdot)$ are functions.

- Parameter θ is called the 'natural parameter' and ϕ is a scale parameter.

Example: Poisson distribution

Poisson(μ) has pmf

$$\begin{aligned} f_Y(y) &= \frac{\mathrm{e}^{-\mu}\mu^y}{y!} \qquad \text{for } y = 0, 1, 2, \ldots \\ &= \exp\left\{\log\left(\frac{\mathrm{e}^{-\mu}\mu^y}{y!}\right)\right\} \\ &= \exp\left\{-\mu + y\log(\mu) - \log(y!)\right\} \\ &= \exp\left\{\frac{y\log\mu - \mu}{1} - \log(y!)\right\} \end{aligned}$$

So the Poisson distribution belongs to the exponential family, with natural parameter $\theta = \log\mu$ and functions $a(\phi) = 1$, $b(\theta) = \mu = \mathrm{e}^\theta$, $c(y, \phi) = -\log(y!)$

Scale parameter ϕ is not needed here, conventional to set $\phi = 1$.

Example: Normal distribution

$N(\mu, \sigma^2)$ pdf is

$$\begin{aligned} f_Y(y) &= \frac{1}{\sigma\sqrt{2\pi}}\exp\left\{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)\right\} \\ &= \exp\left\{\frac{\mu y - \frac{1}{2}\mu^2}{\sigma^2} + \log\left(\frac{1}{\sigma\sqrt{2\pi}}\right) - \frac{y^2}{2\sigma^2}\right\} \end{aligned}$$

So the Normal distribution belongs to the exponential family, with natural parameter $\theta = \mu$, scale parameter $\phi = \sigma^2$, and functions $a(\phi) = \phi$, $b(\theta) = \frac{1}{2}\theta^2$, $c(y, \phi) = \log\left(\frac{1}{\sqrt{2\pi\phi}}\right) - \frac{y^2}{2\phi}$

Exercise: check that Binomial distribution belongs to the exponential family.

- The functions $a(\phi)$, $b(\theta)$, $c(y, \phi)$ are not uniquely defined. For a distribution to belong to the exponential family, we just need some such functions to exist.
- If the function defining the natural parameter θ is used as the link function in a GLM, then the link is known as the **canonical link**, e.g.
  - $N(\mu, \sigma^2)$: natural parameter is mean μ
    canonical link is $g(\mu) = \mu$ (identity)
  - Poisson(μ): natural parameter is $\log(\mu)$
    canonical link is $g(\mu) = \log(\mu)$ (log link)
  - Binomial(E, q): natural parameter is $\log\left(\frac{q}{1-q}\right)$
    canonical link is $g(q) = \log\left(\frac{q}{1-q}\right)$ (logit link)

  Note: the canonical link is just an algebraic convenience, it's fine to use other link functions.

8.3 Fitting a GLM

- In general, to fit a GLM we need an iterative procedure.
- Iterative weighted least squares can be used to obtain estimates (MLEs) for $\boldsymbol{\beta}$
- Use appropriate computer software, e.g. in R use `glm(y ~ x, ...)`.

  See examples below for more details.

Example 8.1-continued

Recall the data of example 8.1: numbers of stressful events reported by a sample of people for the previous 18 months. We will fit a GLM with Poisson response and log link.

First, read the data into R.

```
> x = 1:18
> n = c(15, 11, 14, 17, 5, 11, 10, 4, 8, 10, 7, 9, 11, 3, 6, 1, 1, 4)
```

Fit the model using the `glm` command.

```
> stressmodel = glm(n ~ x, family = poisson)
```

Note that the `family = poisson` subcommand assumes by default a **log link**. We can specify a different link if appropriate. (Take a look at R help using ?`glm`)

```
> stressmodel=glm(n~x, family=poisson)
> summary(stressmodel)

Call:
glm(formula = n ~ x, family = poisson)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.9886  -0.9631   0.1737   0.5131   2.0362

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.80316    0.14816  18.920  < 2e-16 ***
x           -0.08377    0.01680  -4.986 6.15e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 50.843  on 17  degrees of freedom
Residual deviance: 24.570  on 16  degrees of freedom
AIC: 95.825

Number of Fisher Scoring iterations: 5
```

The fitted model is

$$\begin{cases} N_i & \sim & \text{Poisson}\,(\mu_i) \\ \text{with } \hat{\mu}_i & = & \exp\left(2.80316 - 0.08377 x_i\right) \end{cases}$$

So, for instance, 6 months prior to interview, $x = 6$ and the fitted value of expected number of stressful events is

$$\hat{\mu}_6 \quad = \quad \exp(2.80316 - 0.08377 \times 6) \quad = \quad 9.98$$

Fitted values can be obtained in R using

```
> fitted.values(stressmodel)
```

and residuals (by default, *deviance residuals* — see later) using

```
> residuals(stressmodel)
```

We can also calculate confidence intervals for the coefficients in the usual way. eg using `summary` output from before, a 95% CI for the slope parameter b is

$$\begin{aligned} \hat{b} \pm z_{0.025} \times s.e.(\hat{b}) &= -0.08377 \pm 1.96 \times 0.01680 \\ &= (-0.117, -0.05) \end{aligned}$$

Alternatively, use the R command `confint()`

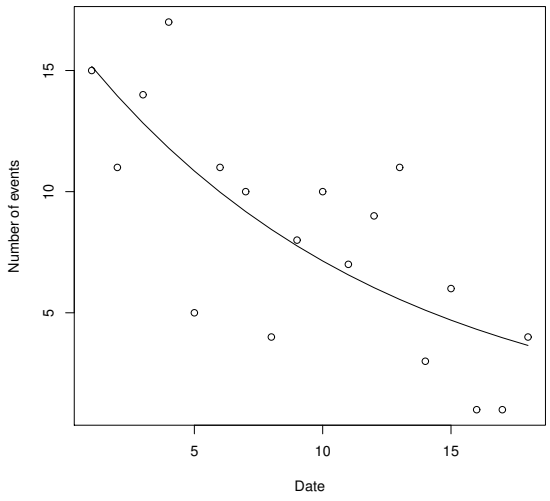
```
> confint(stressmodel)
```

```
Waiting for profiling to be done...
                 2.5 %      97.5 %
(Intercept)  2.5044611  3.08575752
x           -0.1172036 -0.05125814
```

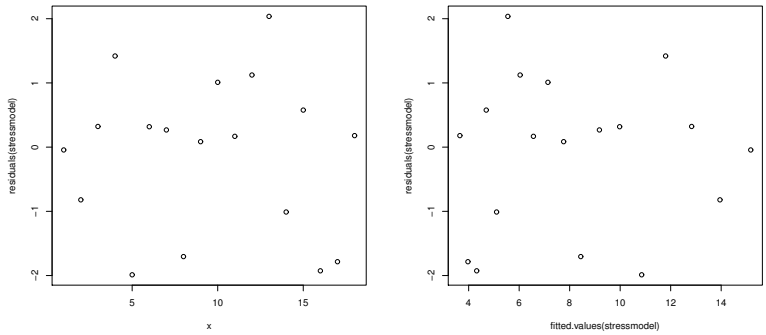
Check `?confint` for details.

The fitted curve can be drawn with the data using

```
> plot(n~x, xlab="Date", ylab="Number of events")
> lines(fitted.values(stressmodel))
```




```
> plot(residuals(stressmodel)~x)
> plot(residuals(stressmodel)~fitted.values(stressmodel))
```



Residuals (deviance residuals) look randomly distributed — no obvious problem with the model.

Example 8.3

- 10 groups of subjects (insects) are exposed to different doses of a drug.
- The response variable is *propdead*, the proportion of subjects in group killed ($y = d/n$), whilst the explanatory variable is *dose*, the dose of drug administered (in suitable units).

Group	Group size (n)	Number killed (d)	Proportion (y)	Dose (x)
1	64	7	0.1094	3.689
2	66	13	0.1970	3.912
3	59	15	0.2542	4.094
4	62	25	0.4032	4.248
5	55	27	0.4909	4.382
6	60	43	0.7167	4.500
7	63	52	0.8254	4.605
8	57	51	0.8947	4.700
9	60	57	0.9500	4.787
10	61	61	1.0000	4.868

Model: suppose that number killed (D_i) in group of size n_i satisfies

$$D_i \sim Bin(n_i, \pi_i),$$

where the D_i are independent.

So proportion killed Y_i in group of size n_i satisfies

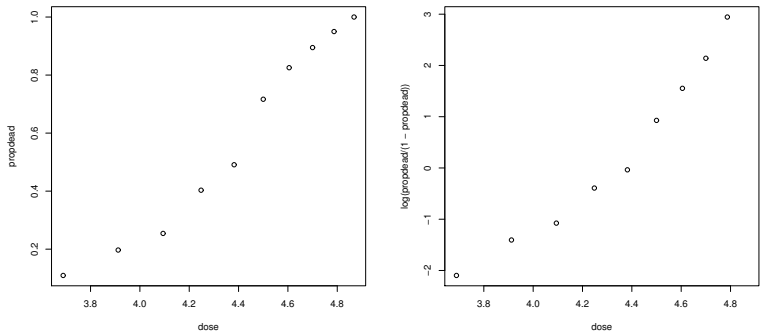
$$n_i Y_i \sim Bin(n_i, \pi_i)$$

With logit link:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i = b_0 + b_1 x_i$$

where b_0 is the intercept and b_1 is the dose effect.

```
plot(propdead ~ dose)
plot(log(propdead/(1-propdead)) ~ dose)
```



Plotting *propdead*, relationship appears non-linear (sigmoid). Becomes more linear when logit transformation is applied. We will proceed with a linear predictor (but may need to revise later).

One way to apply logistic regression in R (see `?glm`):

```
drugmodel = glm(propdead ~ dose, weights = groupsize, family = binomial)
```

Note that in R we

- declare the *proportion* of "successes" (here, proportion of subjects killed) as the response
- include the group sizes as a set of weights
- use binomial distribution
- for binomial, logistic is the default link function

  [otherwise use option `family = binomial(link = "logit")`]

```
> summary(drugmodel)


Call:
glm(formula = propdead ~ dose, family = binomial, weights = groupsize)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.8635  -0.5465   0.2424   1.0777   2.7251

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -20.2724     1.5956  -12.71   <2e-16 ***
dose          4.7331     0.3661   12.93   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 298.067  on 9  degrees of freedom
Residual deviance:  17.234  on 8  degrees of freedom
AIC: 57.3

Number of Fisher Scoring iterations: 4
```

The fitted model is

$$\begin{cases} D_i & \sim & Bin\,(n_i,\,\pi_i) \\ \text{with } \operatorname{logit}(\hat{\pi}_i) & = & (-20.2724 + 4.7331 x_i) \end{cases}$$

So, for instance, for dose $x = 4.0$, the linear predictor is

$$\hat{\eta} = -20.2724 + 4.7331 \times 4.0 = -1.34$$

Now $\eta = \log\left(\frac{\pi}{1-\pi}\right)$, and so (inverting logistic function) the fitted proportion killed for dose $x = 4.0$ is

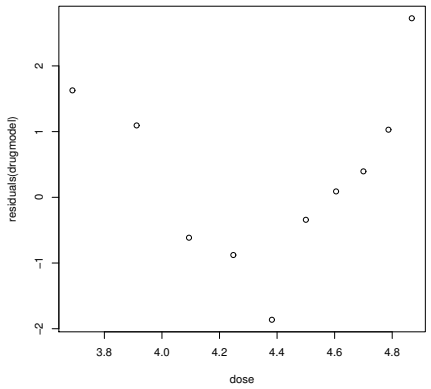
$$\hat{\pi} = \frac{\exp(\hat{\eta})}{1+\exp(\hat{\eta})} = \frac{\exp(-1.34)}{1+\exp(-1.34)} = 0.2075$$

Or directly in R (see `?predict.glm`):

```
> predict(drugmodel, data.frame(dose=4), type="response")
```

0.2075261

```
> plot(residuals(drugmodel) ~ dose)
```



Plot of deviance residuals shows that the model needs to be improved.
To obtain Pearson residuals (see later), would use `residuals(drugmodel, type="pearson")`

8.4 Model Checking: Analysis of Deviance and Residuals for GLMs

- To fit a GLM to data (that is, estimate the regression parameters), we use Maximum Likelihood.
- That is, maximise the log-likelihood $l(\boldsymbol{\beta})$.
- To assess goodness-of-fit, compare model of interest with the 'saturated model', which has one parameter for each observation.
- If l(fitted model) is close to l(saturated model), then the model is a good fit.
- We can assess the fit of the model using the **deviance**.
- The deviance of the model is given by

$$D \quad = \quad -2\,(\,l(\text{fitted model}) - l(\text{saturated model})\,)$$

where

- l(fitted) is the **log-likelihood evaluated at the estimated parameter(s)** (e.g. at the MLE $\hat{\boldsymbol{\beta}}$)
- the **saturated** (or full, or maximal) model is the **model with a different parameter for each observation**

- D is equal to $-2 \times$ log(likelihood ratio statistic)
- The deviance can be used to check the fit of a model, and also to compare the fit of two nested models.

Example: Deviance for logistic regression

Consider the model

$$D_i \sim \text{Bin}(E_i, q_i), \quad \text{independent.}$$

Likelihood function is

$$L(\boldsymbol{q}; \boldsymbol{d}) = \prod_{i=1}^{n} \binom{E_i}{d_i} q_i^{d_i} (1-q_i)^{E_i - d_i}$$

so log-likelihood is

$$l(\boldsymbol{q}; \boldsymbol{d}) = \sum_{i=1}^{n} \left\{ \log \binom{E_i}{d_i} + d_i \log q_i + (E_i - d_i) \log(1-q_i) \right\}$$

To find MLEs, differentiate w.r.t. each q_k:

$$\frac{\partial l}{\partial q_k} = \frac{d_k}{q_k} - \frac{E_k - d_k}{1-q_k} \text{ for each } k$$

At a maximum point, $\frac{\partial l}{\partial q_k} = 0$ for $k = 1, 2, \ldots, n$, and so we find

$$\hat{q}_k = \frac{d_k}{E_k} \text{ for } k = 1, 2, \ldots, n.$$

(Natural estimate — use observed proportion to estimate death probability.)

Note: So far, we have not introduced any link function. This is the *maximal model*, or *saturated model* — a separate parameter q_k for each observation, with no assumed relationship between them.

Now suppose we measure some explanatory variable x_i, and assume the logistic link

$$\log\left(\frac{q_i}{1-q_i}\right) = \beta_0 + \beta_1 x_i$$

Log-likelihood is

$$\begin{aligned} l(\boldsymbol{q}; \boldsymbol{d}) &= \sum_{i=1}^{n} \left\{ \log \binom{E_i}{d_i} + d_i \log q_i + (E_i - d_i) \log(1-q_i) \right\} \\ &= const + \sum_{i=1}^{n} \left\{ d_i \log\left(\frac{q_i}{1-q_i}\right) + E_i \log(1-q_i) \right\} \end{aligned}$$

With logistic link,

$$l\left(\beta_0, \beta_1; \boldsymbol{d}\right) \;=\; const + \sum_{i=1}^{n} \left\{ d_i \left(\beta_0 + \beta_1 x_i\right) - E_i \log\left(1 + \mathrm{e}^{\beta_0+\beta_1 x_i}\right) \right\}$$

Differentiate w.r.t. each parameter:

$$\begin{aligned} \frac{\partial l}{\partial \beta_0} &= \sum_{i=1}^{n} \left\{ d_i - \frac{E_i \, \mathrm{e}^{\beta_0+\beta_1 x_i}}{1 + \mathrm{e}^{\beta_0+\beta_1 x_i}} \right\} \\ \frac{\partial l}{\partial \beta_1} &= \sum_{i=1}^{n} \left\{ d_i x_i - \frac{E_i \, \mathrm{e}^{\beta_0+\beta_1 x_i} x_i}{1 + \mathrm{e}^{\beta_0+\beta_1 x_i}} \right\} \end{aligned}$$

To find MLEs, solve $\frac{\partial l}{\partial \beta_0} = \frac{\partial l}{\partial \beta_1} = 0$ for $\hat{\beta}_0, \hat{\beta}_1$. Needs to be done numerically, in R.
Recall definition of deviance:

$$D \;=\; -2 \left(l(\text{fitted model}) - l(\text{saturated model}) \right)$$

For maximal model, have MLEs $\hat{q}_i = d_i / E_i$, and

$$l\left(\hat{\boldsymbol{q}}; \boldsymbol{d}\right) \;=\; \sum_{i=1}^{n} \left\{ \log \binom{E_i}{d_i} + d_i \log \hat{q}_i + (E_i - d_i) \log(1 - \hat{q}_i) \right\}$$

For fitted model of interest, MLEs $\hat{\beta}_0, \hat{\beta}_1$ must be computed numerically, and then

$$\begin{aligned} l\left(\hat{\beta}_0, \hat{\beta}_1; \boldsymbol{d}\right) \;=\; \sum_{i=1}^{n} \Bigg\{ & \log \binom{E_i}{d_i} + d_i \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right) \\ & - E_i \log\left(1 + \mathrm{e}^{\hat{\beta}_0+\hat{\beta}_1 x_i}\right) \Bigg\} \end{aligned}$$

So putting this together, for the logistic regression model we find

$$\begin{aligned} D \;=\; -2 \sum_{i=1}^{n} \Bigg\{ & d_i \left(\hat{\beta}_0 + \hat{\beta}_1 x_i\right) - E_i \log\left(1 + \mathrm{e}^{\hat{\beta}_0+\hat{\beta}_1 x_i}\right) \\ & - d_i \log\left(\frac{d_i}{E_i - d_i}\right) - E_i \log\left(1 - \frac{d_i}{E_i}\right) \Bigg\} \end{aligned}$$

Could try to simplify this further, but not a lot of point — need to use R to numerically compute $\hat{\beta}_0, \hat{\beta}_1$ anyway, and then R will also compute the deviance.

Exercise: find an expression for the deviance for the log-linear trend model with Poisson response. (You can check your answer by evaluating your expression with the data of example 8.1 and comparing with the deviance value computed by R.)

Analysis of deviance

Test the model fit:

H_0: model fit is adequate vs. H_1: fit not adequate

- Under the hypothesis that model fit is good (H_0), it can be shown that

$$D \sim \chi^2_{n-p} \qquad \text{(approximately)}$$

where n is the number of observations and p the number of parameters fitted.

- **Good models have small deviance.**

Therefore, reject H_0 for large D (one-sided test).
Use critical values from χ^2_{n-p}, or use `R` to compute p-value.

Example 8.1 (Poisson log-linear model)

Recall the data of example 8.1: numbers of stressful events reported by a sample of people for the previous 18 months.

```
> stressmodel=glm(n~x, family=poisson)
> summary(stressmodel)

Call:
glm(formula = n ~ x, family = poisson)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.9886  -0.9631   0.1737   0.5131   2.0362

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.80316    0.14816  18.920  < 2e-16 ***
x           -0.08377    0.01680  -4.986 6.15e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 50.843  on 17  degrees of freedom
Residual deviance: 24.570  on 16  degrees of freedom
AIC: 95.825
```

- From `summary` output, Residual deviance = 24.570. Have 18 observations, fitting a model with $p = 2$ regression parameters, so deviance df $= 18 - 2 = 16$.

- To test goodness-of-fit, compare deviance with χ^2_{16} distribution, so p-value is `1 - pchisq(24.570, df=16) = 0.07777325`. Reject goodness-of-fit at the 10% level, but not at the 5% level.

- That is, at the 5% level, the fitted log-linear model is not a significantly worse fit than the maximal model.

- This model is a reasonably good fit to the data ($p = 0.078$ for deviance goodness-of-fit test).

Example 8.3 (Binomial logistic model)

Recall the data of example 8.3: 10 groups of insects exposed to different doses of a drug.

```
    > summary(drugmodel)
Call:
glm(formula = propdead ~ dose, family = binomial, weights = groupsize)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.8635  -0.5465   0.2424   1.0777   2.7251

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -20.2724     1.5956  -12.71   <2e-16 ***
dose          4.7331     0.3661   12.93   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 298.067  on 9  degrees of freedom
Residual deviance:  17.234  on 8  degrees of freedom
AIC: 57.3
```

- From `summary` output, Residual deviance = 17.234. Have 10 observations, fitting a model with $p = 2$ regression parameters, so deviance df $= 10 - 2 = 8$.
- For testing goodness-of-fit, p-value is `1 - pchisq(17.234, df=8) = 0.02776262`. Reject goodness-of-fit at the 5% level (although not at the 1% level).
- That is, at the 5% level, the fitted logistic regression model is a significantly worse fit than the maximal model.
- Model is not a very good fit to the data ($p = 0.028$ for deviance goodness-of-fit test).
- Recall we previously saw that a plot of residuals for this fitted model showed a clear pattern. Both the residuals plot and the deviance goodness-of-fit test suggest that this model can be improved.

Model residuals

Let $\mu_i = \mathrm{E}[Y_i|X_i]$, and $\hat{\mu}_i$ the corresponding fitted value.

Ordinary ('`response`') residuals $(y_i - \hat{\mu}_i)$ are not generally used in GLMs (as they have non-constant variance). Instead use:

- **Pearson residuals**

$$r_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{\mathrm{var}\,(\hat{\mu}_i)}}$$

- **Deviance residuals**

$$r_i^D \;=\; \text{sign}\,(y_i - \hat{\mu}_i)\,\sqrt{d_i}$$

where d_i is the term of the deviance, D, associated with y_i

To understand the definition of deviance residuals, note that the log-likelihood for the whole sample is the sum of the log-likelihoods of individual observations,

$$l(\boldsymbol{\beta};\boldsymbol{y}) \;=\; \sum_{i=1}^{n} l\,(\boldsymbol{\beta};y_i)$$

and the deviance may be written as

$$D \;=\; \sum_{i=1}^{n}\left\{-2\left(l\left(\widehat{\boldsymbol{\beta}};y_i\right) - l\,(\widehat{\boldsymbol{\mu}};y_i)\right)\right\} \;=\; \sum_{i=1}^{n} d_i$$

Recall Multiple Linear Regression: estimate parameters by minimising Sum of Squared residuals

$$SS \;=\; \sum_{i=1}^{n} e_i^2$$

By analogy with MLR, write

$$D \;=\; \sum_{i=1}^{n} d_i \;=\; \sum_{i=1}^{n}\left(r_i^D\right)^2$$

and choose the sign of r_i^D to agree with that of the raw residual $(y_i - \hat{\mu}_i)$
That is, $r_i^D = \text{sign}\,(y_i - \hat{\mu}_i)\,\sqrt{di}$. `R` gives deviance residuals by default.

Note: for a GLM with non-normal response variable,

- Do not expect residuals to be normally distributed, so no point looking at QQ plot of residuals.
- Do not expect homoscedasticity. For instance, with Poisson response variable, variance increases as mean response μ_i increases (variance = mean for Poisson distribution).

8.5 Model comparison: Analysis of deviance and Akaike Information Criterion

- Recall the deviance (sometimes called the *residual deviance*) is defined as

$$D \;=\; -2\left(l_{\widehat{\boldsymbol{\beta}}} - l_{sat}\right)$$

where

  - $l_{\widehat{\boldsymbol{\beta}}}$ is the maximised log-likelihood for the regression model of interest
  - l_{sat} is the maximised log-likelihood for the saturated (or maximal) model

- Note that we always have $l_{\widehat{\boldsymbol{\beta}}} \leq l_{sat}$, so that deviance values are non-negative.

- The closer the model of interest is to the full (saturated) model, the smaller the deviance will be, and the better the fit.

- **Good models have small deviance**

- How do we check if deviance is "small enough"? If model fits adequately, then

$$D \sim \chi^2_{n-p} \quad \text{(approximately)}$$

  where n is the number of observations and p the number of fitted parameters.

Null deviance

- Consider now the *null model*, in which all mean response are assumed to be equal, so $\mu_i = \mu$ for all i.

- The deviance for this model is called the *null deviance*.

- Notice that in general,

$$\begin{aligned} & l_{null} \leq l_{\widehat{\boldsymbol{\beta}}} \leq l_{sat} \\ \Rightarrow \quad & -2\, l_{sat} \leq -2\, l_{\widehat{\boldsymbol{\beta}}} \leq -2\, l_{null} \\ \Rightarrow \quad & D_{null} \geq D_{\widehat{\boldsymbol{\beta}}} \geq 0 \end{aligned}$$

- If the response is **not related** to the covariate, the model (residual) deviance will not be much smaller than the null deviance.

Example 8.3

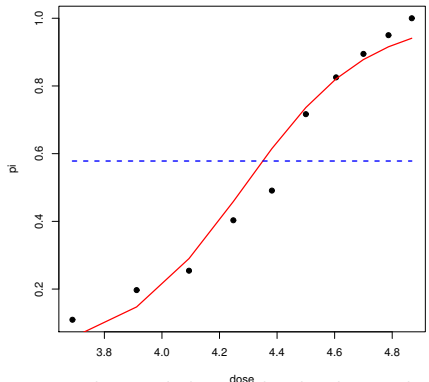
Recall the data of example 8.3: 10 groups of insects exposed to different doses of a drug. We have 3 *nested* models:

- The saturated (full) model
  Fitted values are $\hat{\pi}_i = d_i/n_i$

- The logistic model
  Fitted values obtained by using the logistic function with dose as a covariate.

- The null model
  Fitted values are equal to the common (mean) estimate $\hat{\pi} = (\sum_i d_i) \,/\, (\sum_i n_i)$

```
> pi.mean = sum(numdead)/sum(groupsize)
> plot(propdead~dose, ylab = "pi", pch=19)
> lines(dose, fitted(drugmodel), col="red", lwd=1.8)
> lines(dose, rep(pi.mean,length(dose)), col="blue", lwd=2.0, lty=2)
```



Plot shows all 3 models. Maximal model = black dots; logistic regression model = red curve; null model = blue dashed line.

Recalling that Binomial log-likelihood is

$$l\left(\boldsymbol{\pi};\boldsymbol{d}\right) \;=\; \sum_i \left(\log \binom{n_i}{d_i} + d_i \log\left(\pi_i\right) + \left(n_i - d_i\right) \log\left(1-\pi_i\right) \right)$$

we can compute

$$\begin{aligned} l_{sat} &= -18.033 \\ l_{\widehat{\boldsymbol{\beta}}} &= -26.65 \\ l_{null} &= -167.065 \end{aligned}$$

Deviances are thus

$$\begin{aligned} \text{(Residual) deviance} &= -2\left(l_{\widehat{\boldsymbol{\beta}}} - l_{sat}\right) \\ &= -2\left(-26.65 - (-18.033)\right) \;=\; 17.234 \\ \text{Null deviance} &= -2\left(l_{null} - l_{sat}\right) \\ &= -2\left(-167.065 - (-18.033)\right) = 298.07 \end{aligned}$$

Confirm deviance values from R output:

```
> drugmodel
Call:  glm(formula = propdead ~ dose, family = binomial, weights = groupsize)

Coefficients:
(Intercept)         dose
    -20.272        4.733

Degrees of Freedom: 9 Total (i.e. Null);  8 Residual
Null Deviance:      298.1
Residual Deviance: 17.23        AIC: 57.3
```

Analysis of deviance: model comparison

- Compare the fit of two *nested* models:
  M_1 with variables $x_1, \ldots, x_p$
  M_2 with variables $x_1, \ldots, x_p, x_{p+1}, \ldots, x_{p+k}$

- Test the significance of the additional k parameters in M_2:

$$\begin{aligned} H_0 &: \beta_{p+1} = \beta_{p+2} = \cdots = \beta_{p+k} = 0 \\ \text{vs. } H_1 &: \beta_{p+j} \neq 0 \text{ for at least one } j \in \{1, \ldots, k\} \end{aligned}$$

- We can test the significance of the additional k parameters in M_2 using the change in deviance. Under H_0,

$$D = D_1 - D_2 \sim \chi^2_k \qquad \text{(approximately)}$$

- Reject H_0 for large D (one-sided test)

- The 'bigger' model M_2 will always fit better than the 'smaller' nested model M_1. Not rejecting H_0 means that M_2 does not give *significantly* better fit than M_1, hence prefer M_1.

Example 8.3 (continued)

- Recall for example 8.3, for logistic regression model M_2 and null model M_1, we computed
  residual deviance $D_2 = 17.234$ on 8 degrees of freedom,
  null deviance $D_1 = 298.07$ on 9 degrees of freedom.

- Comparing the two models, change in deviance is $D_1 - D_2 = 298.07 - 17.234 = 280.836$ on $9 - 8 = 1$ d.f.

- Testing whether logistic regression model fits significantly better than the null model, p-value is
  `1 - pchisq(280.836, df=1) = 0` (to machine precision).

- Logistic regression model fits very significantly better than the null model. It seems that the covariate (dose of drug) does have an effect upon the response (insect mortality rate).

Although our linear logistic model fits better than the null model, we previously saw that it is still not a very good fit. Try quadratic model.

```
> drugmodel2 =glm(propdead~dose+I(dose^2), weights=groupsize, family=binomial)
> summary(drugmodel2)
Call:
glm(formula = propdead ~ dose + I(dose^2), family = binomial,
    weights = groupsize)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-0.79342  -0.33844  -0.00908   0.34745   1.68311

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   47.154     18.821   2.505 0.012232 *
dose         -26.989      8.913  -3.028 0.002461 **
I(dose^2)      3.713      1.051   3.531 0.000414 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

    Null deviance: 298.0666  on 9  degrees of freedom
Residual deviance:   4.5903  on 7  degrees of freedom
AIC: 46.656
```

Now consider models M_1 (linear) and M_2 (quadratic).

$$M_1 : \eta = \beta_0 + \beta_1 x$$

$$M_2 : \eta = \beta_0 + \beta_1 x + \beta_2 x^2$$

('Nested': start with M_2, impose the constraint $\beta_2 = 0$, obtain model M_1 nested within M_2.)

Deviance: $D_1 = 17.23$ on 8 df, and $D_2 = 4.590$ on 7 df

Change in deviance:

$$D_1 - D_2 \quad = \quad 17.23 - 4.59 \quad = \quad 12.64 \text{ on 1 d.f.}$$

Testing whether M_2 provides significantly better fit than M_1,

$$\text{p-value} \quad = \quad \Pr\left(\chi^2_1 > 12.64\right) \quad = \quad 0.0004$$

Addition of quadratic term provides significant improvement in the fit of the model.

```
> anova(drugmodel, drugmodel2, test="Chi")

Analysis of Deviance Table

Model 1: propdead ~ dose
Model 2: propdead ~ dose + I(dose^2)
  Resid. Df  Resid. Dev  Df  Deviance  Pr(>Chi)
1         8    17.2340
2         7     4.5903   1   12.644    0.0003768 ***
```

See `?anova.glm`

Using quadratic model M_2, for dose x we have

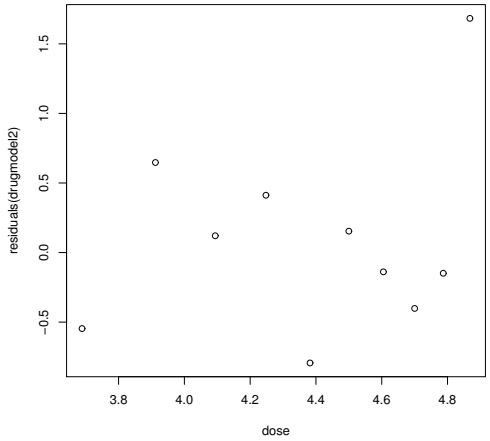
$$\hat{\eta} = 47.15 - 26.99x + 3.71x^2$$

$$\hat{\pi} = \frac{e^{\hat{\eta}}}{1+e^{\hat{\eta}}}$$

For instance, for dose $x = 4.0$, fitted insect death probability is

```
> predict(drugmodel2, data.frame(dose=4), type="response")
0.1978144
```

Plot residuals versus dose for quadratic model.



The residual plot looks much more satisfactory than before.
Plot suggests that this quadratic model provides a better description of our data.

Analysis of deviance

Note on use of deviance in R:

The deviance, as defined for GLMs here, includes the *scale* parameter ϕ

R provides in its output the **residual deviance**, where

$$D = \frac{\text{Residual Deviance}}{\phi}$$

When ϕ is unknown, use an estimate of it for computing D
(e.g. the '`dispersion parameter`' in R.)
You can verify that:

- For Binomial(n, p), $\phi = 1$
- For Poisson(λ), $\phi = 1$
- For N(μ, σ^2), $\phi = \sigma^2$
- For Gamma(a, a/λ), $\phi = 1/a$

Akaike Information Criterion (AIC)

- Another measure of fit is the **Akaike Information Criterion**:

  $$AIC = -2\, l_{\widehat{\boldsymbol{\beta}}} + 2p$$

  where p is again number of parameters fitted.

- Use AIC to compare the fit of different models.
- **Good models should have small AIC**
- AIC also accounts for number of parameters in model; it penalises models with large number of parameters

Consider two nested models M_1 and M_2, with M_2 having more parameters. Rule of thumb:

- If $AIC_{M_1} - AIC_{M_2} > 2$ conclude that M_2 provides substantial improvement as compared to M_1

Only useful for **comparing** models. In example 8.3, for our linear logistic model M_1 we have

$$AIC = -2\,(-26.65) + 4 = 57.28$$

For the quadratic logistic model M_2, we have $AIC = 46.7$. Comparing the two models,

$$AIC_{M_1} - AIC_{M_2} = 57.3 - 46.7 = 10.6,$$

suggesting M_2 should be preferred.

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Chapter 9

Discrete and categorical data analysis: Goodness-of-fit

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9.1 Goodness-of-fit

We have already considered the dependence of discrete measurements (e.g. Poisson, binomial data) to covariates through GLMs. We will now consider the **goodness-of-fit** of theoretical distributions to observed *discrete* and *categorical* data.

Categorical data arise whenever counts (as opposed to measurements) are made. Subjects (sample items) are classified as belonging to one of a set of categories and the numbers in the categories (the frequencies) are recorded.

Example 9.1 (discrete data)

Accidents of 414 workers over a 3 month period.

Number of accidents	0	1	2	3	4	5	≥ 6	Total
Frequency observed	296	74	26	8	4	6	0	414

Example 9.2 (categorical data)

Eye colours of males visiting an optician, in four categories.

Colour	A	B	C	D
Frequency observed	89	66	60	85

Example 9.3 (categorical data)

In a genetic plant breeding experiment a total of 1,500 plants were categorised into one of four classes, with the following results.

Class	C1	C2	C3	C4
Frequency	1071	62	68	299

Both examples 9.2 and 9.3 are examples of *one-way classifications* of categorical data.

Goodness-of-fit

We usually think of the category as the explanatory variable and the distribution of the items among the categories as the response variable. Classifications can be regarded as one of:

1. qualitative (no structural relation between categories)
2. ordered
3. quantitative

Classifications with only two categories (male/female; yes/no; on/off) are called dichotomous or binary.In this case, the distinctions (1)–(3) above make no difference to the analysis.

We can test the goodness of fit of various models (theoretical distributions) to such data using the χ^2 g-o-f test considered previously (for continuous data). In general we consider the probability p_i of observations to belong to category i, and we have the following hypotheses:

$$\begin{aligned} H_0 &: p_i = \pi_i \text{ for } i = 1, 2, \ldots, r \\ \text{versus } H_1 &: p_i \neq \pi_i \text{ for some } i \end{aligned}$$

for some theoretical probabilities $\pi_1, \ldots, \pi_r$

We use the usual (Pearson) χ^2 statistic

$$X^2 = \sum_{i=1}^{r} \frac{(o_i - e_i)^2}{e_i}$$

Under H_0 its value should come from a

$$\chi^2 \textbf{ distribution with } r - p - 1 \textbf{ df}$$

Here

$$\begin{aligned} o_i &= \text{observed frequencies} \\ e_i &= \text{expected frequencies} \\ r &= \text{number of 'cells' (categories)} \\ p &= \text{number of parameters estimated} \end{aligned}$$

As before, we reject H_0 for large values of the X^2 statistic.

As with continuous data, caution is required with sparse data leading to small expected frequencies. Remember that in practice we require $e_i \geq 5$. An alternative rule often used is to ensure that:

$e_i \geq 1$ for all cells, and

$e_i \geq 5$ for at least about 80% of the cells.

If not, **combine adjacent cells** sensibly.

Example 9.1 (continued)

Accidents of 414 workers over a 3 month period.

Number of accidents	0	1	2	3	4	5	≥ 6	Total
Frequency observed	296	74	26	8	4	6	0	414

This is a *quantitative* classification. We want to investigate the fit of a Poisson distribution.

$$\begin{aligned} H_0 &: \text{data come from a Poisson distribution} \\ \text{versus } H_1 &: \text{Poisson not appropriate} \end{aligned}$$

Note that the parameter of the distribution is NOT specified in H_0

First compute expected frequencies using $\Pr(X = x) = \dfrac{\mathrm{e}^{-\lambda}\lambda^x}{x!}$, where λ is estimated from the data as $\hat{\lambda} = 0.47343$.

Number of accidents, x	0	1	2	3	4	5	≥ 6	Total
Observed frequency, o_x	296	74	26	8	4	6	0	414
Expected frequency, e_x	257.9	122.1	28.9	4.6	0.54	0.05	0.004	414

We need to combine cells to achieve $e_x \geq 5$:

Number of accidents, x	0	1	2	≥ 3	Total
Observed frequency, o_x	296	74	26	18	414
Expected frequency, e_x	257.9	122.1	28.9	5.2	414

Calculate $X^2 = 56.87$, and under H_0 should be from χ^2 distribution with $4 - 1 - 1 = 2$ df. p-value $= 4.5 \times 10^{-13}$

Therefore, we have overwhelming evidence against the null hypothesis.

Example 9.2 (continued)

Eye colours of males visiting an optician, in four categories.

Colour	A	B	C	D
Frequency observed	89	66	60	85

Classification here is *qualitative*.

Does the eye colour of males visiting this optician belong to categories A, B,C or D with equal probabilities?

$$\begin{aligned} H_0 &: p_i = 0.25, \quad i = 1, 2, 3, 4 \\ \text{versus } H_1 &: p_i \neq 0.25 \text{ for some } i \end{aligned}$$

The theoretical model (distribution) is *fully specified* by H_0

Theoretical frequencies are easy to compute ($300 \times 1/4$) and we obtain:

$X^2 = 8.03$ which under H_0 should be from χ^2 distribution with $4 - 1 = 3$ df

p-value = 0.045

Therefore, we have some evidence against the hypothesis that there is the same chance for each eye colour.

Example 9.3 (continued)

In a genetic plant breeding experiment a total of 1,500 plants were categorised into one of four classes, with the following results.

Class	C1	C2	C3	C4
Frequency	1071	62	68	299

We now wish to test the fit of the following model:

Class	C1	C2	C3	C4
Probability	$(2+\theta)/4$	$(1-\theta)/4$	$(1-\theta)/4$	$\theta/4$

where θ is an unknown parameter such that $0 < \theta < 1$

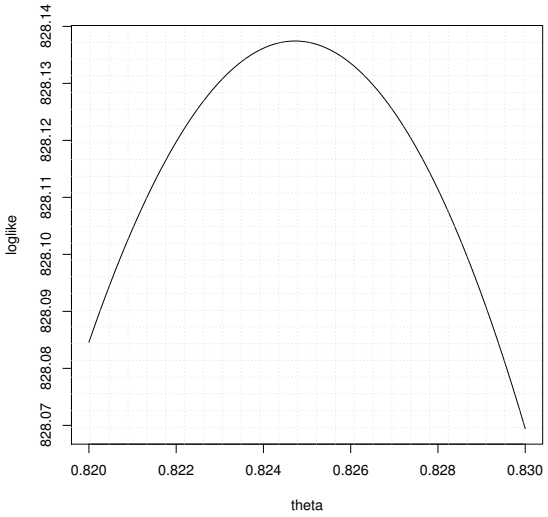
$$\begin{aligned} H_0 &: \text{above model is appropriate for the observed data} \\ \text{versus } H_1 &: \text{model not appropriate} \end{aligned}$$

The model is not fully specified by H_0 and we need to estimate the parameter θ

We will use MLE. Log-likelihood is:

$$l(\theta) = 1071 \log(2+\theta) + 130 \log(1-\theta) + 299 \log(\theta) + c$$

In this case an *analytical solution* can be found (differentiate etc – try this as an exercise). However we will demonstrate how the MLE can be found using the plot of $l(\theta)$ in `R`. First plot $l(\theta)$ on an appropriate range of θ. Then identify in `R` the θ value that corresponds to the maximum.



Plot shows that MLE is given by $\hat{\theta} = 0.8247$. Note that other, more formal numerical methods can also be used here – e.g. Newton-Raphson.

Compute expected frequencies.

Class	C1	C2	C3	C4
Frequency o_x	1071	62	68	299
Expected frequency e_x	1059.3	65.7	65.7	309.3

$X^2 = 0.761$ which under H_0 should be from χ^2 distribution with $4 - 1 - 1 = 2$ df

p-value = 0.684

No evidence at all against the hypothesis that plants are classified according to the suggested model.

Note that in `R`, the command

`chisq.test(x, p = probs)`

where `x` are observed frequencies and `probs` are theoretical probabilities, can **sometimes** be used for a g-o-f test.

However, two important limitations:

- It does not automatically deal with small e_i. This will need to be checked by the user and appropriate action taken.
- It does not allow the number of estimated parameters to be accounted for in d.f.

Chapter 10

Categorical data analysis: Two-way classifications

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Example 10.1: Cinema goers

	Male	Female
Cinema often	22	21
Not often	20	12

- A two-way classification arises when each item may be classified according to two separate criteria. We get a two-way table or cross-tabulation.
- Data can be qualitative, ordered, quantitative.
- Analysis the same – interpretation may be different.
- A two-way table is often called a 'contingency table'

10.1 Notation (2×2 case)

Consider a classification with one **factor** having 2 **levels** (F_1 and F_2), and two different responses (R_1 and R_2):

	F_1	F_2	Total
R_1	n_{11}	n_{12}	$n_{1\bullet}$
R_2	n_{21}	n_{22}	$n_{2\bullet}$
Total	$n_{\bullet 1}$	$n_{\bullet 2}$	$n_{\bullet\bullet} = n$

This is the case in example 10.1.

Can also have a classification with 2 factors.

Denote by p_{ij} the probability that a sample unit belongs to row i and column j

10.2 Usual hypotheses

(a) Multinomial data (*table total n fixed*)

H_0: variables/responses are independent
$\Pr(\text{row } i \text{ and column } j) = \Pr(\text{row } i) \times \Pr(\text{column } j)$

That is,

$$\begin{aligned} H_0 &: p_{ij} = p_{i\bullet} \times p_{\bullet j} \text{ for all } i, j \\ \text{versus } H_1 &: p_{ij} \neq p_{i\bullet} \times p_{\bullet j} \text{ for some } i, j \end{aligned}$$

(b) Condition on n and $n_{\bullet j}$ (*fixed column totals*)

$N_{ij} \sim Bin\,(n_{\bullet j},\, p_{ij}) \quad j = 1, 2, \ldots, c;$ independent
H_0: response is homogeneous ($p_{ij} = p_i$ for all levels j

i.e. null hypothesis is that response has the same distribution for all levels of the factor.

Analysis for both cases is the same – relies on the usual χ^2 statistic.

10.3 Test of no association (independence)

Test independence of response and factor (or two factors) using the Pearson χ^2 statistic:

$$X^2 = \sum_{i=1}^{r}\sum_{j=1}^{c} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

which under H_0 has a χ^2 distribution with $(r-1) \times (c-1)$ df.

Here: r is number of rows
c is number of columns
o_{ij} and e_{ij} are observed and expected frequencies.

In this context,

$$\boxed{e_{ij} = \frac{n_{i\bullet} \times n_{\bullet j}}{n}}$$

Example 10.2

A consumer association has done some research on customer views on the reliability of budget airlines. The results are tabulated below.

Airline	Good	Average	Poor
High life	50	40	30
Sky Flying	40	50	80
Up and Away	35	55	50

Carry out an appropriate hypothesis test to determine if there is an association between the airline and reliability.

H_0: There is no relationship between airline and reliability (i.e. the distributions are independent)

H_1: There is a relationship between airline and reliability

First add marginal totals to the table.

Airline	Good	Average	Poor	Total
High life	50	40	30	120
Sky Flying	40	50	80	170
Up and Away	35	55	50	140
Total	125	145	160	430

Then calculate row and column probabilities and hence expected frequencies under H_0:

Airline	Good	Average	Poor	Total
High life	34.9	40.4	44.7	120
Sky Flying	49.3	57.2	63.5	170
Up and Away	40.8	47.4	51.8	140
Total	125	145	160	430

$$\begin{aligned} \text{Degrees of freedom} &= (\text{rows} - 1) \times (\text{columns} - 1) \\ &= (3-1) \times (3-1) \;=\; 4 \end{aligned}$$

Value of test statistic is

$$X^2 \;=\; \frac{(50-34.9)^2}{34.9} + \frac{(40-40.4)^2}{40.4} + \cdots + \frac{(50-51.8)^2}{51.8} \;=\; 20.425$$

Under H_0 should be from χ^2 distribution with 4 df

p-value= $\Pr(\chi^2_4 \geq 20.425) = 0.0004$

Strong evidence against H_0. Seems reliability is not independent of airline. In `R`, can use `chisq.test`

10.4 Yates' continuity correction

- Used in 2×2 contingency tables for tests of independence.
- We want to apply correction for approximating binomial probabilities with a continuous χ^2 distribution.
- Correction:

  Subtract 0.5 from $|o - e|$ before squaring it.

- This produces p-values that are closer to an 'exact' solution.
- But can be overly conservative.

Example 10.3

In a clinical trial, 50 patients are given a drug (treated group) and 50 are given a placebo (control group); the table gives the numbers suffering particular side-effects.

	Drug	Placebo	Total
Side-effects	15	4	19
No side-effects	35	46	81
Total	50	50	100

H_0: Suffering side effects is independent of choice of drug or placebo

H_1: Not independent

Expected frequencies computed under H_0 are

	Drug	Placebo	Total
Side-effects	9.5	9.5	19
No side-effects	40.5	40.5	81
Total	50	50	100

Under H_0, test statistic is from χ^2 distribution with 1 df

Without Yates' correction, test statistic is $X^2 = 7.8622$, and so p-value $= 0.005$

With Yates' correction, test statistic is $X^2 = 6.4977$, and so p-value $= 0.0108$

In this case, the evidence against H_0 is less strong with correction than one would think without correction.

10.5 Fisher's exact test

- With small data sets the χ^2 approximating distribution is not valid.
- For 2×2 tables we can use **Fisher's test** to calculate exact p-values.

- It relies on the *hypergeometric distribution* for the counts in the table when the row and column margins are considered to be fixed.

Consider the following general case:

	F_1	F_2	Total
R_1	a	b	$a+b$
R_2	c	d	$c+d$
Total	$a+c$	$b+d$	$n=a+b+c+d$

Suppose the smallest observed count is b, say. Then exact probability for observed value b is

$$\Pr(B=b) = \frac{\binom{a+b}{b}\binom{c+d}{d}}{\binom{n}{b+d}} = \frac{(a+b)!\,(c+d)!\,(a+c)!\,(b+d)!}{a!\;b!\;c!\;d!\;n!}$$

The p-value is then given by the probability to observe a value at least as extreme as b

$$\begin{aligned}\text{p-value} &= \Pr(B \le b) \\ &= \Pr(B=b) + \Pr(B=b-1) + \cdots + \Pr(B=0)\end{aligned}$$

We can get a 2-sided p-value as

$$\text{p-value} = \Pr(B \le b) + P(B \ge a)$$

Example 10.3 (continued)

Clinical trial data.

	Drug	Placebo	Total
Side-effects	15	4	19
No side-effects	35	46	81
Total	50	50	100

$$\Pr(B=4) = \frac{\binom{19}{4}\binom{81}{46}}{\binom{100}{50}} = \frac{19!\;81!\;50!\;50!}{15!\;4!\;35!\;46!\;100!} = 0.0039$$

$$\text{p-value} = \Pr(B \le 4) + \Pr(B \ge 15) = 0.0047 + 0.0047 = 0.0094$$

We have evidence against independence. Observe that the exact p-value here is close to the p-value of the χ^2 test with Yates' correction. In `R`, can use `fisher.test`

Example 10.1 (continued)

Cinema goers.

	Male	Female	Total
Cinema often	22	21	43
Not often	20	12	32
Total	42	33	75

$$\begin{aligned} \Pr(\text{Often} \mid \text{Male}) &= 22/42 = 0.524 \\ \Pr(\text{Often} \mid \text{Female}) &= 21/33 = 0.636 \end{aligned}$$

Significant difference?

Is there an association between gender and cinema attendance?

H_0: No association between gender and cinema attendance.

Carry out χ^2 test in R.

```
> matcinema=matrix(c(22,20,21,12),2,2)
> chisq.test(matcinema)

        Pearson's Chi-squared test with Yates' continuity correction

data:  matcinema
X-squared = 0.5522, df = 1, p-value = 0.4574
```

No evidence of association between gender and cinema attendance.

```
> chisq.test(matcinema)$expected

      [,1]  [,2]
[1,] 24.08 18.92
[2,] 17.92 14.08
```

Example 10.4

Consider now a larger sample with same proportions.

	Male	Female	Total
Cinema often	110	105	215
Not often	100	60	160
Total	210	165	375

$$\begin{aligned} \Pr(\text{Often} \mid \text{Male}) &= 110/210 = 0.524 \\ \Pr(\text{Often} \mid \text{Female}) &= 105/165 = 0.636 \end{aligned}$$

Significant difference? Perform χ^2 test in R

```
> matcinema2=matrix(c(110,100,105,60),2,2)
> chisq.test(matcinema2)
        Pearson's Chi-squared test with Yates' continuity correction

data:  matcinema2
X-squared = 4.3361, df = 1, p-value = 0.03731
```

Evidence (at 5% level) to reject H_0

Evidence that there is an association between gender and cinema attendance.

```
> chisq.test(matcinema2)$expected
      [,1] [,2]
[1,] 120.4 94.6
[2,]  89.6 70.4
```